

High-performance heterogeneously integrated photodiodes for microwave photonics and quantum measurements

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The first time I heard the word photodiode was five and half years ago in Prof. Beling's Solid State Devices class when I was a second-year undergraduate student. The following semester, I enrolled in his Photonics Fabrication Lab and that was my first time entering a cleanroom. These two classes had larger impact to my career path than I imagined and to date, I still go back to the lecture notes and learn something new every time. One thing after another, I ended up making one of the best photodiodes in the world on different photonic platforms under the guidance of my advisor, Professor Andreas Beling. It was Prof. Beling who sparked my interest in photonics and gave me the opportunities to explore this field. I would like to thank my advisor Prof. Beling for his guidance, support, patience and understanding. My experience in his group is invaluable and to him, I am forever grateful.

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1 Abstract

Light is widely used for communications, sensing, and, more recently, for quantum applications. However, signal processing is still heavily dependent on electrical components. Therefore, the need for high performance photodiodes, that can convert optical signals into electrical signals, is inevitable. Over the last 20 years, integrated photonics has drawn attention owing to advantages of low loss, scalability, cost, functionality, and performance when compared to discrete components. By heterogeneous integration, advantages from different material systems become accessible all at once. This work presents high-performance (bandwidth, responsivity and saturation power) scalable heterogeneously integrated group III-V photodiodes on non-native platforms including thin-film lithium niobate (TFLN) and silicon nitride.

There are four main aspects of characterizing the performance of a photodiode: bandwidth, responsivity, saturation power, and noise. The noise of the photodiode comes from thermal noise, and shot noise that originates from both, dark current and photocurrent. The scaling of noise is insignificant compared to the signal photocurrent and therefore if the photodiode can work at high photocurrent the signal-to-noise ratio (SNR) typically improves. With larger absorber volume, typically, the responsivity increases and the bandwidth decreases. This work investigates high-speed and high-power devices, drawing connections between physical parameters with device performance, and methods for improvements.

With the realization of heterogeneous integration of high-performance photodiodes, integrated photonics systems for different applications become available. This work also presents photonic systems enabled by heterogeneously integrated photodiodes.

In **Chapter 3**, details on the heterogeneous integration process of photodiodes are discussed along with the waveguide chip design and fabrication process optimization. With the process optimization in multiple fabrications runs, the yield of photodiodes integrated on the photonic TFLN chip increased from 25% to 87%.

In **Chapter 4**, the first heterogeneously integrated photodiode on TFLN with record-high bandwidth of 80 GHz is demonstrated. This was achieved by optimizing the design of the III-V epitaxial layer structure to reach high transit-time limited bandwidth while maintaining high responsivity (0.6 A/W). Low dark current and an open eye diagram at 40 Gbit/s are achieved.

In **Chapter 5**, the high-power performance of photodiodes on TFLN is presented at 48 GHz with -1.9 dBm radio frequency (RF) output power and 10 mA photocurrent. The limitations of saturation power are analyzed and quantified. The space-charge impedance of the photodiode was determined to be $434\ \Omega$, extracted from multiple bandwidth measurements with different current levels and bias conditions. Simulation results for improving the absorption profile are presented, showing the potential to reach the limit as of top illuminated photodiodes for these waveguide integrated photodiodes.

In **Chapter 6**, the first integrated on-chip TFLN-based photonic RF-link is demonstrated enabled by heterogeneous integration of the photodiode and a TFLN modulator. Gain and noise figure measurements are presented.

In **Chapter 7**, the first integrated high-quantum efficiency on-chip homodyne detector is demonstrated on a SiN photonic platform with high quantum clearance of 19 dB.

In **Chapter 8**, methods to achieve photon number resolution single photon avalanche photodiodes (APDs) are discussed with preliminary results presented including the first heterogeneously integrated separate absorption charge and multiplication (SACM) APD on a SiN waveguide.

2 Introduction

2.0.1 Motivations

With the exponentially increasing amount of data from the booming developments and broad implementations of artificial intelligence, the need for developments in data transport and data processing is unavoidable. Electronics solutions have been the main workhorse for the past decades but they also have been showing physical limitations in power consumption, size, and speed [1–3]. Huge amount of efforts have been put in looking for the next generation of integrated platforms that can overcome the limitations of traditional electronic systems. To this end, light, or photons, has attracted attention in the recent decades. Free-space and fiber optics have contributed many breakthroughs across various fields including LIDAR and long-haul and data communications. With progress in material quality and their fabrication processes, ultra-low loss fibers for 1550 nm and 1310 nm wavelengths light came to reality and have gradually replaced data

communications using electrons in copper wires with photons in optical fibers. With the development in nano-scale fabrication, low loss optical waveguides have been realized on miniature photonic chips that are replacing free-space optics. Photonics has become the hope to overcome the fundamental limitations of traditional electronic systems and to support the development for not only handling data but also boosting computational force [4, 5].

Unlike electronics, there are currently many different material platforms for photonic systems and each of them has its own unique trade-offs. Silicon on insulator (SOI) with silicon optical waveguides has a large share in commercial applications due to its low cost, relatively low propagation loss, and complementary metal-oxide semiconductor (CMOS) compatibility [6]. On the other hand, lithium niobate (LN) has become one of the materials of choice due to its material property of large electro-optic coefficient, the large second-order nonlinear susceptibility, and a wide optical transparency window [7]. LN-based devices, such as electro-optic modulators, have been the workhorse of the optoelectronic industry for decades. Recently, the revolutionary technology of thin-film lithium niobate (TFLN) and breakthroughs in nanofabrication techniques have made high-performance-integrated nanophotonic components possible and the manufacturing of wafer-scale high-quality devices on TFLN has become fully commercialized. Moreover, these nanostructure devices, such as optical modulators and nonlinear wavelength converters have already outperformed legacy devices in bulk form [8]. Nanofabrication breakthroughs enabled ultra-low-loss resonators to be built on TFLN which realized novel applications, such as optical frequency combs and quantum transducers [9]. TFLN high-speed electro-optic modulators have become commercialized and work fully compatibly with CMOS voltages. Silicon nitride (SiN) is another promising platform due to its ultra-low propagation loss, low two-photon absorption, and low temperature sensitivity [10]. It has become a popular material platform especially for quantum applications such as squeezing of light [11, 12]. This thesis focuses on heterogeneous photodiodes on TFLN and SiN photonic platforms. Comparing to traditional silicon photonic chips, TFLN and SiN both offer distinct features and advantages which silicon can not offer.

Since different material platforms offer different utilities, heterogeneous integration of bringing all components and functionalities onto a single photonic miniature chip has drawn more attention with the maturity of single component development. All photonic

	Refractive index at 1550 nm	Waveguide height	Single mode condition
TFLN	2.21	0.25 μm	0.8 μm
SiN	1.974	0.8 μm	1 μm

Table 1: Material and optical waveguide properties of TFLN and SiN used in this thesis

circuits still rely on electronic readout circuits and therefore photodiodes that can convert optical signals into electrical signals, are essential and irreplaceable components in all photonic systems. Instead of coupling the optical signal to an off-chip discrete photodiode, having a heterogeneously integrated on-chip photodiode reduces optical loss and increases packaging density. In industry today, most heterogeneous integrated photodiodes (PDs) for the detection of 1310 - 1550 nm wavelength light are made of Ge by growing it on Si [6]. These Ge photodiodes offer high speed [13], relatively low cost, and they are compatible with the foundry fabrication processes. However, it is difficult to engineer Ge PDs to target different wavelengths and they typically have higher dark current which is critical for applications that require low noise or high signal-to-noise ratio. In addition, their high-power handling capability is limited. In contrast to Ge, group III-V PDs can have direct bandgaps and can be engineered for different purposes using heterostructures and thus resulting in high-performance active device including photodiodes [14]. This dissertation presents works on heterogeneous integration of III-V photodiodes on TFLN and SiN platforms for 1550 nm wavelength.

2.0.2 Fundamentals of photodiodes

Photodiodes convert optical power to electrical current by the photogeneration of electron-hole pairs when photons with equal or larger energy than the material bandgap are being absorbed. For 1550 nm wavelength, the InGaAsP/InAlGaAs/InP material system was chosen for its appropriate range of bandgaps, lattice match to InP substrate, and its wide availability from commercial crystal growers. This work use InGaAs as the absorber for 1550 nm for heterogeneously integrated photodiode and InP as the drift layer. Material constants that I used in my simulations are shown in table 2. More details about the band structure design and influence on bandwidth will be discussed in chapter 3.

The external responsivity R_{ext} quantifies the ability of the photodiode to transform optical power into electrical current from the experimental data and is defined as the measured average photocurrent I_{ph} generated per optical power which is incident to the

	InGaAs	InP
Bandgap	0.75 eV	1.35 eV
Refractive index at 1550nm (n _{ik})	(3.5307, 0.076399)	(3.1649, 0)
Dielectric constant	13.8	12.5
Electron saturation velocity	$1 \times 10^7 \text{ cm/s}$ [15]	$1 \times 10^7 \text{ cm/s}$ [16]
Hole saturation velocity	$0.45 \times 10^7 \text{ cm/s}$	

Table 2: Material properties of InGaAs and InP

photonic chip (measured at the tip of the input fiber) P_{opt} with the unit of A/W:

$$R_{ext} = \frac{I_{ph}}{P_{opt}} \quad (1)$$

The external quantum efficiency η_{ext} is defined as the number of charge carrier pairs generated per incident photon:

$$\eta_{ext} = \frac{I_{ph}}{q} \cdot \frac{hv}{P_{opt}} \quad (2)$$

For 1550 nm wavelentgh, the relationship between R_{ext} and η_{ext} is:

$$R_{ext} = \eta_{ext} \cdot 1.25A/W \quad (3)$$

The internal responsivity R_{int} is defined as the measured photocurrent generated per optical power that the photodiode receives $P_{waveguide}$ excluding any system loss such as fiber-to-chip coupling loss, reflection loss at the facet, waveguide loss, and any loss from passive components on the photonic chip before reaching to the photodiode:

$$R_{int} = \frac{I_{ph}}{P_{waveguide}} \quad (4)$$

And similarly internal quantum efficiency at 1550 nm can be related to the internal responsivity:

$$R_{int} = \eta_{int} \cdot 1.25A/W \quad (5)$$

When a photodiode is under reverse bias, the total current through the depletion layer composes the drift current due to carriers generated within the depletion region and the diffusion current due to carriers generated outside the depletion region and can

be expressed as [17]:

$$J_{tot} = q\Phi_0[1 - \frac{\exp(-\alpha W_D)}{1 + \alpha L_p}] + \frac{qp_{n0}D_p}{L_p} \quad (6)$$

where the photocurrent density J_{ph} is:

$$J_{ph} = q\Phi_0[1 - \frac{\exp(-\alpha W_D)}{1 + \alpha L_p}] \quad (7)$$

and the dark current density J_{dark} is:

$$J_{dark} = \frac{qp_{n0}D_p}{L_p} \quad (8)$$

$$L_p = \sqrt{D_p\tau_p} \quad (9)$$

Where Φ_0 is the incident photon flux, W_D is the depletion-layer width, D_p is the diffusion coefficient for holes, p_{n0} is the equilibrium hole density and τ_p is the lifetime of excess minority carriers.

The shot noise from the photocurrent originates from the photon nature and randomness of the light and can be described by a Poisson distribution. The shot noise current I_{shot} of a photodiode is related to the total current, dark current plus photocurrent, at certain measurement resolution bandwidth (RBW) and assuming 50Ω:

$$I_{shot} = \sqrt{2 \cdot q \cdot I \cdot RBW} \quad (10)$$

$$P_{shot} = 10 \cdot \log \left(\frac{I_{shot}^2 \cdot 50\Omega}{1mW} \right) \quad (11)$$

For example, if the resolution bandwidth is 100 kHz, with 100 mA current, the shot noise power P_{shot} is -97.95 dBm which is very close to the noise floor of most measurement equipment. Therefore, usually a low-noise amplifier is used in the system, and the shot noise can be expressed as in current and the amplifier linear gain:

$$P_{shot} = 10 \cdot \log \left(\frac{I_{shot}^2 \cdot 50\Omega \cdot gain}{1mW} \right) \quad (12)$$

One thing to note is that the noise equations are only true when the measurements are

done within the bandwidth of both photodiode and amplifier operation range since above this bandwidth, the combination of these two acts like a low pass filter which results in lower output noise power.

2.0.3 Fundamentals of Mach–Zehnder modulators and radio frequency (RF) optical links

Due to the material property of LN, when an externally applied electric field across the optical waveguide is present, the optical signal changes its phase when passing through this region. When the applied voltage is 0 V, there is no phase difference between both

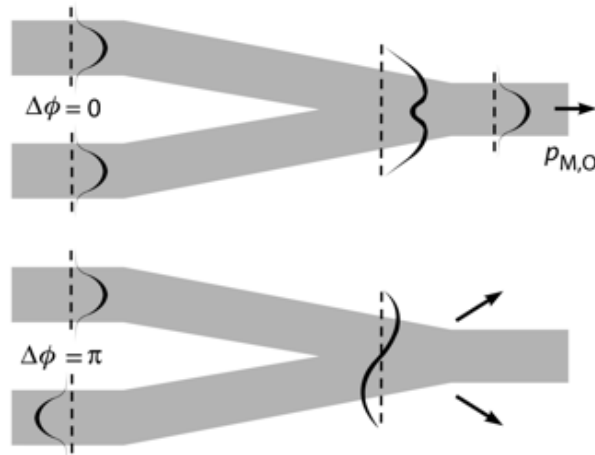
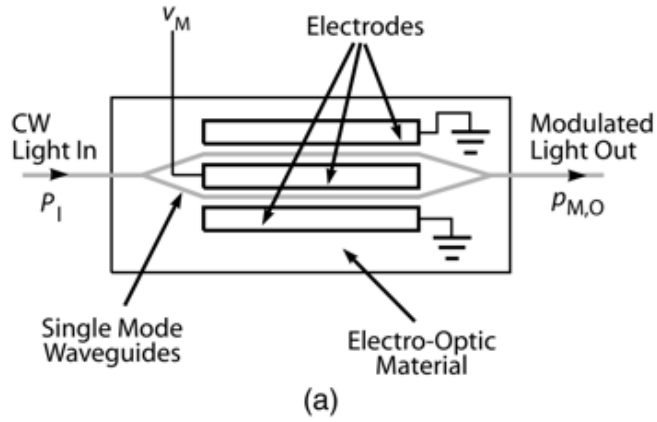


Figure 1: Illustration of optical intensity modulator [18]

arms and light in both arms gets combined resulting in maximum output. When there is a voltage applied, the refractive index changes which in turn changes the optical phase velocity, and therefore changes the phase of the light propagating in both arms but with opposite signs. Since the signals in the two arms are out of phase, when they are combined, they excite higher order modes which are not supported in the single mode waveguide

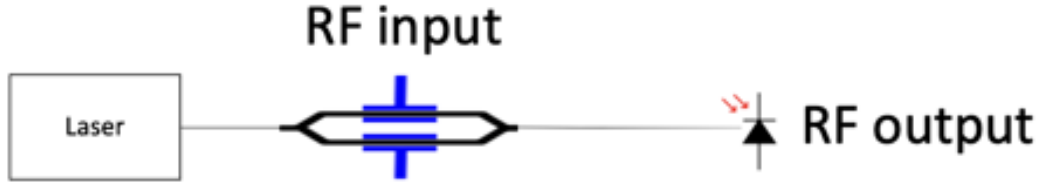


Figure 2: RF Link

at the output and therefore the output decreases as the voltage applied increases. To achieve a phase difference of π , the required external voltage is defined as V_π . The relationship between the optical output power of a MZM modulator and voltage applied can be expressed as:

$$P_{out} = \frac{P_{in}}{2} (1 + \cos(\frac{\pi v_{applied}}{V_\pi})) \quad (13)$$

When a small-signal electrical modulation, $v_{applied}$ is applied in addition to a bias point, $V_{applied}$, the output optical signal can be expressed as [18]:

$$P_{out} = \frac{P_{in}}{2} [1 + \cos(\frac{\pi V_{applied}}{V_\pi}) \cos(\frac{\pi v_{applied}}{V_\pi}) - \sin(\frac{\pi V_{applied}}{V_\pi}) \sin(\frac{\pi v_{applied}}{V_\pi})] \quad (14)$$

In order to achieve the maximum modulation efficiency, meaning the output optical power changes the most with $v_{applied}$, the bias point should be chosen as $V_\pi/2$, this is called the quadrature point of an MZM modulator. Therefore the small signal relationship between the optical output and electrical input, v_m , can be approximated as [18]:

$$p_{output} = \frac{P_{input}}{2} \left(-\frac{\pi v_m}{V_\pi} \right) \quad (15)$$

The electrical power from a modulation voltage source, v_s , on an impedance, R_s , can be expressed as:

$$p_{load} = \frac{v_s^2}{4R_s} \quad (16)$$

When the modulator is terminated with the same impedance as the source, then $v_m = v_s/2$. Therefore the electrical power can be expressed as:

$$p_{load} = \frac{v_m^2}{R_s} \quad (17)$$

For a photonic RF link as shown in fig. 2, the gain is defined as the ratio of output electrical RF power from the photodiode and the input electrical RF power applied to the modulator and therefore can be expressed as:

$$Gain = \left(\frac{P_{in} \pi R_s}{2V_\pi} \right)^2 \cdot R_{ext}^2 \quad (18)$$

Another important figure of merit for an RF link is noise figure (NF). It is defined as the signal-to-noise ratio (SNR) to input thermal noise.

$$NF = 10 \log \left(\frac{N_{out}}{Gain N_{th}} \right) \quad (19)$$

$$N_{th} = k \cdot T \cdot RBW \quad (20)$$

Where N_{th} is the thermal noise power, k is the Boltzmann constant, and T is temperature.

3 Heterogeneous integration process of photodiodes on dissimilar photonic platforms

3.1 Introduction and motivation

With developments in nanofabrications, photonic integration has drawn more attention and achieved further progress. The motivation for photonic integration is to integrate discrete, fiber-coupled photonic components into miniature, scalable chips that are more compact, and potentially lower loss and more resilient to environment. Data communications and optical computing, as well as sensing and quantum applications have been a major driving force for the integrated photonics field because of their need for low loss, scalability, and high stability during measurements due to the sensitive nature of e.g. quantum signals. Another motivation comes from the push of computational power since the low loss and high-bandwidth nature inherited in photons, energy consumption and computational speed can reach another breakthrough by utilizing integrated photonics.

Usually a passive photonic waveguide chip includes a silicon substrate, a buried oxide layer, an optical waveguide layer with higher refractive index, and a top cladding layer. There are many materials of choice for the optical waveguide for different applications such



Figure 3: Heterogeneous integration techniques illustration (a) Cross section of monolithic growth of an InGaAs/InP photodiode on silicon substrate [19] (b) Process illustration of transfer printing [20] (c) Process illustration of bonding [21]

as silicon nitride which has low propagation loss and low-temperature sensitivity, lithium niobate which has low-loss optical modulation and efficient all-optical nonlinearities, and silicon for tight bending radii and relatively low propagation loss and low cost. One thing to note is that low propagation loss has been realized in most materials by efforts in the fabrication process optimization [22, 23]. Group III-V materials have been proven to be the material of choice for active components such as lasers [24] and photodiodes [25]. Conventionally, active components like lasers and detectors are off-chip, and light has to be coupled from the laser to the photonic chip using fiber and the optical signal is then coupled to off-chip detectors experiencing at least two times of coupling loss at the fiber-chip interfaces. In order to minimize this loss especially after the generation of optical signal to detection, it is essential to have photodiodes on-chip. Integrating two different material systems like passive photonic waveguide and active group III-V materials on the same chip usually is considered as heterogeneous integration. There are three main categories of different techniques for heterogeneous integration: monolithic heterogeneous growth [19], transfer printing, and wafer or die bonding. Monolithic growth refers to growth of III-V material onto other material by engineering the intermediate layers to overcome the lattice mismatch. Both lateral and vertical deposition have been demonstrated and impressive results have been showed for active components with these techniques [19]. Transfer printing refers to a pick-and-place method by first fabricating the device and then separate them from the source wafer and place them into the desired place using a transfer printer [20] on the target wafer. Bonding refers to bond an unprocessed

wafer or die of III-V material first and then process them into singular devices [21]. Bonding becomes the choice of heterogeneous integration for this work with the advantage of lithographic alignment accuracy, preserving crystalline quality of the III-V material, and most importantly accessibility.

3.2 Bonding

3.2.1 Introduction

By the difference in chemical bonds formed at the interface, bonding can be categorized into two types, direct bonding and adhesive bonding. Direct bonding frequently involves the two interfaces being silicon dioxide. And by applying high pressure and high temperature under vacuum, molecular bonds between the oxides form. This reaction forms hydrogen gas as a side product [26]. In order to increase the bonding yield, usually patterns are added to the bonding area to release this gas. Direct bonding requires high quality surface in both cleanliness and flatness (0.6 nm [26]). Adhesive bonding refers to using an adhesive in between the two material to be bonded, and the chemical bond is a Van der Waals bond. There are two common polymer adhesives used for bonding: SU8 and Benzocyclobutene (BCB). Since the adhesive is in liquid form, it can also serve as planarization agent which relaxes the restriction on surface condition. By first spin coat the photonic waveguide chip a layer of adhesive and curing them by either or both UV exposure and long time hard bake, the polymer cross links and becomes robust. And then similar to direct bonding, by applying high pressure and high temperature under vacuum condition, the chemical bonds form. In my work, SU8 was used as a bonding adhesive because its longer shelf-time comparing to BCB, lower process temperature required, and similarities with conventional photoresist in processing [27]. A general illustration of the heterogeneous integration process of photodiodes using adhesive bonding is shown in fig4. More details about the process and recipes can be found in the appendix.

3.2.2 Cross section comparison

Due to the different conditions of the photonic waveguide chips over the course of my work, I changed the bonding process to accommodate this. Figure 5 shows three different structures I worked with. For the trench waveguide structure, the optical waveguide can

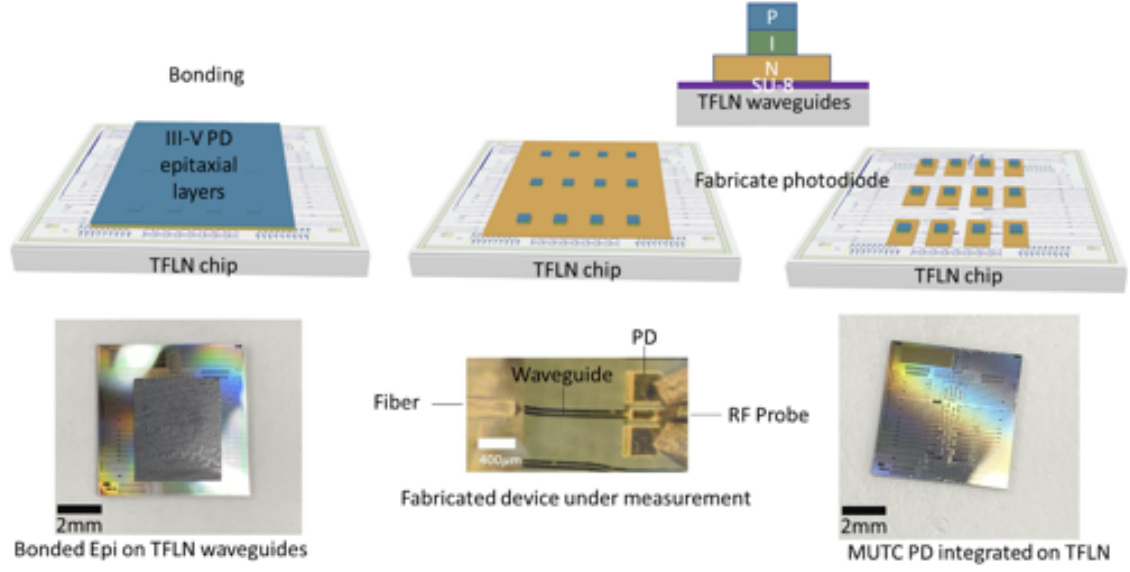


Figure 4: Illustration of heterogeneous integration process of photodiodes using adhesive bonding

be written in positive resist. Small areas in the trenches need to be planarized by SU8 (150 nm) resulting moderate bonding yield. For the ridge waveguide structure, a large area needs to be planarized so the SU8 should be thicker (250 nm) to accommodate this area. For a surface treated by chemical-mechanical polishing (CMP), the bonding yield is the highest and since the surface is flat there is no need for planarization, and the certainty of SU8 thickness is high (100 nm).

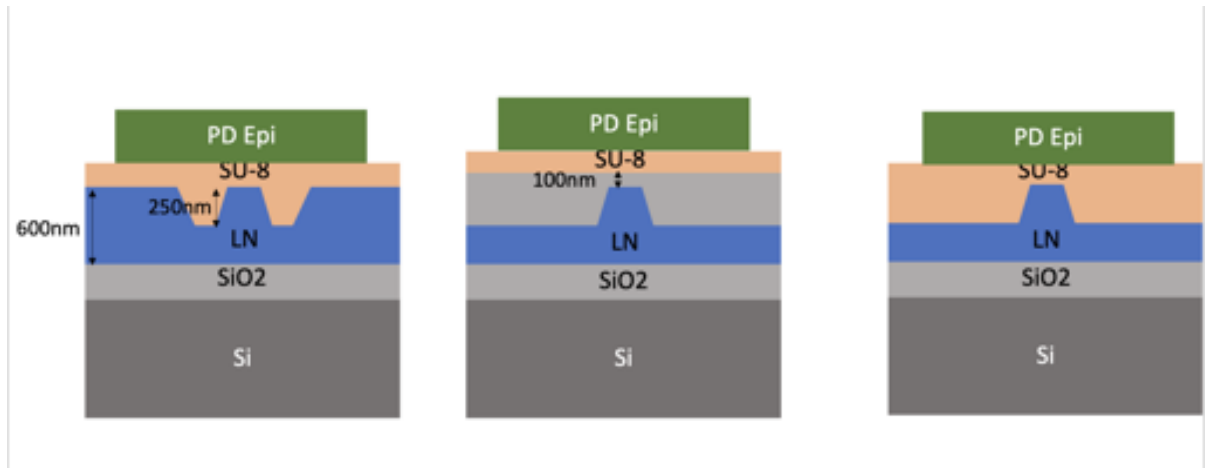


Figure 5: Cross section views for (a) Trench waveguide (b) CMP waveguide (c) Ridge waveguide

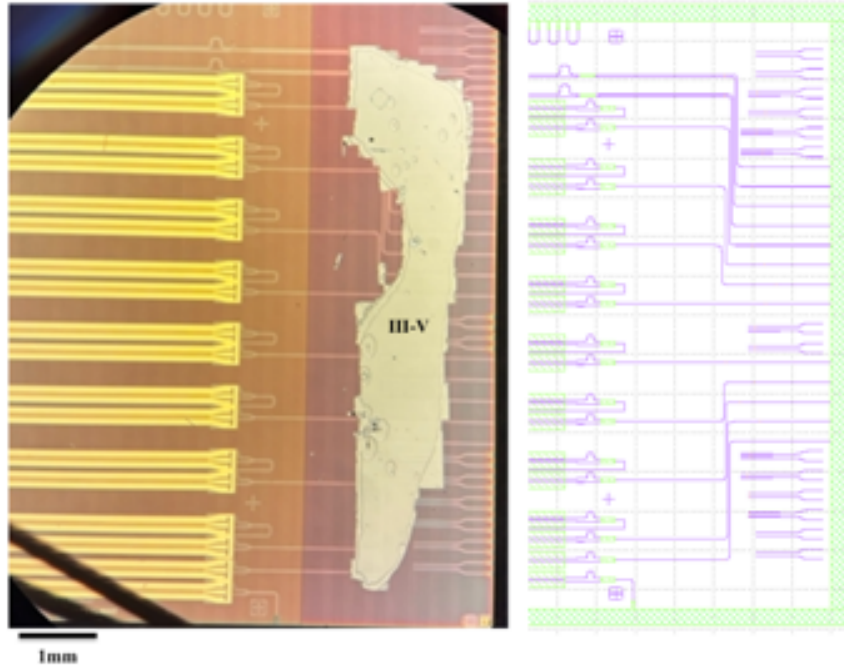


Figure 6: Microscope picture of bonded chip after substrate removal and its waveguide layout design

3.3 Passive waveguide layout

In order to achieve higher device yield, the waveguide layout should be optimized to mitigate the risk of some area debonding after substrate removal. For test structures, the design goal should be to fit as many waveguides as possible. For structures that are connected to other devices, the design goal should follow the principle of one single waveguide goes through the maximum area in the bonding area. This is to maximize the probability of having a piece of III-V material on the waveguide. Figure 6 shows that when some of the bonding material debonded, because of this waveguide design, the device still can be fabricated.

In theory, all of the structures inside of the bonding area are covered by epi and are invisible during the first lithography step to define the p-mesa place so it is essential to add alignment markers both inside and outside of the bonding area. For high alignment accuracy, it is ideal to have alignment markers at the four corners of the entire chip and the four corners of the bonding area as shown in fig 7.

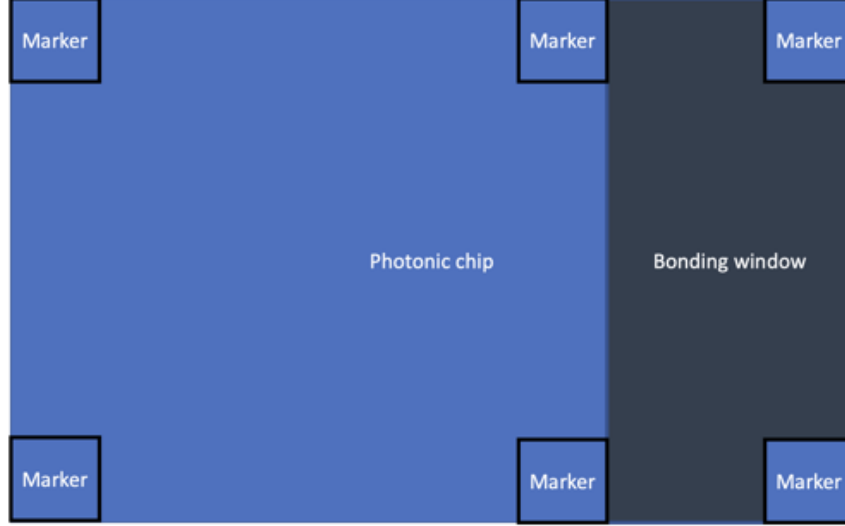


Figure 7: Illustration of marker placements

3.4 Mesa formation and metal contacts

In general there are two ways of forming III-V mesas, wet etch and dry etch. Wet etch typically results in smoother sidewalls and higher selectivity between InGaAs and InP [28]. However, since most wet etchants etch material isotropically resulting in large undercuts, only mesa sizes larger than $20\ \mu\text{m}$ with $1\ \mu\text{m}$ height can be made using wet etchant. For dry etch, a mix of Cl_2 and N_2 was used to chemically etch the III-V material. Both photoresist and metal hardmasks can be used for dry etch with enough thickness and selectivity for mesa thicknesses around $1\ \mu\text{m}$. However, since the temperature can get higher and hard to control inside the dry etch chamber, sometimes photoresist can be burnt. An hour-long acetone bath or O_2 plasma usually follows the dry etch for resist cleaning purposes.

For mesa sizes smaller than $400\ \mu\text{m}^2$, it is hard to probe the mesa contact directly on top of it and therefore a probing metal pad is needed to connect the small mesa contact to a larger contact pad for probing purposes without shorting the device. This can be done in two different approaches. The first one is to form an airbridge structure by electrode plating. A seed layer of Ti and Au is first deposited with the mesa top contact and the probing pad being separated from each other. Then without liftoff, a second plating lithography is added on top of the metal with patterns connecting the mesa contact and probing pad. By the nature of gold electro-plating, all metal exposed to the solution will get plated and the thickness can be controlled by the plating rate. Another approach is

to passivate and planerize the mesa sidewalls using SU8 and directly deposit probing pad patterns directly connected to the mesa top metal contact as shown in fig8. I used both methods in my work and more details can be found in the appendix.

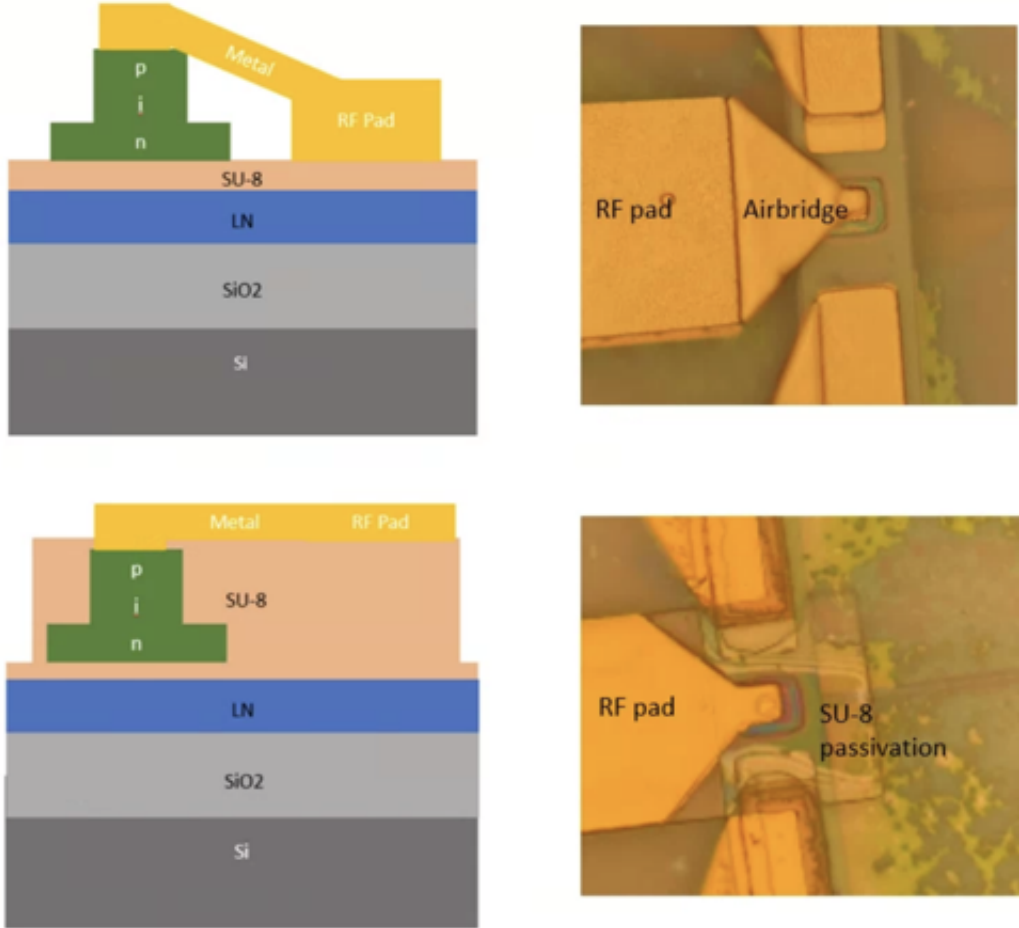


Figure 8: Side view of SU8 passivation (bottom) and traditional air-bridge structure (top)

3.5 Compatibility

Since the goal for heterogeneous photonic integration is to have all different types of devices all functional on the same chip, it is important to have a fabrication process that is compatible with other devices on chip and a robust device structure that can survive other processing steps. To investigate this, a fabricated chip PD on TFLN chip was heated to 330° C for 20 minutes and then cooled to 70° K and the I-V characteristics did not change and no mechanical deformation was found. This proves that devices fabricated are compatible with a wide temperature range from 70 degree Kelvin to 330

degree Celsius allowing for post-processing.

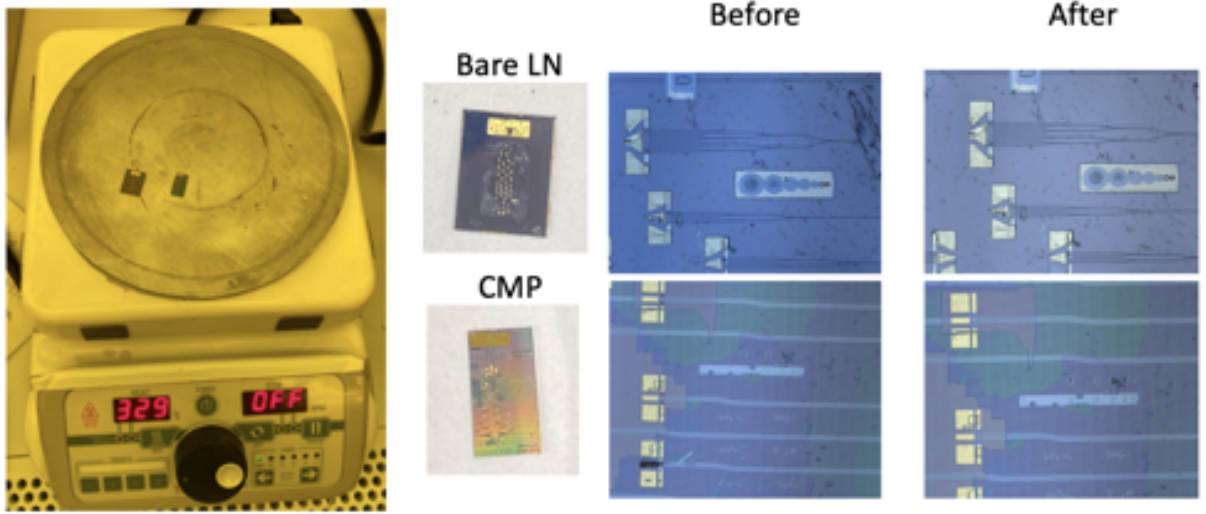


Figure 9: Microscope pictures of devices before and after temperature test

In my work, I integrated the photodiodes on pre-fabricated chips that already included the passive optical waveguides and, in some cases, metal contacts for various functions like heaters, modulator electrodes etc. For chips without top cladding, and exposed metal pads, it is essential to cover those areas with thick photoresist through the entire fabrication process. For processes after photodiode integration, it is essential to ensure that the photodiodes are robust enough to survive in those processes. For photonic wire bonding (PWB) lasers to the TFLN chip, PGEMA (also called AZ EBR) was used by our collaborators at Freedom Photonics as the developer for PWB. Since there is SU8 passivation on the photodiode structure, I tested them in 20 min room temperature bath in AZ EBR on those SU8 passivation structures and there were no deformations observed proving that this photodiode structure is compatible with the PWB of lasers. Results and measurements of heterogeneous PDs on TFLN with PWB laser will be discussed in the RF link chapter.

4 High-speed photodiode

4.1 Introduction

Photodetectors that convert high-speed optical signals into the electrical domain are essential components in optical communication networks and microwave photonics systems. Conventionally, output optical signals from a photonic system are fiber coupled into a discrete photodiode. This, however, introduces additional loss and is not compatible with the need for fully integrated chip-scale photonic solutions. In the past years, various efforts on integrated photodetectors on nonnative substrates have focused on platforms including silicon and silicon nitride [29,30], and bandwidths of PDs on Si approaching 70 GHz have been reported [31].

Lately, lasers were proven to have the compatibility to be integrated on the TFLN platform with low loss [32]. To date, almost all building blocks for a complete photonic system have been realized on the TFLN platform with outstanding performance except one—photodetectors. An early attempt was made on integrating photodiodes on LN waveguides using GaAs [33], and one demonstration of an integrated silicon photodetector on TFLN with 10-MHz bandwidth for visible wavelengths was made [34]. Only very recently, more progress has been made in PD on TFLN integration including Wei. et al ultra-wide band photodiode on TFLN [35].

In this chapter, the developed photodiode epitaxial layer structures will be explained including bandwidth analysis. Design parameter optimization methods will be discussed with the results from the first integrated high-speed photodiode on thin-film lithium niobate platform breaking the record bandwidth of heterogeneously integrated photodiodes on nonnative substrates [36].

4.1.1 Bandwidth of photodiode

Over the years, photodiode structures have evolved to improve their high-speed performance. In a basic p-i-n photodiode, the intrinsic absorber is sandwiched between an n-type doped and a p-type doped material that does not absorb light. When a reverse bias is applied, photogenerated holes drift to the p-side while electrons drift to the n-side under the effect of electric field. Both holes and electrons generated in the absorber region participate equally in the photocurrent. Since holes typically travel at a lower velocity

than electrons, the uni-traveling carrier (UTC) photodiode was developed [37]. In a UTC PD structure, instead of the intrinsic region is the absorber, the p-side becomes the absorber region. This way, holes disappear with their typically short dielectric relaxation time since they are majority carriers and only fast carriers (electrons) limit the output response. Later, modified-UTC (MUTC) structures were proposed to further increase the bandwidth, responsivity, and high power capability. Figure 10 shows a MUTC structure [38]. In general, the response speed of the photodiode depends on the time needed for carriers to travel to the contacts where they are collected in addition to the circuit response which is determined by the resistance-capacitance (RC) time constant.

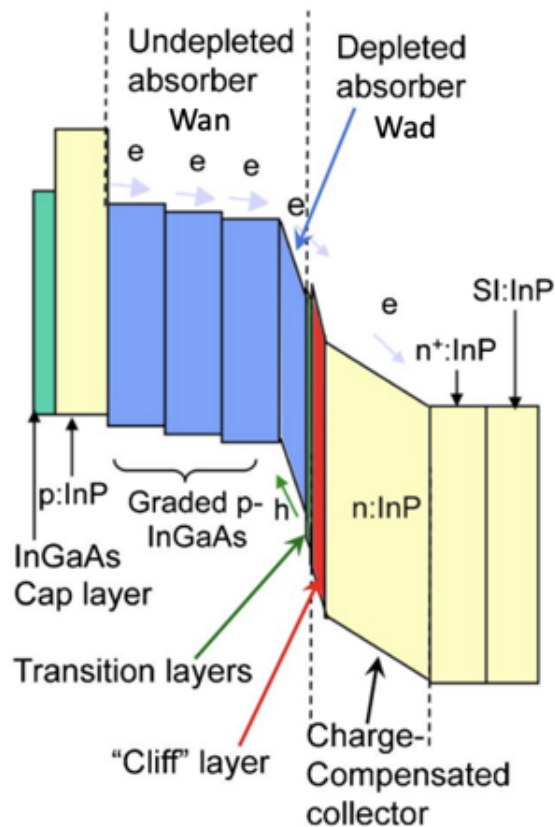


Figure 10: Band structure illustration of an MUTC photodiode with cliff layer [38]

4.1.2 Resistance-capacitance (RC) limitation and transit-time limitation

Photodiodes can be treated as a simple parallel plate capacitor with p-type region and n-type region being the two plates and the depletion region in the middle. Therefore, the

capacitance can be written as

$$C = \frac{\epsilon A}{d} \quad (21)$$

where A is the active area, ϵ is the permittivity and d is the length of the depleted region. The total series resistance of a photodiode comes from the contact resistance, resistances in the quasi-neutral regions, and includes the external load resistance. Photodiodes can be modeled as an RC circuit (R and C are in parallel) and therefore their 3 dB-cutoff frequency can be represented as:

$$f_{RC} = \frac{1}{2\pi R_{total}C} \quad (22)$$

The series resistance in the PD is typically at the magnitude of a few Ohms and the load resistance is usually 50Ω . Since $R_{total} = R_s + 50\Omega$, it is hard to improve the bandwidth by decreasing the resistance whereas the capacitance can be reduced effectively by decreasing the size of the active area or increasing the length of the depleted region. The transit-time limitation can be calculated from the current density in a photodiode at small AC signal. There are the following current densities that dominate the response in a typical UTC or MUTC structure: Drift hole current generated from the depleted absorber region, drift electron current from the depleted absorber region, diffusion electron current from the nondepleted absorber region. They can be expressed as integral forms in respect to position as below [39]:

Diffusive electron current:

$$J_{e1}(\omega, x) = -W_{an}R_{an}(\omega)\exp[-j\omega\frac{x}{v_e}] \quad (23)$$

Drift electron current in depleted absorber:

$$J_{e2abs}(\omega, x) = -\int_0^x \exp[-j\omega\frac{x}{v_e}]dx \quad (24)$$

$$R_{an}(\omega) = \frac{1}{1 + j\omega\tau_{an}} \quad (25)$$

$$\tau_{an} = \frac{W_{an}^2}{3D_e} + \frac{W_{an}}{v_{th}} \quad (26)$$

Drift electron current in drift layer:

$$J_{e2drf}(\omega, x) = - \int_0^{W_{ad}} \exp[-j\omega \frac{x}{v_e}] dx * \exp[-j\omega \frac{x - W_{ad}}{v_e}] \quad (27)$$

Drift hole current in depleted absorber:

$$J_h(\omega, x) = - \int_x^{W_{ad}} \exp[-j\omega \frac{W_{ad}x}{v_h}] dx \quad (28)$$

Where W_{an} is the undepleted absorber thickness, W_{ad} is the depleted absorber thickness.

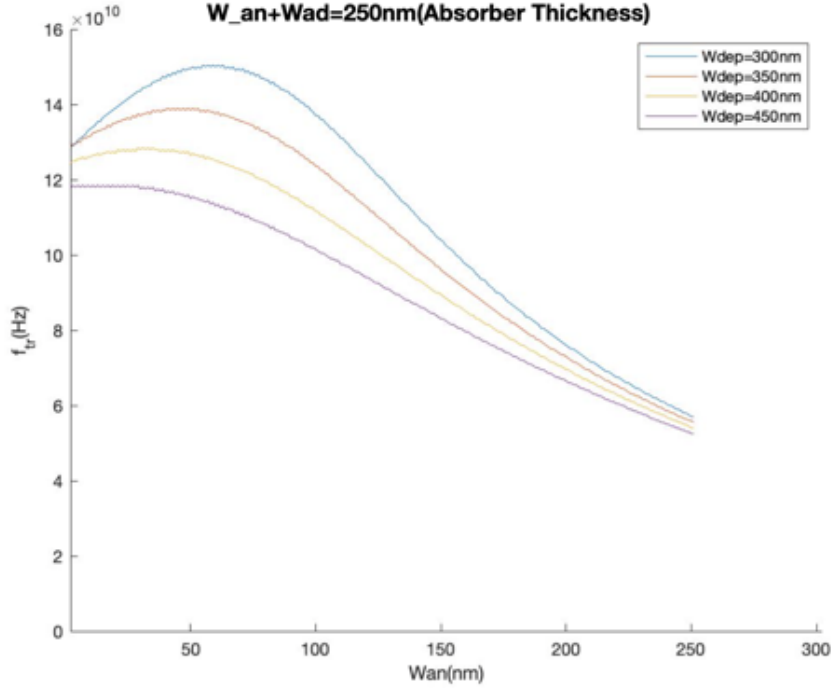


Figure 11: Transit-time bandwidth limitation with different depletion region thicknesses and a fixed total absorber thickness 250 nm where W_{an} is the undepleted absorber thickness and W_{ad} is the depleted absorber thickness

4.2 Design parameters optimization

From equation 21 and the plot in figure 11, it can be seen that by increasing the depletion region W_{dep} the RC limit increases while the transit-time limit decreases. It is obvious that if the depletion region increases, the capacitance decreases but electrons need to travel for a longer distance and therefore the transit time limit decreases. It is hard to

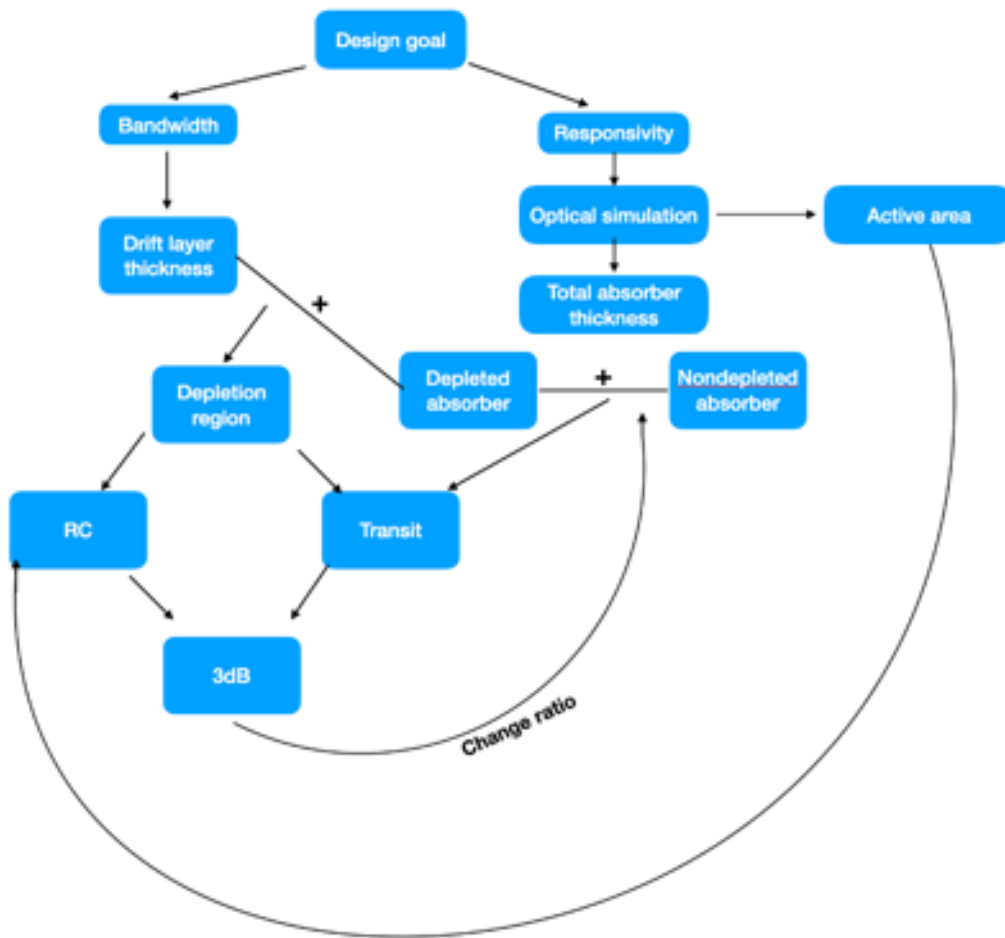


Figure 12: Flow chart of photodiode design optimization procedure

solve for the best parameter combination analytically. To this end I wrote an algorithm to model the carrier transport in photodiode and sweep for different parameters. The process flow of the algorithm is shown in figure 12. First the material parameters of the epitaxial structure were set including: carrier lifetime, carrier velocity, active area, sweep range, drift layer thickness. Then run optical simulation to see what the minimum absorber thickness is needed for the required responsivity and then estimate the drift layer thickness from the bandwidth goal using the parallel plates equation. The algorithm first calculates the RC and transit-time limitations and use these two results to calculate the 3dB bandwidth and then change the thickness of the undepleted absorber and the thickness of the depleted absorber and then update the depletion region length to calculate the RC and transit-time limit again until the sweep is done.

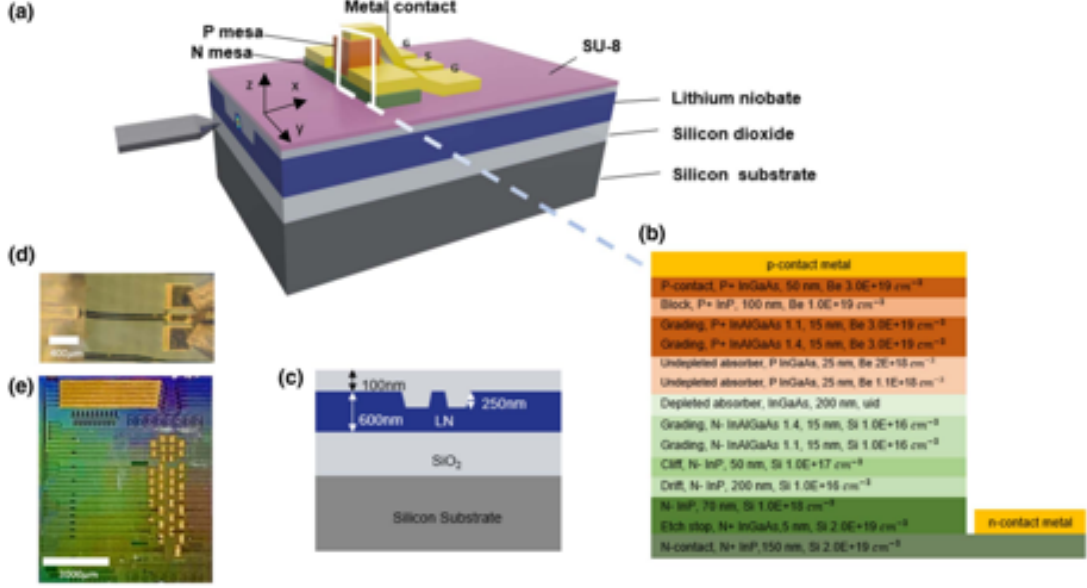


Figure 13: MUTC PDs integrated on TFLN. (a) Structural illustration. (b) Epitaxial structure of the n-down PD. (c) TFLN waveguide cross section. (d) Microscope image of a device under test with a lensed fiber and a radio frequency probe. (e) Microscope image of chip with integrated PDs.

4.3 Results

Figure 13 (b) shows the MUTC PD layers that were grown as a p-down structure lattice-matched on a semi-insulating InP substrate. In this design, the thicknesses of the InGaAs absorber layers were optimized to maximize the carrier transit time-limited bandwidth. To this end, I calculated the frequency response of the sum of the three following current contributions: electron diffusion current, hole drift current, and electron drift current. They can be expressed as integral functions with respect to positions of carriers spanning the region from the absorber to the drift layer [34]. The ratio of the depleted absorber and the undepleted absorber was then chosen to maximize the carrier transit time-limited bandwidth of the photodiode. Figure 14(a) shows the calculation result using the parameters in Table 1. With a fixed total absorber thickness of 250 nm, I found a maximum of 167 GHz for a 50 nm undepleted absorber and a 200-nm depleted absorber. A 50 nm cliff layer made of moderately n-type-doped InP is added to reduce the space charge effect [39]. Since InGaAs and InP have a difference in band gap, InAlGaAs with a 0.86-eV band gap was chosen as the bandgap-grading layer material. Grading layers on both the n-side and the p-side help carriers to overcome the band discontinuities.

As shown in fig 13 and in contrast to our previous work in [30], I designed a structure

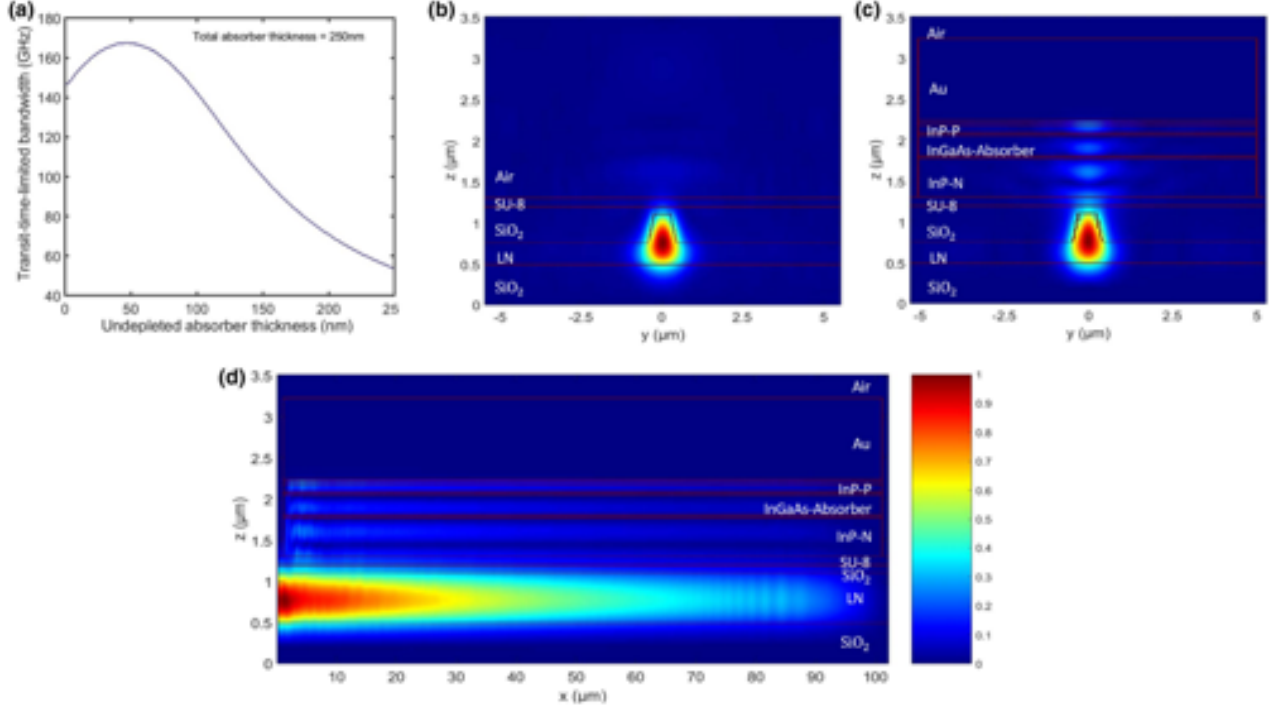


Figure 14: Device simulations. (a) Calculation of transit time-limited bandwidth versus undepleted absorber thickness with total absorber thickness of 250 nm and a fixed drift layer thickness of 200 nm. Optical simulations of E-field magnitude: (b) yz cross section at $x = 0 \mu\text{m}$ (passive waveguide), (c) yz cross section at $x = \mu\text{m}$ (PD), and (d) zx cross section at $y = 0 \mu\text{m}$

that results in n down after bonding with the following benefits: the InP n-contact layer is transparent at the operating wavelength 1550 nm; a low sheet resistance in the contact layer can be achieved since high n-type doping levels ($>10^{19}\text{cm}^{-3}$) can easily be reached in InP together with high electron mobility; formation of n-metal contacts with low contact resistance; and keeping p-type InGaAs as the top contact layer without compromising the responsivity. Although the layer structure was specifically designed for 1550 nm, I expect that the PD has a broad working wavelength range that is ultimately limited by the bandgaps of the highly doped InP contact layer (920 nm) and the absorber InGaAs (1650 nm).

Figure 13 (c) shows a schematic of the TFLN waveguide which was optimized for photodiode integration and fabricated by our collaborator Hyperlight. The total thickness of LN is 600 nm with the waveguide height being 250 nm. A layer of SiO_2 was deposited on top of the trench waveguides followed by chemical-mechanical polishing (CMP) to improve surface flatness. The total thickness of the SiO_2 layer after CMP is 100 nm on top of the waveguide, which is optimal for both bonding and evanescent light input

Table 1. Bandwidth Calculation Parameters

Thermionic emission velocity in InGaAs	2.5×10^5 m/s
Hole saturation velocity	0.45×10^5 m/s
Electron velocity in InGaAs and InP	1×10^5 m/s
Electron mobility in p-type InGaAs	0.5 m ² /(V · s)
Electron diffusivity in p-type InGaAs	1.3×10^{-2} m ² /s
External load resistance	50Ω
Series resistance	5Ω

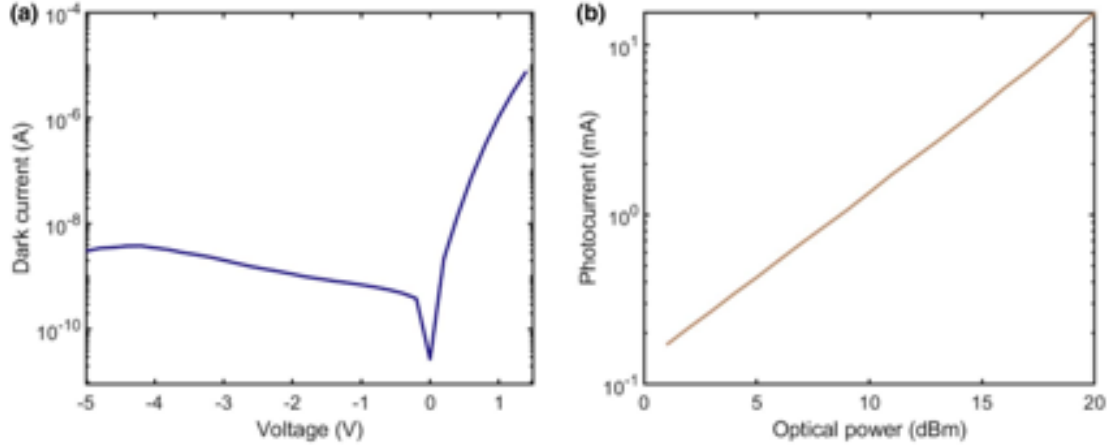


Figure 15: DC measurements of MUTC photodiodes integrated on thin-film LN platform. (a) Dark current characteristics of a $1400\text{-}\mu\text{m}^2$ device. (b) Measurement of photocurrent versus optical power (measured at fiber tip) of a $600\text{-}\mu\text{m}^2$ device.

coupling from the wave- guide into the PD. With this standard TFLN platform, other optical devices, such as ring resonators and modulators can be simultaneously fabricated on the same chip as shown in Fig. 13(e).

Photodiodes were fabricated by first bonding a $4\text{-mm} \times 8\text{-mm}$ InGaAs/InP MUTC wafer die on TFLN waveguides covering the area where PDs were then fabricated. With sufficient margins on all sides, the alignment accuracy for this step was $500 \mu\text{m}$. I used SU-8 as the adhesive bonding agent. The thickness of SU-8 was controlled by spin coating speed and designed to be 100 nm for a high optical coupling efficiency. Figures 14(b)–14(d) show a simulation of the light coupling from the LN waveguide into the photodiode. Figure 16 shows the relationship between internal quantum efficiency and photodiode length. SU8 was both UV and high temperature cured to achieve higher stability. After a 40-min outgas, the PD epitaxial material die and the TFLN chip were brought in contact and cured at 130°C under 10 psi for 60 min. The InP substrate of the III–V photodiode die was removed after bonding using wet etch (HCl). For the n-mesa fabrication, I used inductively coupled plasma reactive ion etching (Cl_2 and N_2) followed

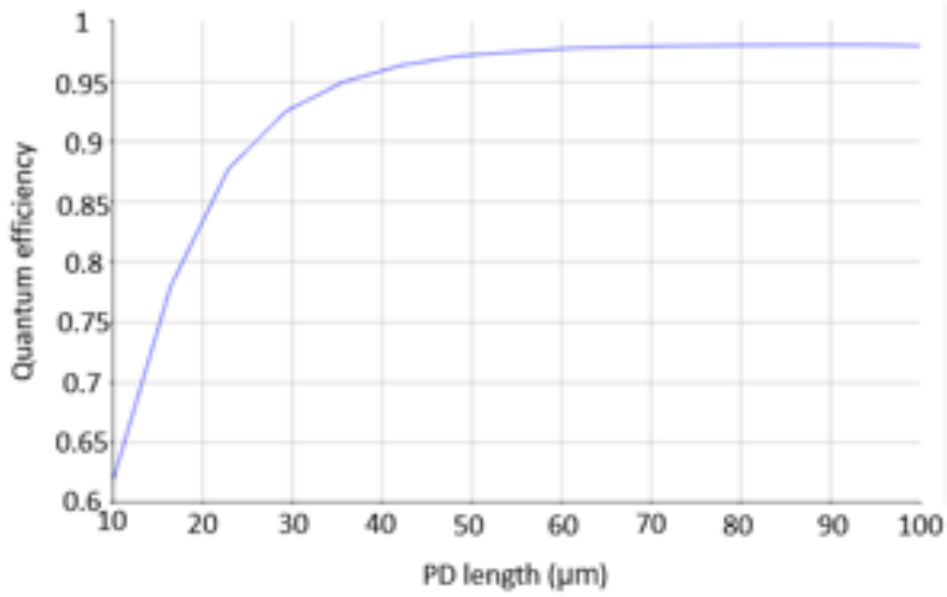


Figure 16: Simulation of internal quantum efficiency and photodiode length

by a short selective wet etching (HCl) to expose the InP contact layer. The combination of the two methods allowed us to largely avoid undercut which can be detrimental for small mesa areas, whereas, ensuring to stop the etching above the InP. The p-mesa formation was performed by wet etching (H_3PO_2 and HCl). Figure 17 and figure 18 show the illustration of the fabrication process. I fabricated PDs with the following active areas (width in $y \times$ length in x): $10 \mu\text{m} \times 10 \mu\text{m}$, $20 \mu\text{m} \times 10 \mu\text{m}$, $15 \mu\text{m} \times 20 \mu\text{m}$, $20 \mu\text{m} \times 30 \mu\text{m}$, and $20 \mu\text{m} \times 50 \mu\text{m}$. TFLN waveguides outside the bonding area were protected by photoresist during the fabrication process. An air-bridge structure was adopted to connect the metal on top of an n mesa and a p mesa to the ground-signal-ground (GSG) radio frequency (RF) pads. Those RF pads were designed to yield a characteristic impedance of 50Ω .

Dark current is an important figure of merit for PDs as it is one of the major sources for noise. The fact that the low dark current of our MUTC PD on a thin-film lithium niobate (TFLN) of 3 nA [Fig. 15(a)] is comparable to or lower than the dark current from similar PDs on a native substrate [38], indicates that the integration process did not degrade the performance. The responsivity of the photodetectors was measured at 1550 nm by edge-coupling light into the TFLN waveguides. The external responsivity of the PDs was calculated by dividing the photocurrent by the optical power at the lensed fiber tip (measured by an optical power meter). For a $10 \mu\text{m} \times 10 \mu\text{m}$ - sized

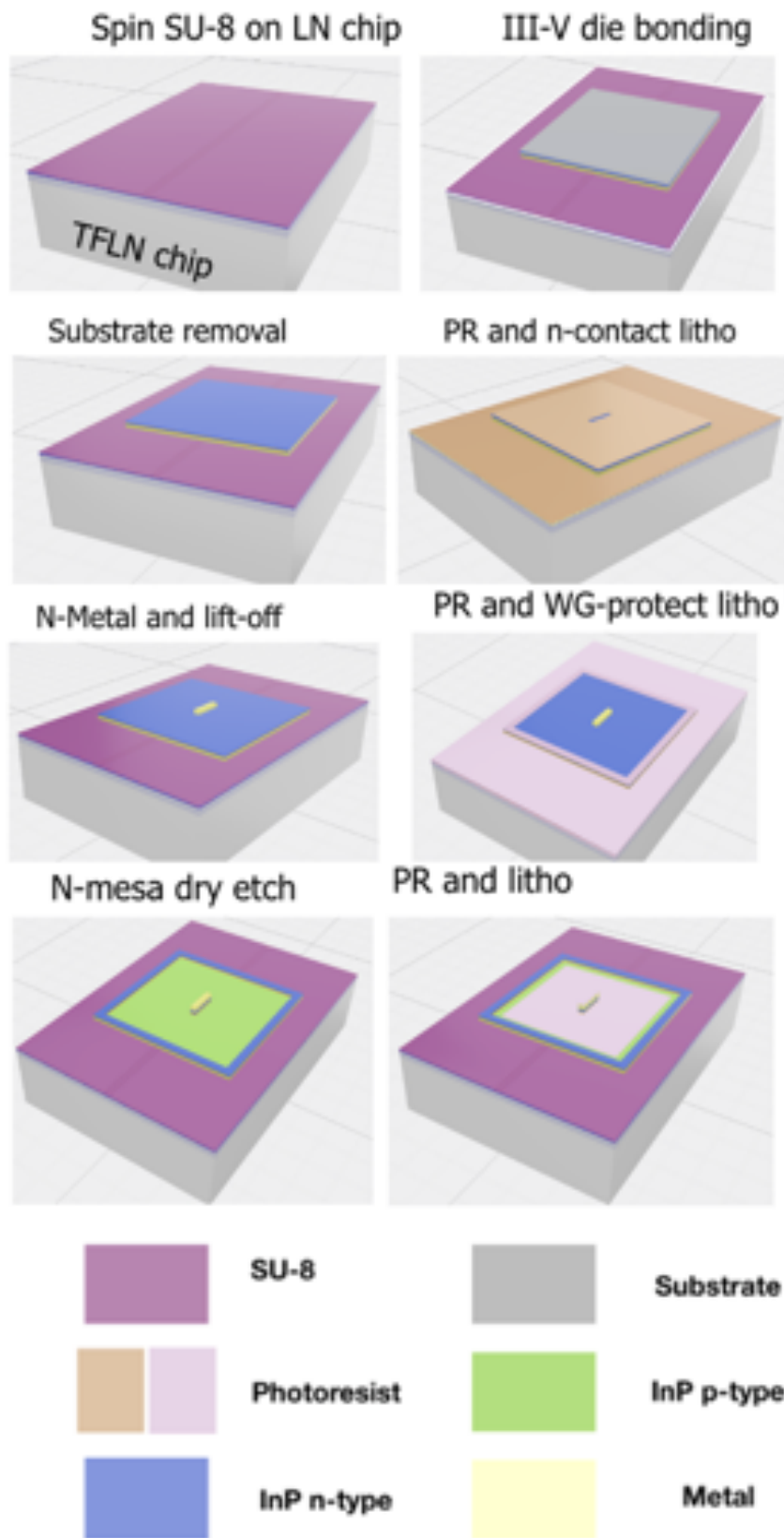


Figure 17: Illustration of fabrication process flow

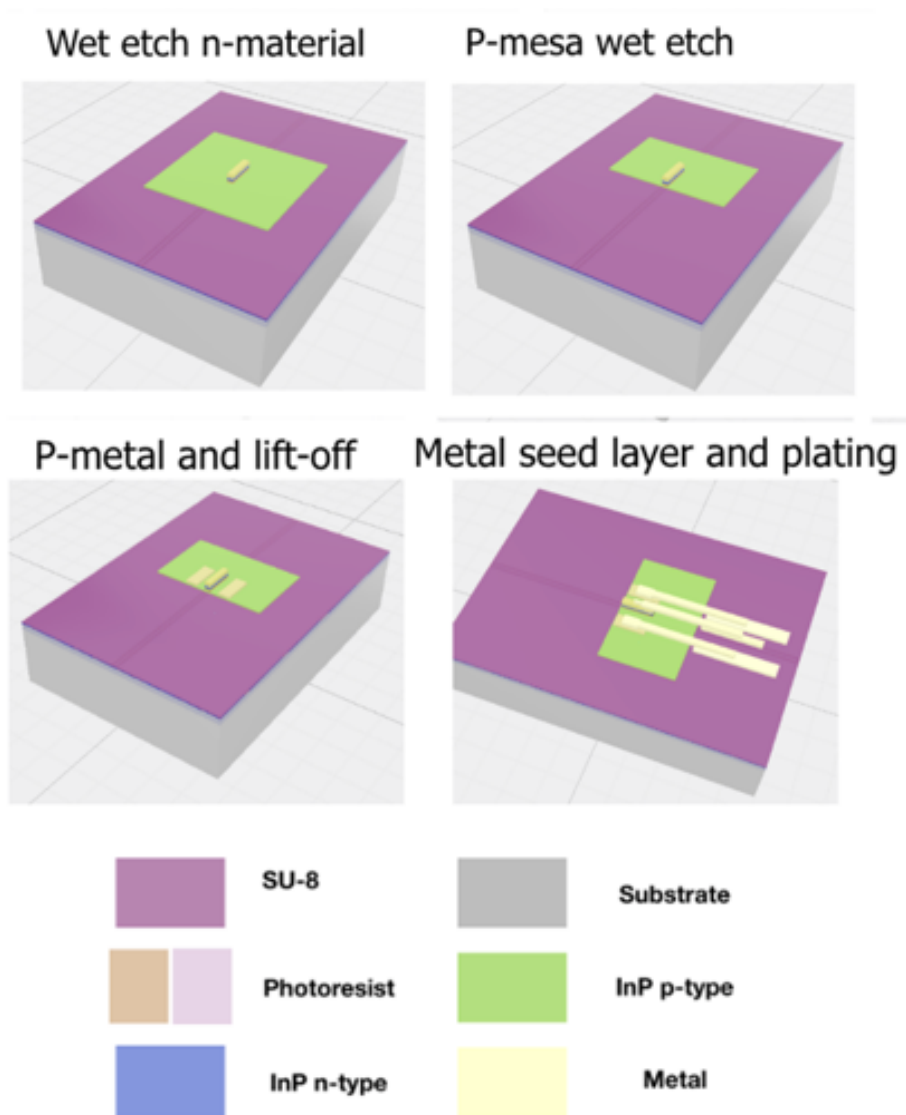


Figure 18: Illustration of fabrication process flow

photodiode, the external responsivity was 0.2 A/W. It should be mentioned that external responsivity is significantly affected by the fiber-to-waveguide coupling loss and does not reflect device performance well in this integration scheme. Thus, internal responsivity was characterized by investigating the ratio of photocurrent and the light coupled into waveguides. The average fiber-to-waveguide coupling loss was 5 dB. The loss was measured on a straight waveguide with a visibly good facet on the same chip using a conventional fiber-to-waveguide-to-fiber setup. This coupling loss measurement also agrees with the simulation result using the commercial software numerical finite difference eigenmode to calculate mode overlap between the fiber mode and the transverse electric mode in the TFLN waveguide. It should be noted that we did not use an antireflection coating. With taking coupling loss into account, the internal responsivity of a $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$ device is 0.6 A/W which agrees with optical simulation using the numerical FDTD module. The responsivities for PDs with areas of $20\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$, $15\text{ }\mu\text{m} \times 20\text{ }\mu\text{m}$, $20\text{ }\mu\text{m} \times 30\text{ }\mu\text{m}$, and $20\text{ }\mu\text{m} \times 50\text{ }\mu\text{m}$ are 0.17, 0.24, 0.52, and 0.4 A/W, respectively. It should be mentioned that we diced the waveguide facets along the edge of the TFLN chip after photodiode integration, and there was visible nonuniformity across different facets. The fact that facet loss varies across different waveguides can explain the spreading of the measured responsivity data. We also measured photocurrent versus optical power on a 20- μm wide and 30 μm -long PD. As shown in Fig. 15(b), the photocurrent was linear for input powers as high as 20 dBm.

The 3-dB bandwidth was measured by a heterodyne setup. Two distributed feedback lasers with wavelengths near 1550 nm were used to generate a frequency tunable optical signal with near 100% modulation depth. The frequency of the beat signal is controlled by thermally tuning the wavelength of one laser, whereas the other laser remains at one stable wavelength. This optical signal was then directed to an erbium doped fiber amplifier (EDFA) followed by a variable optical attenuator to control the power of the optical signal. A single-mode tapered fiber was used to couple light into the TFLN waveguide. The fiber-to-waveguide alignment was performed using a six-axial stage. A GSG RF probe was used to extract the photocurrent by probing at the designed RF metal pads [Fig. 13(d)]. A semirigid coaxial cable was used to connect the RF probe with the bias tee. The RF signal was detected using a calibrated 50- Ω RF power meter with frequency range from DC to 110 GHz. The losses of the RF probe, semirigid coaxial

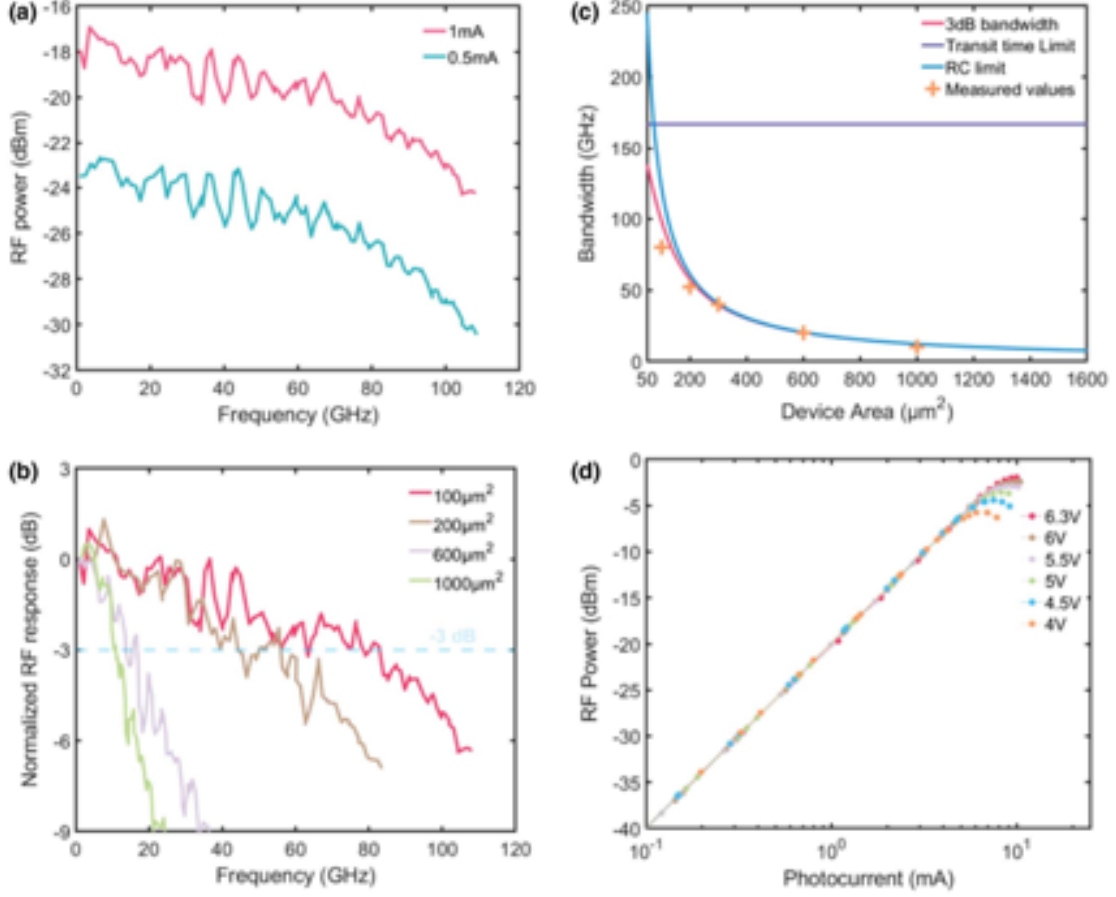


Figure 19: AC measurements of MUTC photodiodes integrated on thin-film LN. (a) Frequency responses of a $100\text{-}\mu\text{m}^2$ PD at different photocurrents. (b) Frequency responses of various sized photodiodes at 1-mA photocurrent and 3-V reverse bias. (c) Simulations and measurements of bandwidth for different device areas. (d) Measurement of RF power of a $300\text{-}\mu\text{m}^2$ PD at 48 GHz at different reverse biases.

cable, and bias tee were determined with a 110-GHz network analyzer and then subtracted from the measured data.

Figure 19(a) shows the bandwidth measurements of a $100\text{-}\mu\text{m}^2$ PD at photocurrents of 0.5 and 1 mA at the same 3-V reverse bias. The 3-dB bandwidth of 80 GHz did not change with different photocurrents, and the output RF power from the photodiode matches the expectation. Figure 19(b) shows the bandwidth measurement results of the photodetectors with 100, 200, 600, and $1000\text{-}\mu\text{m}^2$ device areas. Photocurrent was kept the same during measurements by adjusting the fiber alignment and the polarization of the optical signal controlled by a fiber polarization controller. The highest 3-dB bandwidth of 80 GHz was achieved with the $100\text{-}\mu\text{m}^2$ photodetector, and the 3-dB bandwidth decreased as the device active area increased. If the PD is significantly shorter than the electrical signal wavelength, the photodetector can be treated as a lumped circuit

element. Therefore, the resistance-capacitance (RC) time and carrier transit time are the two primary bandwidth limitations. Figure 19(c) shows the simulated transit time limit, the RC limit, and the 3-dB bandwidth for this epitaxial structure, and the measured bandwidth values agree well with the calculation. In this calculation, we use the calculated transit time-limited bandwidth of 167 GHz and a total resistance of $55\ \Omega$ ($50\text{-}\Omega$ external load and $5\text{-}\Omega$ series resistance).

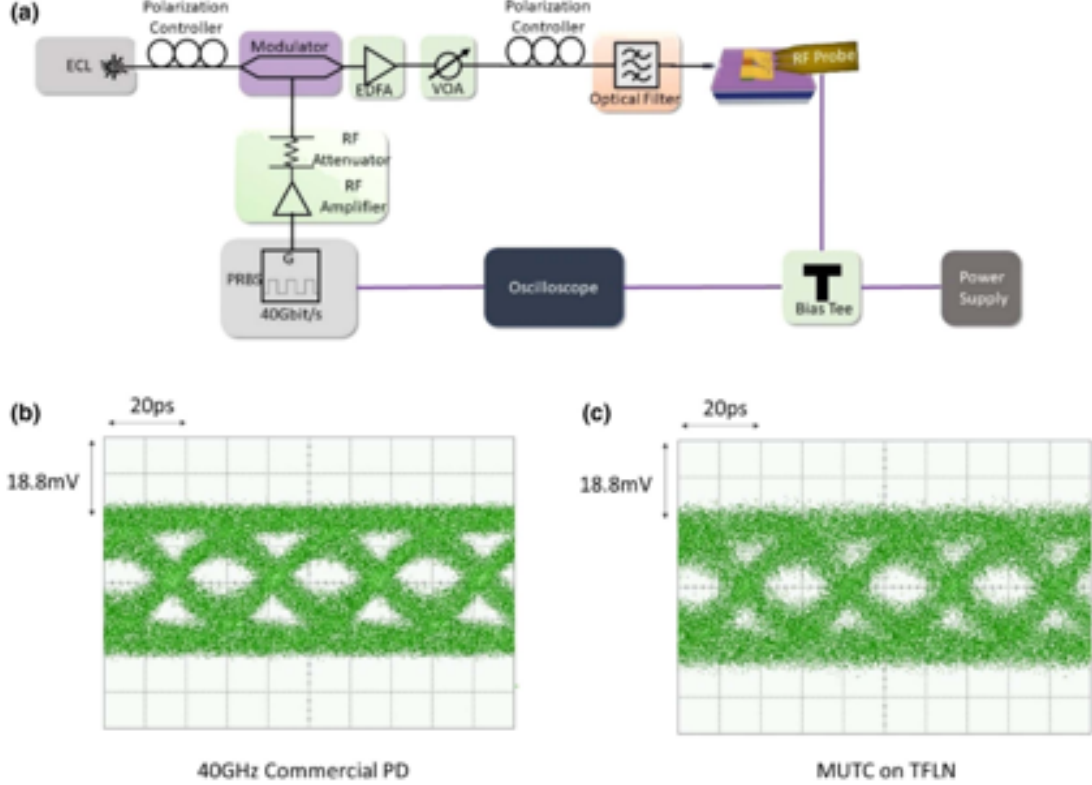


Figure 20: Eye diagram measurement. (a) Measurement system setup. (b) Measured eye diagram for a commercial photodiode. (c) Measured eye diagram for an MUTC photodiode on TFLN with $200\text{-}\mu\text{m}^2$ PD area.

Digital data detection performance of the photodetectors was characterized by an eye diagram measurement. A non-return-to-zero on-off keyed pseudorandom bit sequence (PRBS) data pattern was generated by a PRBS generator at 40 Gbit/s. Figure 20(a) illustrates our experimental setup. An external cavity laser provided the optical signal for a commercial 40-GHz modulator. The 550-mV peak-to-peak signal from the PRBS generator was fed into the modulator biased at its quadrature point. The optical signal then went through an EDFA, a variable optical attenuator, and an optical filter with a 1-nm passband before being coupled into the TFLN waveguides. The electrical signal was extracted by an RF probe and went through a semirigid cable, a bias tee, and

a coaxial cable before entering the 50-GHz sampling oscilloscope for the readout. It is worth mentioning that our PD handled sufficient photocurrent and provided a high signal-to-noise ratio for the oscilloscope, so no additional RF amplifier to boost up the electrical signal was necessary in the measurement. We first measured the eye diagram of a commercial PD with a 40-GHz bandwidth shown in Fig. 20(b) as a reference. Figure 20(c) shows the measurement result of our $200\text{-}\mu\text{m}^2$ PD with the measured 3-dB bandwidth of 52 GHz. It should be mentioned that the commercial PD was connected directly on the input port of the oscilloscope, whereas the on-chip measurement of our photodetector required additional RF parts that were not calibrated out from the eye diagram. Even with the extra loss in the system, our photodetector performs similarly to the commercial PD. The clear open eye diagram proves the high-speed performance of our photodetector and demonstrates its capability for digital communication applications.

5 High-power photodiode

5.1 Introduction

In the previous chapter, MUTC PDs on TFLN has been demonstrated heterogeneously integrated InAlGaAs/InP that achieved low dark current (0.3 nA), high responsivity (0.6 A/W), and bandwidths up to 80 GHz at 1 mA [40]. In this chapter their saturation characteristics will be analyzed, and a prediction is made that it is possible to improve performance by further minimizing space charge resistance and flattening the optical intensity profile in the absorber.

There are two limiting factors in achieving high saturation power for photodiodes. One is thermal failure, and the other one is electric field collapsing due to space charge. Figure 19 (d) shows the previous measurement revealing that the maximum RF output power is -1.9 dBm at 10 mA. In general, higher reverse bias led to higher maximum output power which indicates the presence of the space charge effect. Reverse voltages above 6.5 V led to PD failure at DC dissipated powers of >70 mW, most likely due to Joule heating in the PD. This chapter discuss methods in optical design and electric field design to improve saturation power of heterogeneously integrated MUTC photodiodes on waveguides with low index (n approx. 2).

5.2 Space-charge effect

When a junction is formed, free carriers will flow from high concentration to low, leaving fixed charges (ions) left. These fixed charges form the space charge region. The space charge can be represented as $Q = (p - n + N_D - N_A)q$ with n and p being the free electrons and holes. Using Poisson's equation, the electric field can be obtained by integrating the charge over distance. Far away from the junction interface, the free carrier concentrations are approximately equal to the doping concentration and therefore the space charge is zero and the electric field is also zero. We call this the quasi-neutral region. In the middle, the free carrier concentrations are low and approximately equal to zero. Therefore, the space charge is equal to the doping concentration and the electric field is relatively high. We call this the depletion region because it is depleted of free carriers.

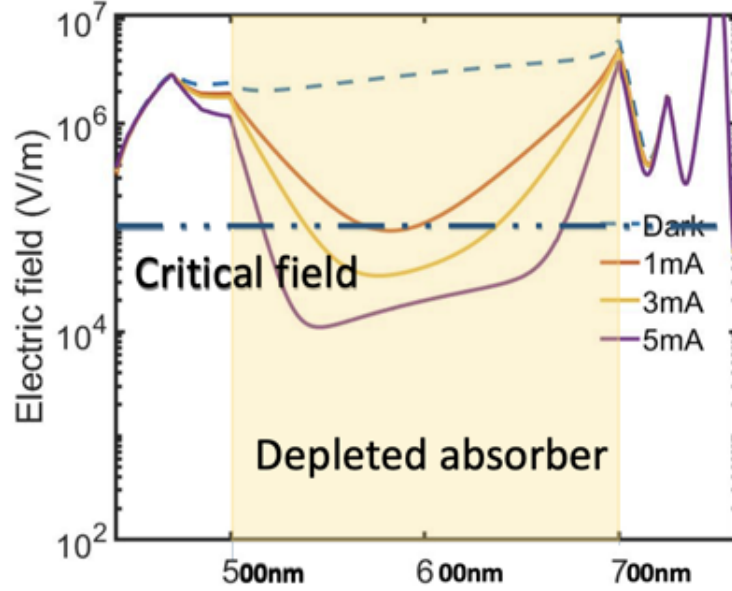


Figure 21: Electric field simulation of an MUTC photodiode absorber under different photocurrent levels.

By applying an external reverse bias, free carrier concentration is further reduced which widens the depletion region. This is when we call the junction is fully depleted and the space charge density in the intrinsic layer of a MUTC photodiode equals the doping concentration. There are two ways in device performance reflects on this effect. One is C-V measurements, at the beginning of adding reverse bias, the capacitance decreases until the junction reaches full depletion. This is because by adding the reverse bias voltage, the width of the depletion region changes and thus change the distance in the two parallel

plate model, resulting the capacitance to decrease until it is fully depleted. When we keep increasing the reverse bias, the depletion region can extend slightly into the highly doped layers at the two ends but this changes the capacitance insignificantly. Because the bandwidth is also related to the capacitance, only when C is at its minimum, the RC bandwidth is at its maximum. Therefore, there is a trend of bandwidth enhancement at the beginning of adding reverse bias until it hits the fully depleted bias. It also changes the transit-time bandwidth because it changes the depletion region length as shown in figure 11.

The space charge effect refers to when photogenerated carriers are injected into the depletion region, the free carrier concentration changes and therefore the total charge changes. This leads to the change in electric field. Figure 21 shows the electric fields with different photocurrents of a $100\ \mu\text{m}^2$ junction area. Since the free carrier concentration increases in the depletion region due to optical generation, the space charge concentration decreases which leads to the bending in the electric field profile. In order to fully deplete the junctions, a higher bias is needed in comparison with zero injection carriers. This means with higher photocurrent, the device needs to be operated at a higher bias to be fully depleted and maintain an electric field sufficient for saturated carrier velocities. At higher photocurrent, the device needs a higher bias to achieve its maximum bandwidth. Therefore the minimum required bias changes with different photocurrent levels to achieve maximum bandwidth.

It is well known that saturation in photodiodes, caused by the space-charge effect, occurs when high photogenerated carrier densities in the depletion region create a space-charge field that counteracts the dark electric field [41]. In order to maintain a sufficiently high electric field and thus saturated carrier velocities, a higher reverse bias is needed, which is ultimately limited by device thermal failure. To quantify space-charge, I measured the frequency responses of a 35 GHz MUTC PD on a $2\text{-}\mu\text{m}$ -wide TFLN waveguide at different photocurrents and reverse biases (figs. 23(a)-(d)). The measurements were done using an optical heterodyne setup with a modulation depth close to 100%. As expected, at higher photocurrents, the PD requires a higher reverse bias to achieve its maximum bandwidth. I also measured the RF output power at a fixed frequency of 48 GHz and determined a 1-dB compression point of 9 mA at -6.3 V. (fig. 22. Higher reverse voltage led to PD failure and we estimated the maximum dissipated DC power to be 70

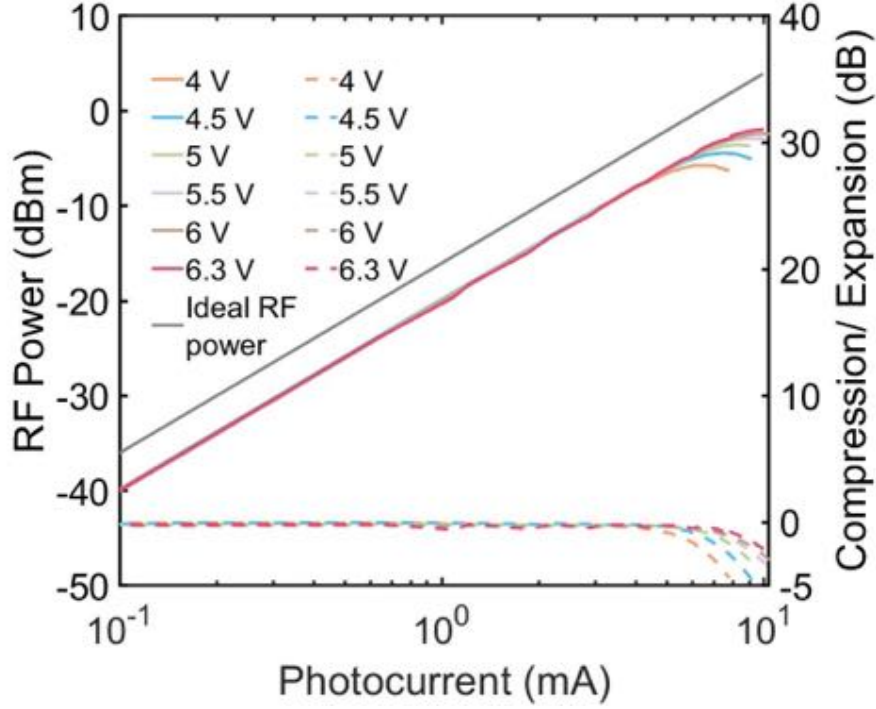


Figure 22: RF output power and RF compression at 48 GHz

mW.

We used a heterodyne setup to characterize the space-charge impedance on a $450\text{-}\mu\text{m}^2$ device that has a 30 GHz bandwidth at 1550 nm. I measured the frequency response of this photodiode at 0.1 mA, 1 mA, 2 mA, 3 mA and 5 mA with increasing bias until the device reaches its highest 3-dB as shown in fig.23(a)-(e). At higher photocurrent, the device needs a higher bias to achieve its maximum bandwidth. Therefore the minimum required bias changes with different photocurrent levels to achieve maximum bandwidth. At 1 mA photocurrent, the device needs 2 V to achieve its maximum bandwidth which also agrees with the electric field simulation in fig.21 indicating 2 V bias is needed for the structure to be fully depleted. Figure 23(f) shows the summary of the required bias vs. different photocurrent as a linear relationship.

In conclusion, we measured a $434\ \Omega$ space-charge impedance on an integrated MUTC photodiode that has 30 GHz bandwidth. This reflects the required bias dependence on input optical power. To improve saturation power of a device, a lower space-charge impedance is more ideal since the extra bias voltage required for high optical input power to maintain high bandwidth will result in more joule heating. In order to reduce the space charge effect, more electric field simulations can be done to investigate different

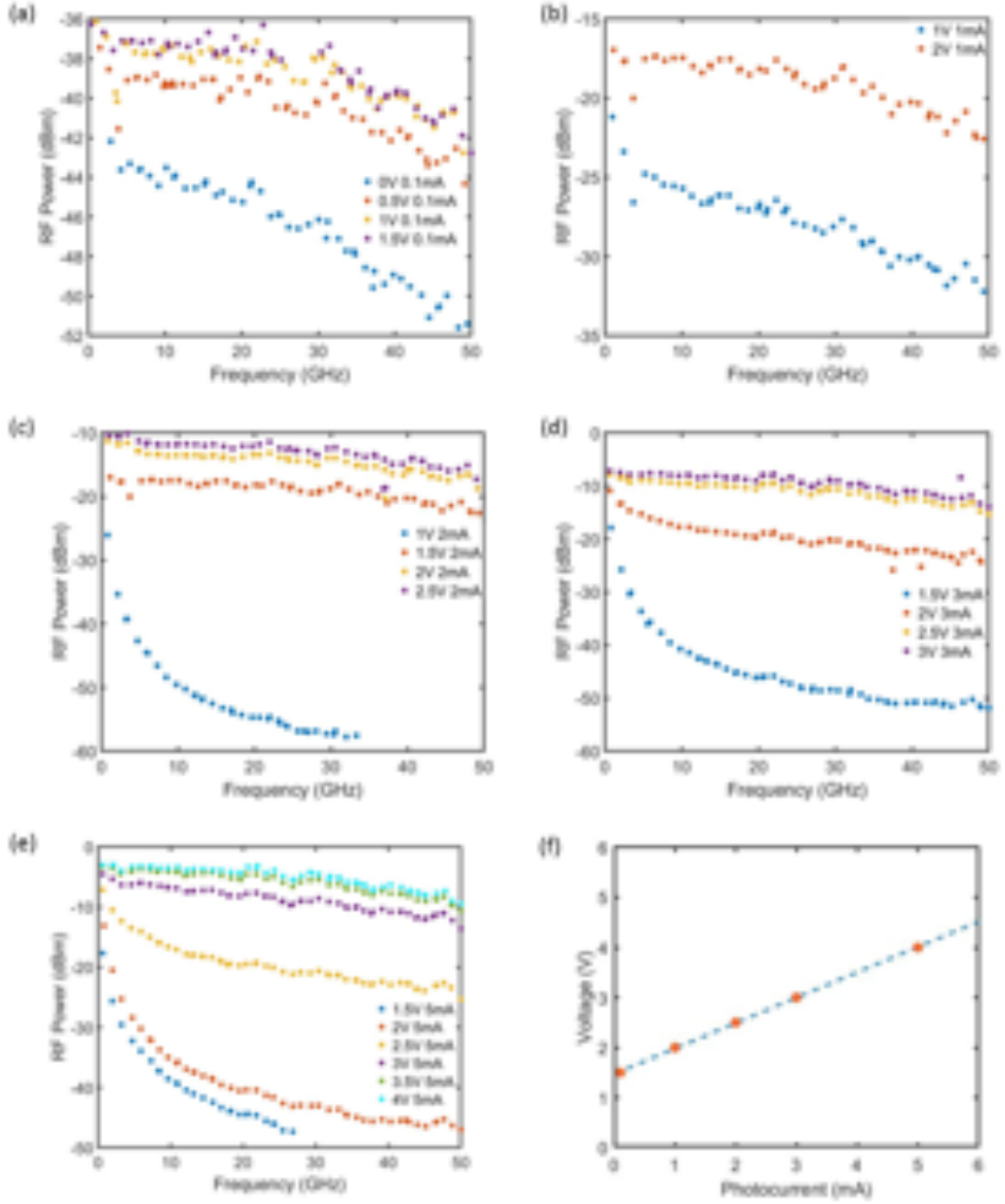


Figure 23: Measurements of space-charge impedance. (a)-(e) Bandwidth measurements at 0.1 mA, 1 mA, 2 mA, 3 mA and 5 mA accordingly with different reverse biases (f) Extracted space-charge impedance

structures. One direction is to increase the background doping level in the depletion region, this way the effect of high injection carriers would become relatively small in comparison to the one that has lower background doping level.

5.3 Uniform absorption profile

In addition to space charge, waveguide PDs face an additional challenge in achieving high saturation photocurrent compared to their back or front-illuminated counterparts due to their nonuniform illumination profile. For a back illuminated photodiode, the light is uniformly incident on the PD and thus all area becomes effective and contributes to the photocurrent whereas in waveguide PDs, light is evanescently coupled and high optical intensity are usually observed in first few micrometers in the absorber. This means, not all absorber area contributes to the photocurrent equally and some area is under high optical intensity of light which can cause premature saturation. This limitation is further exacerbated in heterogeneous waveguide PDs with a large refractive index difference between the waveguide (2-2.2) and PD (3.2-3.5), preventing a more gradual coupling that is advantageous for high-photocurrent operation.

In order to improve this, the optical intensity in the absorber should be as uniform as possible in order to achieve high saturation power. By changing the waveguide structure underneath the photodiode, the absorption profile can be changed.

One way to approach this is with inverse design, using an optimization algorithm to generate design parameters for a desired optimization goal [42]. In this case, the optimization goal is chosen to be the uniformity of optical intensity in the absorber which can be quantified as the maximum amplitude of the optical field at each mesh point inside the absorber. The design parameter is chosen to be the coordinates of the waveguide outline underneath the photodiode. First, a straight waveguide which is singlemode is input to the simulation engine (Lumerical) and then the amplitude of the field in the absorber is calculated and recorded. A gradient descent method is used as the optimizer and the output design parameters are in the form of coordinates of the waveguide shape. Figure 24 shows the comparison of straight waveguide and optimized waveguide. As shown in figure 24, there are acute angles and small feature size in the waveguide shape and therefore the fabrication accuracy and tolerance need to be considered into the design process.

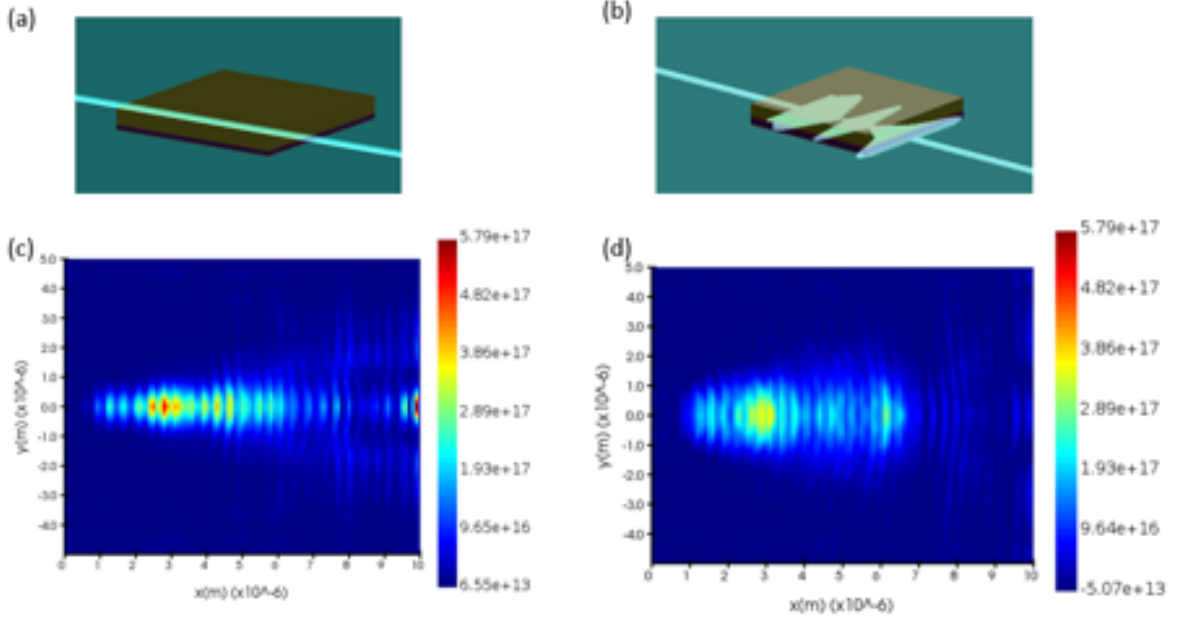


Figure 24: Uniform absorption profile simulation (a) & (c) Schematic of a single mode waveguide under a $10 \mu m \times 10 \mu m$ photodiode and E-field magnitude in the middle of the absorber. (b) & (d) Schematic of an optimized waveguide under a $10 \mu m \times 10 \mu m$ photodiode and E-field magnitude in the middle of the absorber

Another, simpler way to approach this is to make the waveguide as wide as the photodiode. By tapering a singlemode waveguide gradually to tens of micrometers wide, matching the width of photodiode, the fundamental mode in the waveguide becomes larger which results in the optical intensity distribution in the absorber spreading out. This method is more intuitive and has a larger tolerance on fabrication process because the shape of the waveguide is typically a trapezoid with larger feature size comparing to the result of inverse design. By tapering the width of the input singlemode waveguide gradually to tens of micrometers, the mode expands which results in a flat optical intensity distribution in the absorber. Figure 25 shows the E-field profiles in the center of a PD's absorber with different TFLN waveguide widths w underneath. While the normalized maximum E-field magnitude is 0.395 for $w = 2 \mu m$ (fig. 25a), maximum E-field magnitudes are only 0.268, 0.232 and 0.241 for $w = 5 \mu m$, $10 \mu m$, and $15 \mu m$, respectively (fig. 25b-d). The simulated quantum efficiency is 62% for $2 \mu m$, whereas 68%, 67% and 62% with wider waveguides, the mode becomes larger leading to lower maximum e-field magnitude in the absorber. However, if the waveguide is wider than the photodiode, not all light can be coupled to the absorber leading a decrease in the quantum efficiency.

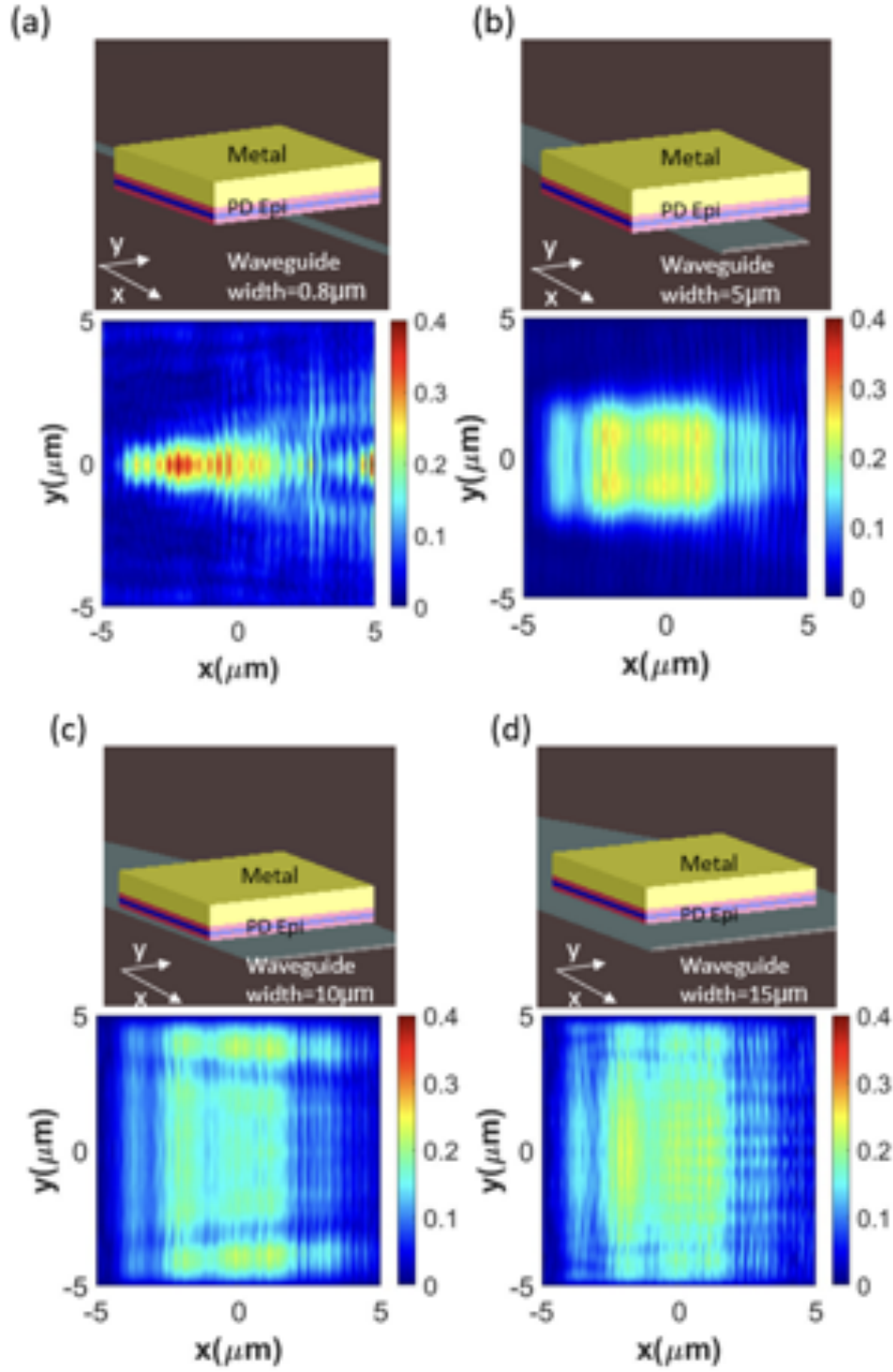


Figure 25: (a)-(d) Simulated E-field magnitude in the center of the absorber for different TFLN waveguide widths underneath a $10\ \mu\text{m} \times 10\ \mu\text{m}$ MUTC PD.

5.4 Conclusion

In conclusion, there are two aspects in improving the saturation power for photodiode integrated on non-native platform with large index difference. One is to reduce the space-charge effect. A lower space-charge impedance is more ideal since the extra bias voltage required for high optical input power to maintain high bandwidth will result more joule heating. One direction is to increase the background doping level in the depletion region, this way the effect of high injection carriers would become relatively small in comparison to the one that has lower background doping level. The other one is to achieve a more evenly distributed e-field profile in the absorber to increase the saturation power while maintaining similar quantum efficiency.

6 Integrated photonic RF-link

6.1 Introduction

The interest in optical links for analog RF signal transmission has continued to expand for the past decade owing to their low energy consumption, broad bandwidth, high speed, and low loss. The idea of using analog optical links first widespread in commercial applications for the distribution of cable television signals [43]. Recently, the technology has shown potential to improve the performance of broadband wireless access networks, remote microwave antenna systems, optical computing, digital-to-analog converter, and RF signal processing and filtering [44].

At first, a photonic RF link was simply connecting a discrete diode laser to a photodiode by fiber. By directly modulating the diode laser using an RF analog signal, the photodiode output can produce an RF output signal with the output power level determined by the laser optical power, fiber loss, and PD responsivity. However, since the diode laser modulation bandwidth is usually low (< 40 GHz), this direct modulation method limits the performance of the link. Therefore, external modulation of the light was naturally employed to improve the operation speed of photonic RF link by using an electro-optic modulator in between the laser and the photodiode. This way, the light is modulated by the modulator and the link can achieve higher bandwidth since the limitation is now the bandwidth of the photodiode and the modulator and both devices have

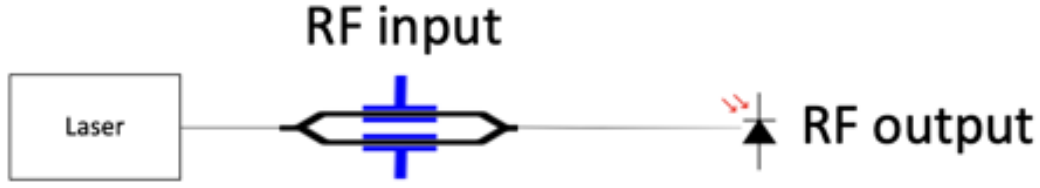


Figure 26: RF Link

been demonstrated with bandwidths well above 50 GHz [45] [46] . Figure 26 shows a basic photonic RF link consisting a laser, an electro-optic modulator and a photodiode.

The input electrical RF signal is applied to the modulator and then detected by the photodiode as electrical RF output power. Since the output RF electrical signal is the photocurrent and therefore by increasing the optical power in the system, the electrical output power can be increased to higher value than the electrical input power to the system. Therefore, in order to have positive gain in the system, it is essential to have high optical power handling capability. There are two crucial figure of merits for a link performance (and most microwave photonic applications), one is the RF link gain and the other one is the noise figure of the link.

Link gain can be expressed by equation 29 that is directly related to the photocurrent of the photodiode and the V_π of the modulator. Therefore, in order to increase the gain, it is important to have high saturation power photodiodes and small V_π modulators. This is also shown in figure 27, in order to have positive gain larger than 0 dB, at least 10 mA photocurrent and V_π smaller than 4 V are required. Since both V_π and photocurrent have the same effect on gain in the equation, changing either of them has the same effect on gain.

$$Gain = \left(\frac{V_{out}}{V_{in}} \right)^2 = \left(\frac{\pi I_D R_0}{2V_\pi} \right)^2 \quad (29)$$

The noise figure (NF) of a link can be expressed as the equation below.

$$NF = 10 \cdot \log \left(\frac{N_{out}}{Gain \cdot N_{th}} \right) \quad (30)$$

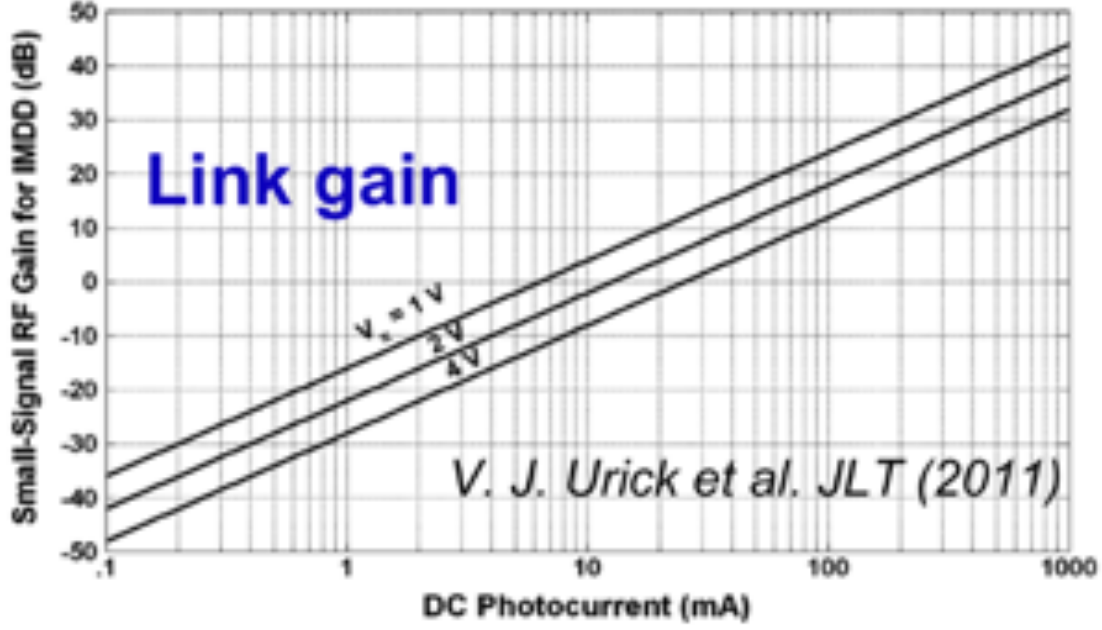


Figure 27: RF link gain calculation [47]

$$N_{th} = k \cdot T \cdot RBW \quad (31)$$

There are five different techniques in system design to improve the link performance without pushing individual device performance [48]. The first one is resonant impedance matching which means to add impedance matching circuits to match the link's input and output ports. This will increase the link's intrinsic gain [49]. The second one is reflective external modulator. Since the gain is inversely proportional to the V_π , by using a reflective design that doubled the duration of interaction between the photons and RF signals, V_π can be reduced by half and therefore increase the gain [50]. The third one is cascaded directly modulated lasers. By arranging several lasers electrically in series and optically in parallel, the total noise out of a link with an N-laser cascade is N times of a single laser system with the output signal is increased by N^2 due to photons are mutually uncorrelated. The fourth one is balanced differential modulation and detection. By using a balanced MZM to produce two identical optical carriers modulated in antiphase, and detecting using balanced photodiodes, the link's output signal is doubled while optical intensity noise is canceled out [48]. The fifth one is low biasing. By biasing at a lower point of the transmission curve of the modulator results in lower noise figure because the dominant output noise terms are proportional to the average detected power while

the link output signal is proportional to the square of the transmission curve's slope and changing the bias point downward from the quadrature point decreases the average power by more than the square of the slope [48]. While this method results in larger nonlinearities originating from the modulator, it can be applied to octave-spanning photonic RF links.

Many photonic RF links have demonstrated high performance using discrete components by connecting laser, modulator and photodiode using optical fibers [18, 51, 52]. Figure 28 shows a summary of reported photonic RF links gain versus frequency plot. With the development of photonic integrated circuits and heterogeneous integration, several on-chip integrated photonic RF links were reported. Xu et al. showed a down-conversion photonic RF link using monolithic integration with both modulator and photodiode integrated on the same chip [53]. Later, they presented a wideband ultrahigh dynamic range photonic RF link using the same InP substrate based material system [54]. Lately, our group has demonstrated wide-bandwidth photonic combiner utilizing silicon photonic platform with Ge photodiode [55]. However, to date, few results above 20 GHz have been reported and achieving acceptable RF link gain and acceptable noise figure remain the biggest challenges in integrated microwave photonics [56].

TFLN offers high-speed modulators with high-power capability and with our success in heterogeneous integration of high-speed MUTC photodiode on TFLN, it becomes an ideal platform for on-chip integrated RF photonic links. This chapter discusses the first integrated photonic RF link on TFLN with an RF link gain bandwidth measurement up to 67 GHz and noise figure along with low-bias operation.

6.2 Design and fabrication

The modulators were designed and fabricated by our collaborator at Hyperlight. The TFLN layout was completed in a collaborated fashion where waveguides and test structures for the photodiode parts were added and connected to the modulators. The length of the modulator is 0.5 cm with traveling wave design electrodes. Because of suspected photorefractive and charge relaxation effects in TFLN [58], there is no reliable DC response on the modulators and therefore, the MZM is imbalanced and can be biased using wavelength because of the length difference in the two different arms of the integrated delay line interferometer that feeds into the MZM. Figure 29 shows a microscope picture

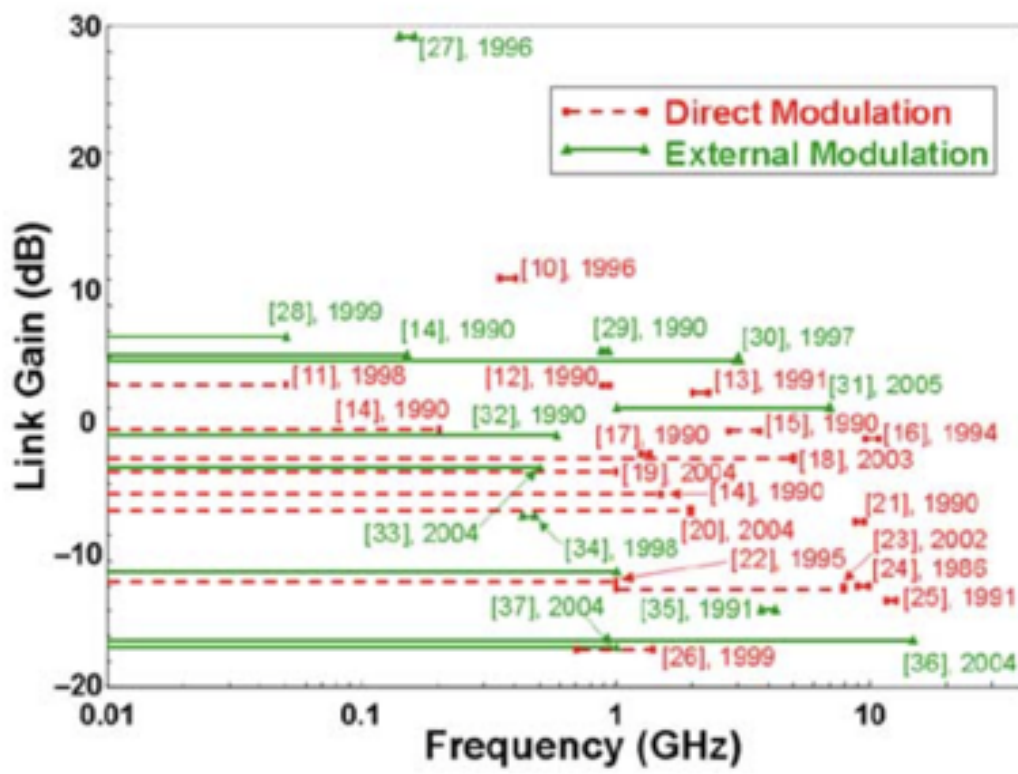


Figure 28: Published RF gains of analog optical links based on discrete components [57]

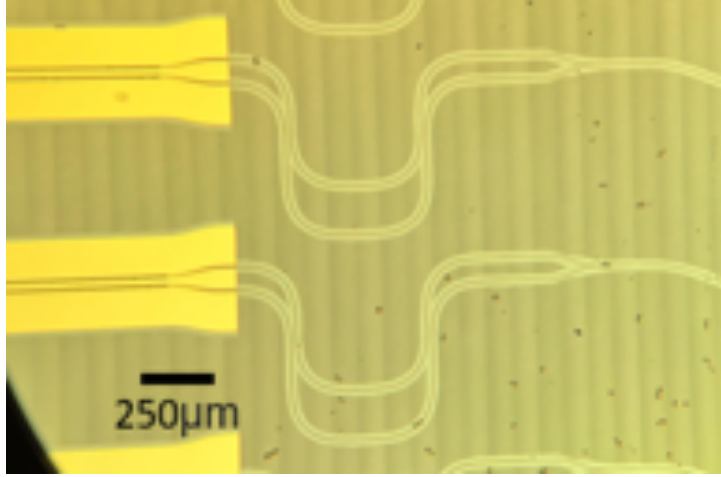


Figure 29: Microscope picture of the imbalanced MZM

of the imbalanced MZM with 250 μm in arm length difference.

The free-spectral-range (FSR) of the modulator was measured by sweeping the wavelength of the optical input and recording the transmitted optical power detected by a photodiode as shown in figure 30.

$$FSR = \frac{\lambda^2}{\Delta L \cdot n} \quad (32)$$

where ΔL is the length difference between the two arms and n is the refractive index.

The epitaxial photodiode structure used was the same as the high-speed PD on TFLN in chapter 4 which was designed to achieve high transit time-limited bandwidth while maintaining high quantum efficiency by optimizing the ratio between the depleted absorber and the undepleted absorber. In order to increase the saturation power performance of the photodiode, the waveguide shape underneath the photodiodes were designed using Lumerical swarm particle algorithm with the optimization goal being the maximum of the magnitude of e-field in the photodiode absorber. Figure 31 shows different combination of shapes for the waveguide underneath a $10\mu\text{m} \times 10\mu\text{m}$ photodiode and the simulated maximum e-field magnitude.

The photonic chip was first fabricated by Hyperlight and electrodes for the modulators were deposited at Harvard University (M. Loncar's group). Figure 32 shows the photonic chip after bonding photodiode epi die. Then, photodiodes were fabricated using the SU8 passivation structure (more details for the fabrication can be found in the appendix). The SU8 was first defined into patterns that cover the sidewall of p-mesa with overlaps

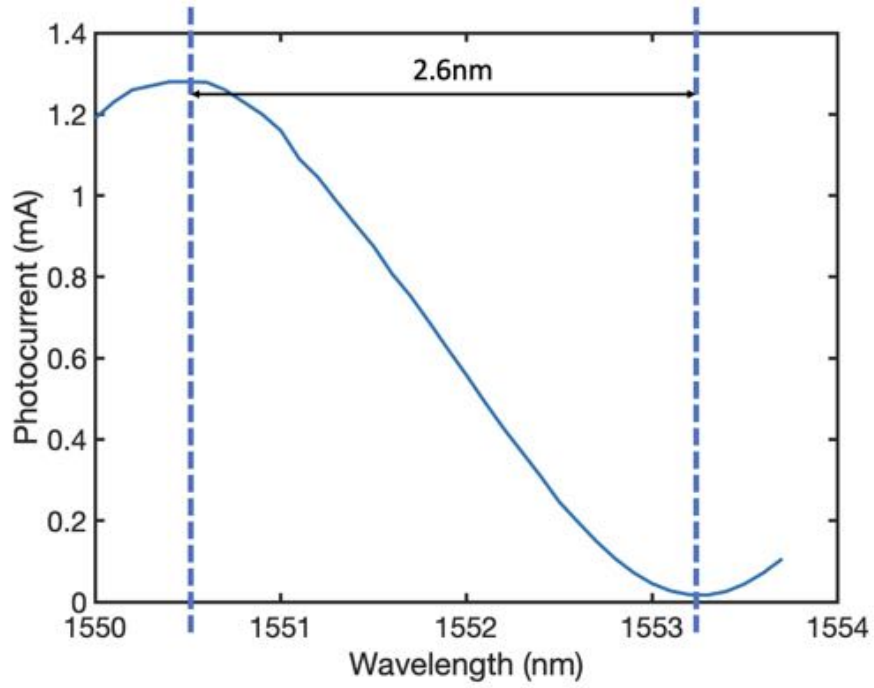


Figure 30: Modulator transmission with 2.6nm FSR/2

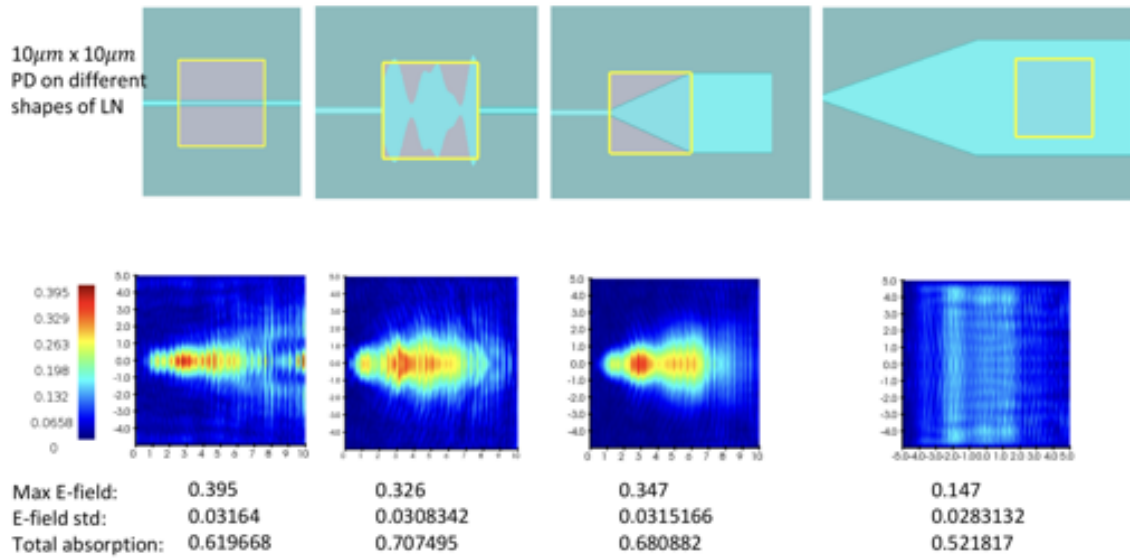


Figure 31: Simulated E-field magnitude in the center of the absorber for different TFLN waveguide shapes underneath a 10 μm x 10 μm MUTC PD.

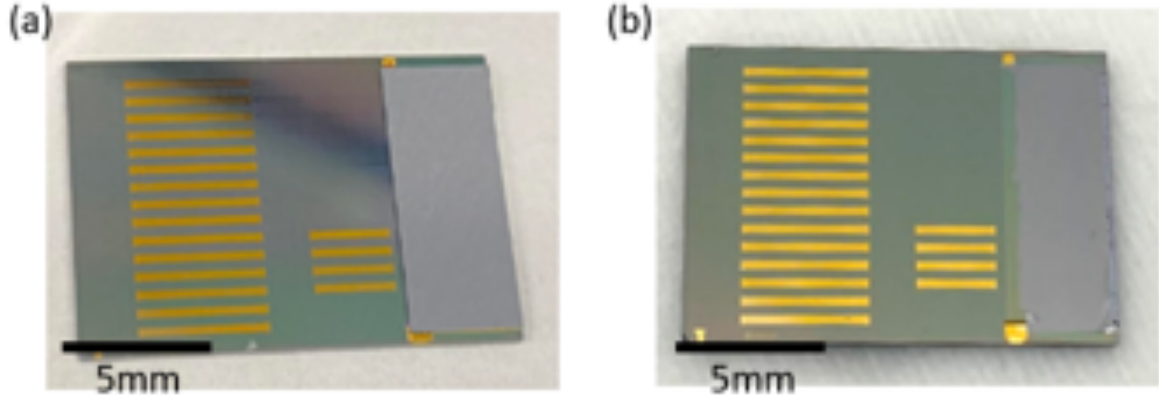


Figure 32: Pictures of RF link chip (a) after bonding (b) after substrate removal. The modulator's traveling wave electrodes are visible as golden rectangles.

on top of the mesa. The thickness of SU8 of this overlap can be as thick as 500 nm and therefore, in order to connect the metals without depositing too thick Au, the SU8 was then etched. Since the etch rate can be hard to control, this process was designed to have larger tolerance on the etch. With 100 nm of Au, the tolerance can be 200 nm for the SU8 etching since it only needs to isolate the contact layer to the rest of the structure. Figure 33 shows this process and figure 34 shows the comparison between traditional air-bridge structure to this SU8 passivation structure. During the fabrication, the modulator part was protected using thick photoresist (AZ4330) through out the process.

Figure 35 shows the entire photonic RF link chip with multiple devices on it and the SEM picture of the photodiode mesa along with the microscope picture of the SU-8 passivation structure underneath the signal pad.

6.3 Measurements

The first attempt for the link-on-chip measurement setup is shown in the figure 36. The RF signal driving the modulator was generated using a signal generator and the output electrical signal from the photodiode was read using an electrical spectrum analyzer (ESA). The DC power supply to the photodiode was provided using a Keithley and RF and DC were separated using a bias-tee. Figure 37 shows the fiber coupled into the modulator and two RF probes for driving the modulator and reading from the photodiode, respectively. In order to find the quadrature bias point of the modulator, a wavelength sweep was done first by sweeping the wavelength and recording both the DC photocurrent

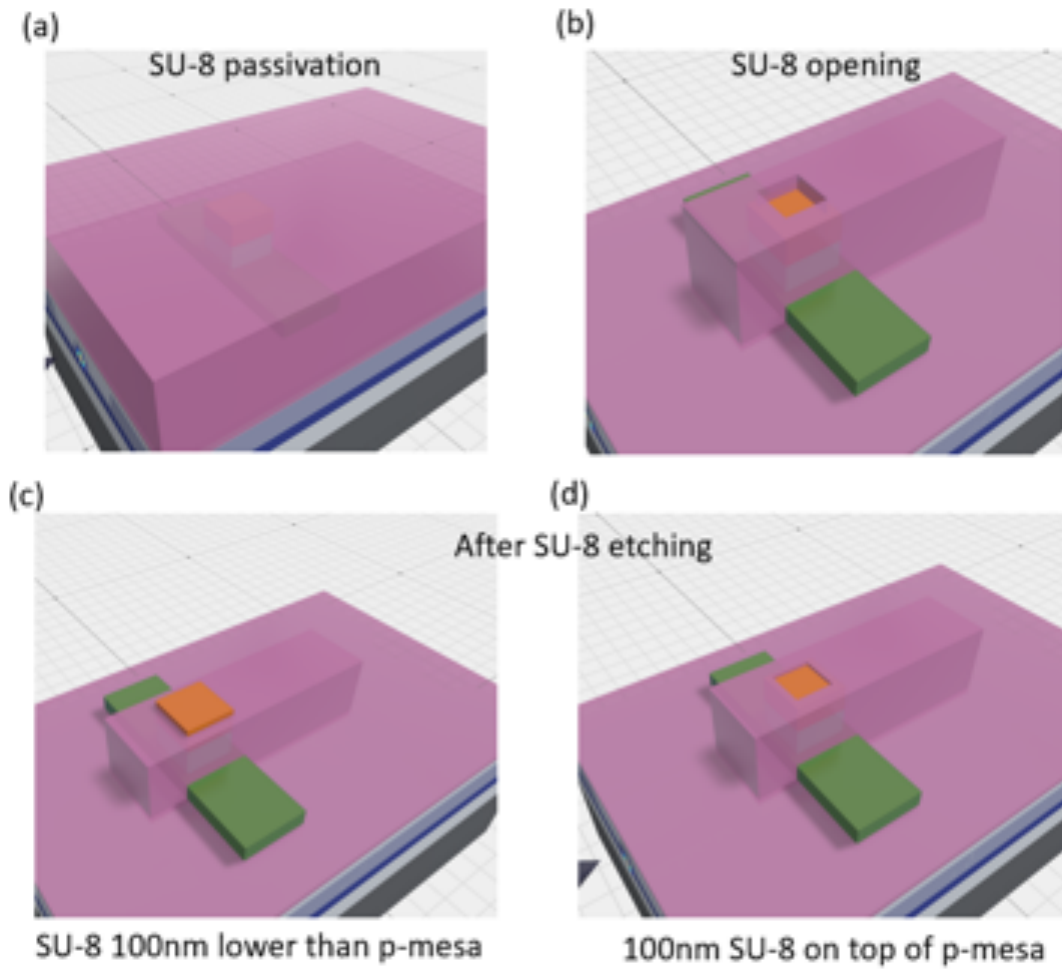


Figure 33: Fabrication process illustration (a)SU8 passivation (b) SU8 opening defined by lithography (c)&(d) show two different situations after SU8 etching

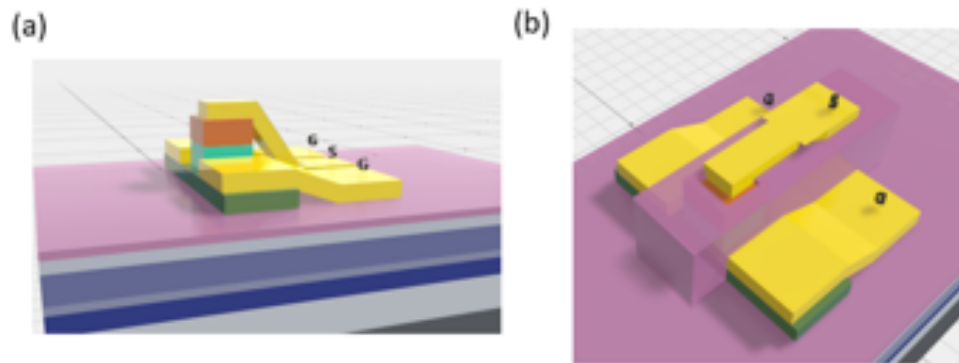


Figure 34: Illustrations of photodiode structure (a) Air-bridge (b)SU8 passivation

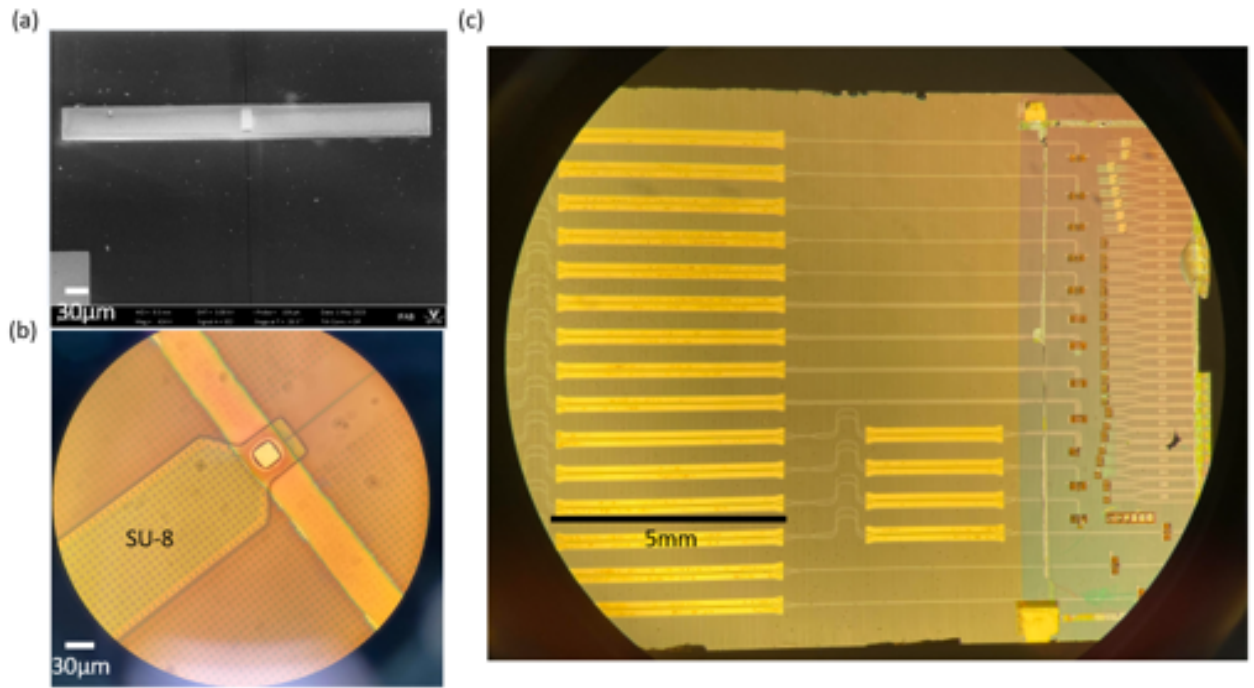


Figure 35: RF Link chip (a) SEM image of a PD mesa (b) Microscope image of SU8 passivation and (c) Microscope picture of the entire chip

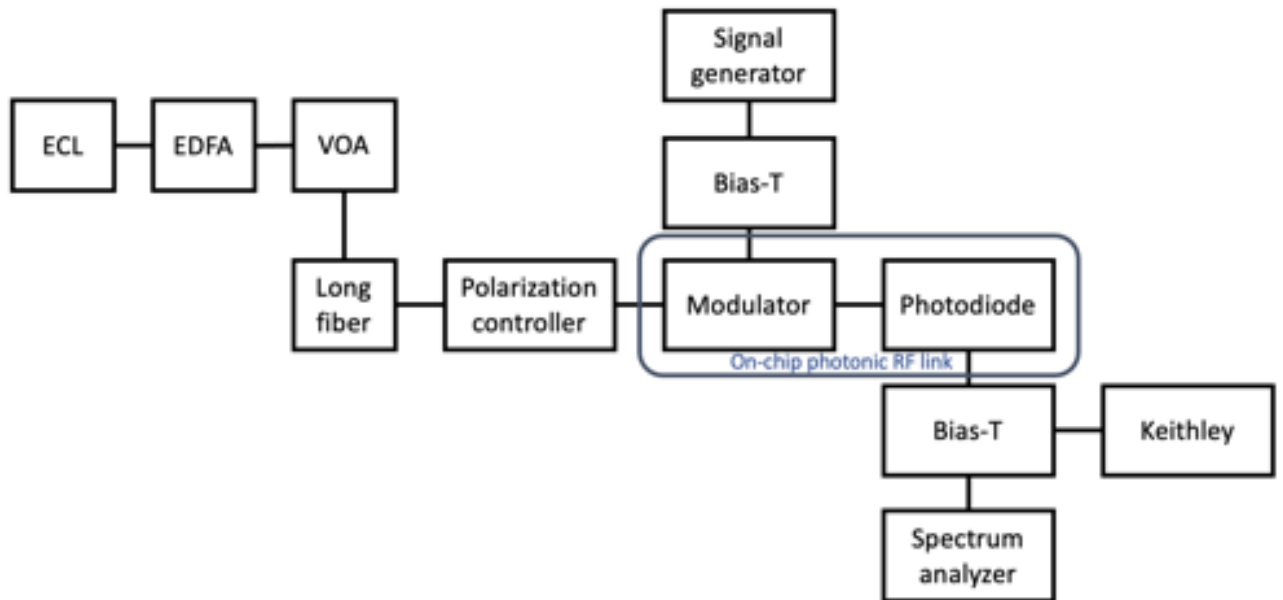


Figure 36: Measurement setup with ESA

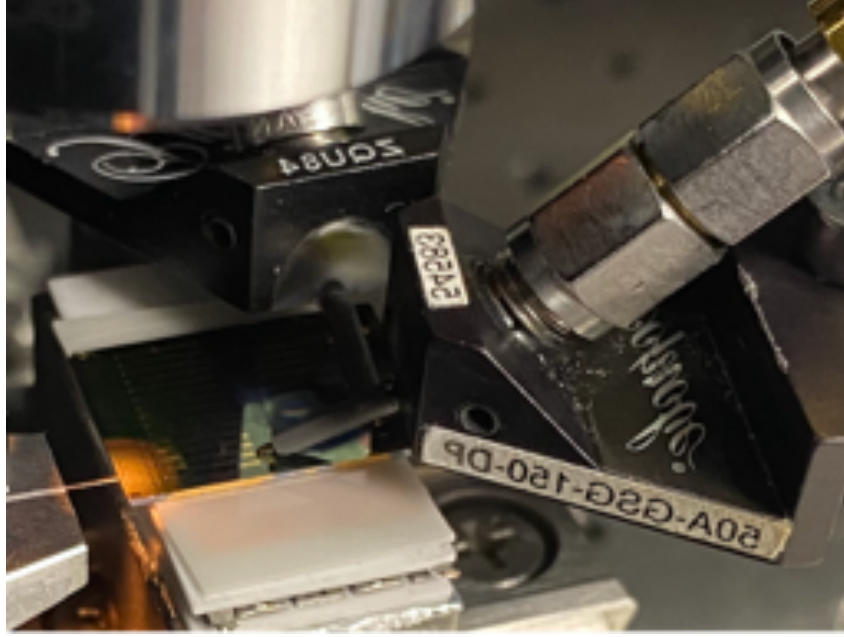


Figure 37: On-chip probing setup

and the RF output power from the photodiode with constant optical power and constant RF input power driving the modulator. The plot in figure 38 shows that the wavelength bias quadrature point of this modulator where the RF power was at its maximum and the corresponding photocurrent was only half of its maximum value. This is because when applying a small AC signal to the modulator, the output power would be maximum where the slope is the largest in the transmission curve where the quadrature point is as discussed in chapter 2.

The RF link frequency response can be measured by supplying a RF signal to the modulator using the signal generator and reading the RF power from the spectrum analyzer with respect to different frequencies as shown in fig 39. This plot shows the raw data (not calibrated for RF loss in the experimental setup) at different input optical power levels.

The loss between the output of the signal generator to the tip of the RF probe and from the RF probe to the input of the spectrum analyzer was calibrated by probing two GSG RF probes onto a short Au coplanar waveguide on a calibration substrate provided by the probe manufacturer GGB as shown in figure 40. This is the loss between two probes and by subtracting this value from the link measurement, we can obtain the RF link gain. As shown in figure 41, the absolute RF output power is calculated by subtracting the calibration data from the raw data.

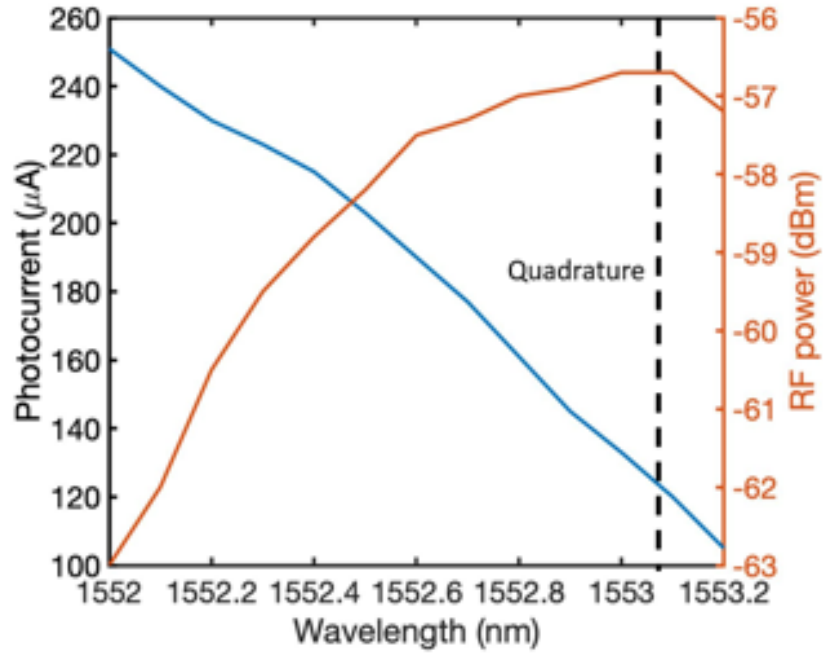


Figure 38: Average photocurrent and RF power from the PD as a function of laser wavelength

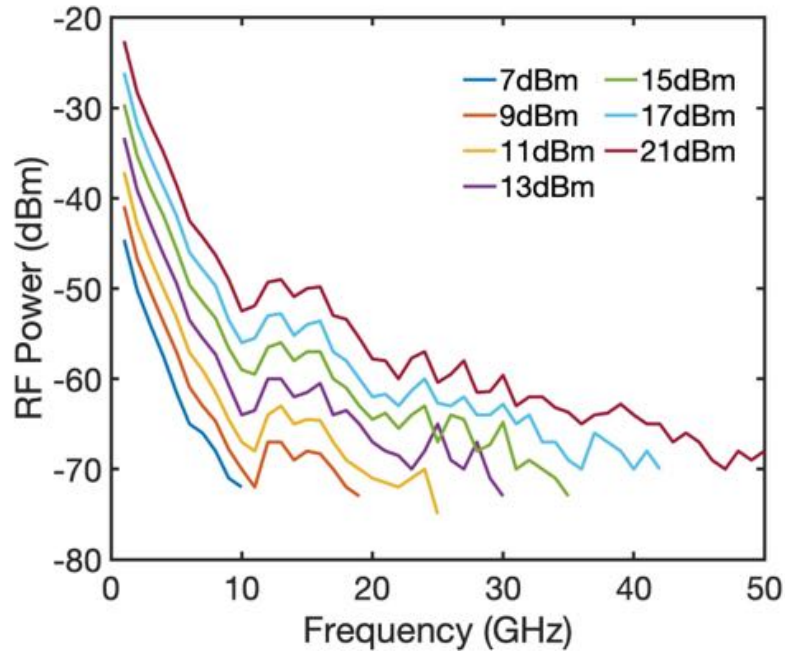


Figure 39: Uncalibrated RF response of the on-chip link at quadrature bias of modulator, 6dBm RF input power, -1.5V reverse bias on photodiode

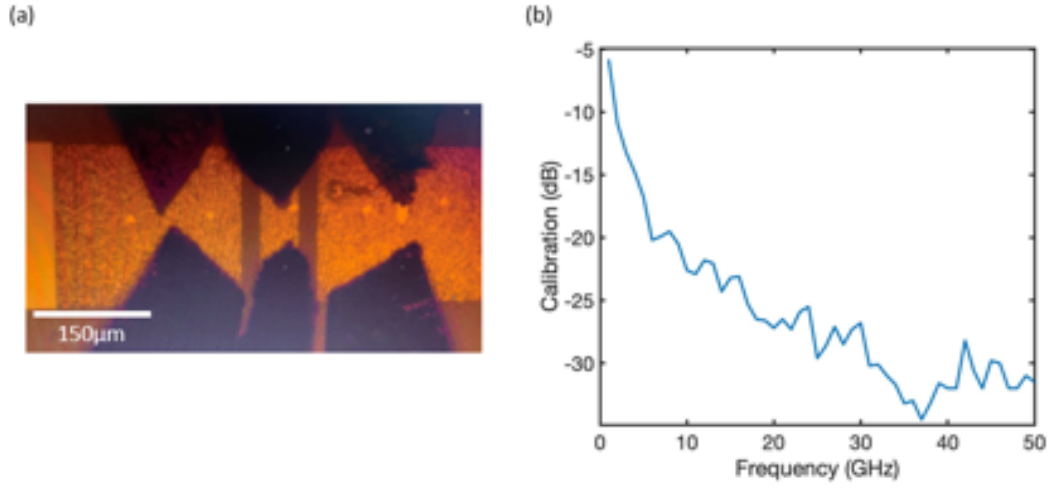


Figure 40: Calibration (a) Microscope picture of the probing setup for calibration (b) Calibration curve

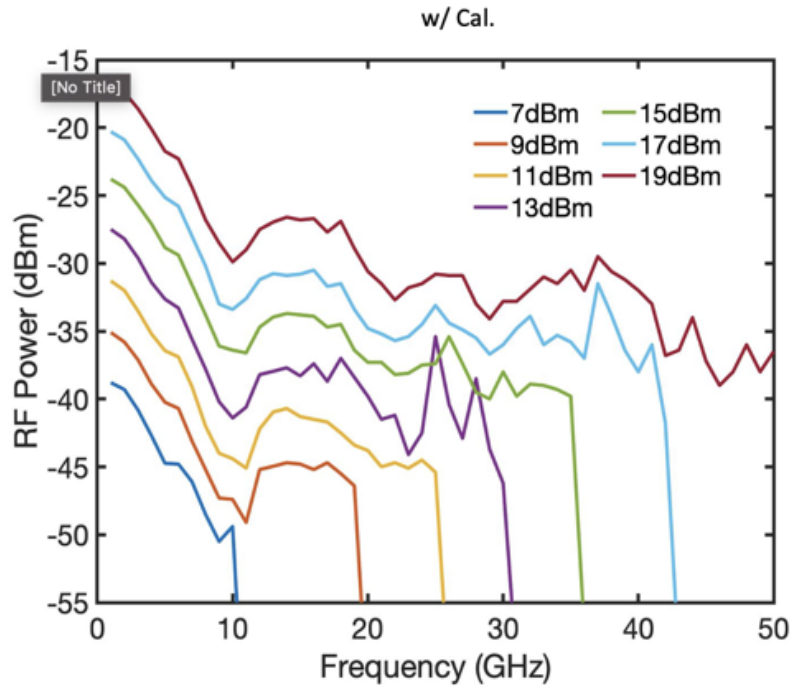


Figure 41: Calibrated RF response of the on-chip link at quadrature bias of modulator, 6 dBm RF input power, -1.5 V reverse bias on photodiode

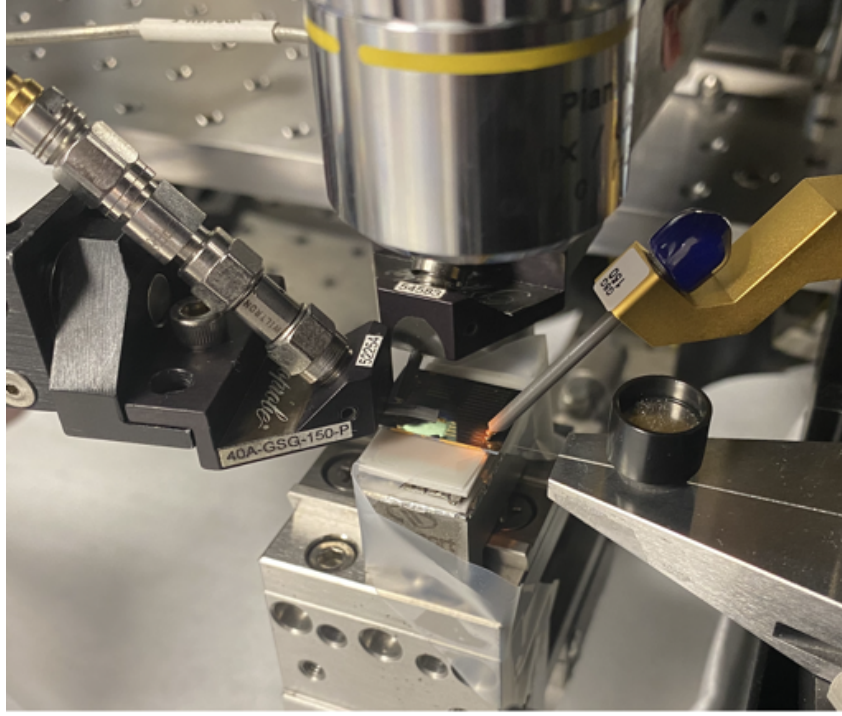


Figure 42: Probing setup with RF probe for modulator termination

For all curves, the difference between adjacent curves is achieved by increasing the optical power by 2 dB which results in 4 dB RF power increase which means the link has not saturated yet and behaves as expected. The sharp drop from DC to 10GHz is likely caused by the modulator not being terminated. With the modulator not being terminated, the reflection is larger resulting lower transmission.

A new RF GSG probe with integrated $50\ \Omega$ resistance was purchased to fit in this setup as shown in figure 42. This probe probed at the opposite end of the modulator and acted as a $50\ \Omega$ termination to suppress reflection of the RF signal.

Since we were primarily interested in the RF link gain which is the S21 parameter of the system, a new setup was built using the network analyzer since it has a readily available quick frequency sweep. This new setup is shown in the figure 43. The bias-tee connected to the modulator was also removed since the modulator was exclusively wavelength-biased.

Figure 44 shows the response with applying the RF signal at the opposite side of the modulator optical input with no termination and figure 45 with termination. At lower frequencies, the gain is higher for the un-terminated system whereas the response is flatter for a terminated system.

Figure 46 and figure 47 show the gain measurements when the input electrical signal is

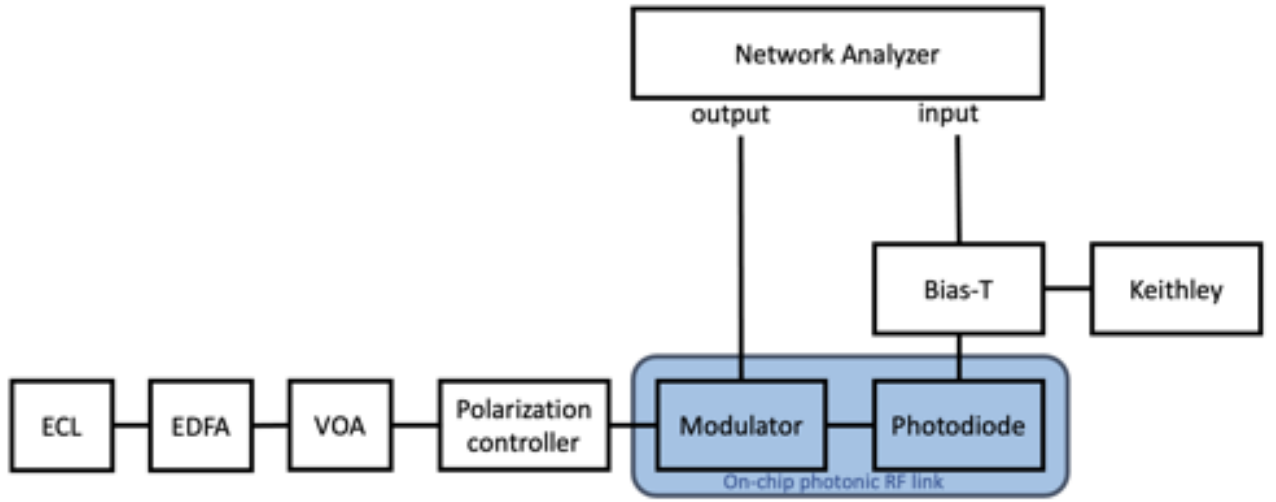


Figure 43: Measurement setup with VNA

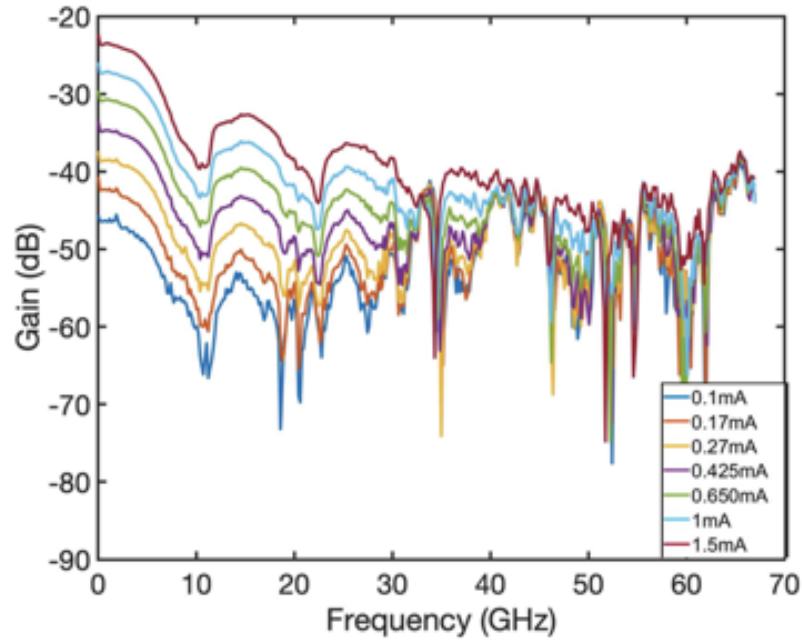


Figure 44: Probe at light output wo/ termination, 1.5V reverse bias on photodiode

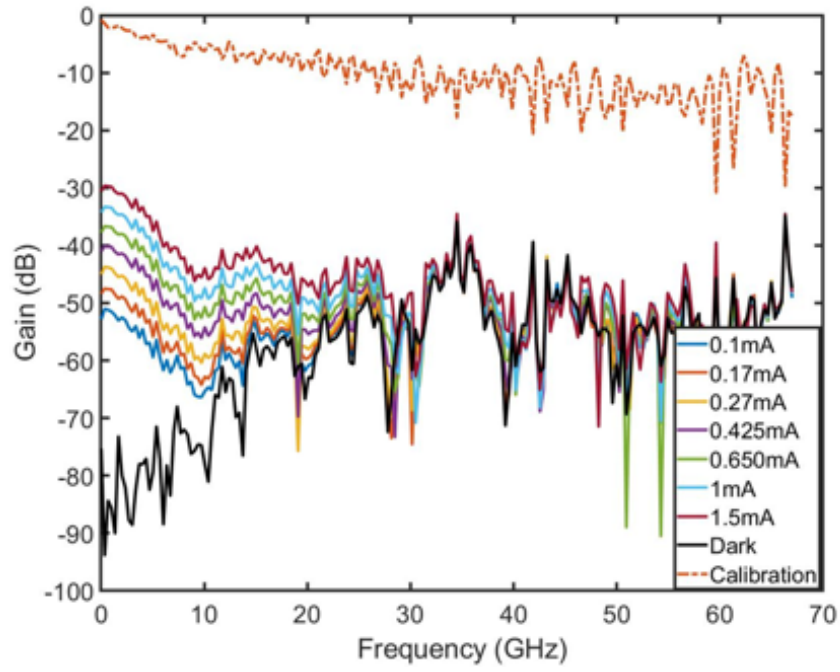


Figure 45: Probe at light output w/ termination, 1.5 V reverse bias on photodiode

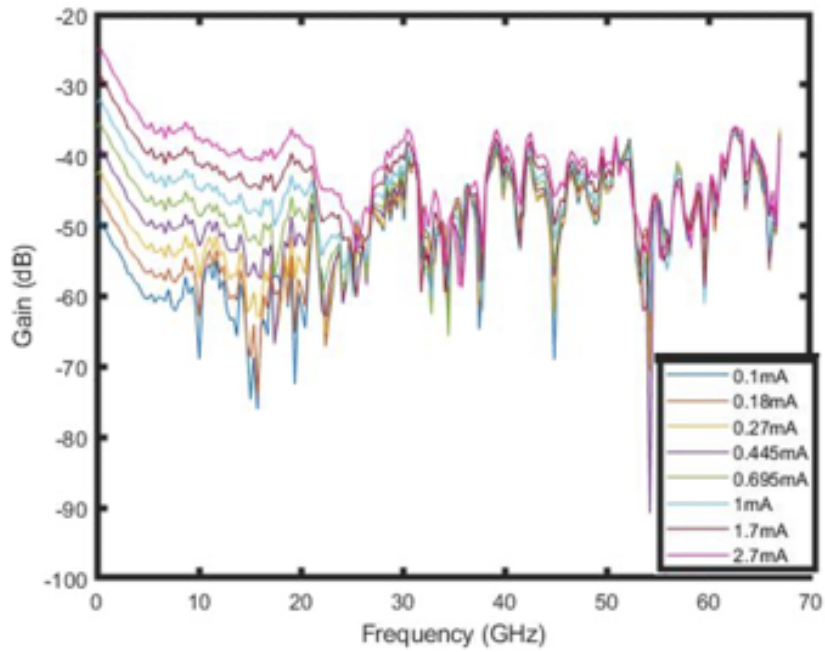


Figure 46: Probe at light input wo/ termination

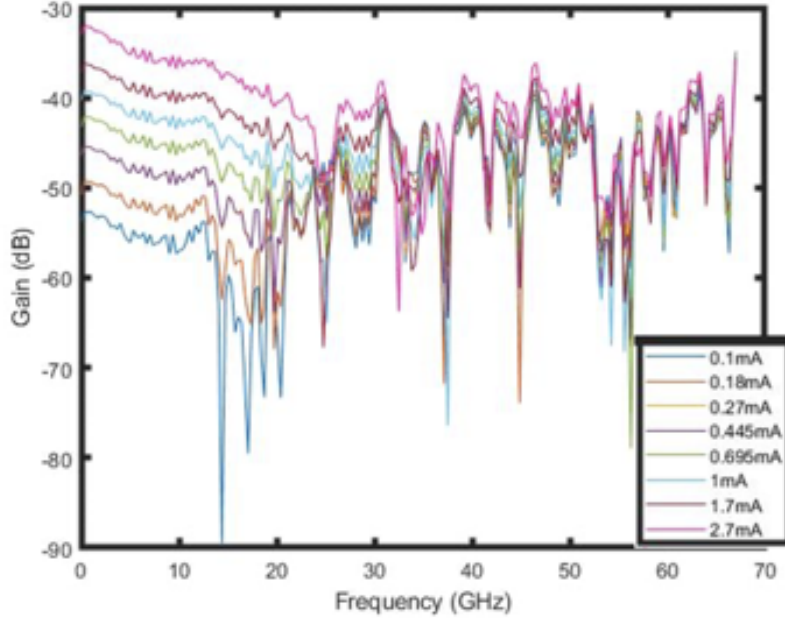


Figure 47: Probe at light input w/ termination

at the same side of the optical input signal of the modulator with and without termination accordingly. Comparing between figure 47 and figure 45, it can be seen that when the electrical signal and optical signal are traveling in the same direction a higher bandwidth is achieved. As expected, this also illustrates the fact that the modulator is based on the traveling wave design.

While there is no DC response on TFLN modulators, there is a need to measure the V_π to calculate the link gain. Methods on measuring the V_π using the RF signal have been demonstrated [59]. One method is to gradually increase the RF power applied to the modulator until the modulator's output signal reaches to a maximum point when measured with a photodiode at a certain frequency as shown in fig. 48. Regardless of the bias condition at the modulator, it will hit a maximum point before dropping to a null.

$$P_{max, in}(f_{RF}) inWatts = \frac{1}{2} \cdot \left(\frac{1.84}{\pi}\right)^2 \cdot \left(\frac{(V_\pi(f_{RF}))^2}{50\Omega}\right) \quad (33)$$

(Ed Ackerman, private communication) Where f_{RF} is the frequency of the RF signal measured and P_{max} is the maximum output RF power detected by the PD. With this method, the V_π was measured and calculated to be around 4.3 V at 1 GHz.

Figure 50 shows the gain measurements with different photocurrent levels from 0.12 mA to 3.44 mA, frequency from DC to 67 GHz at quadrature bias point for the modulator

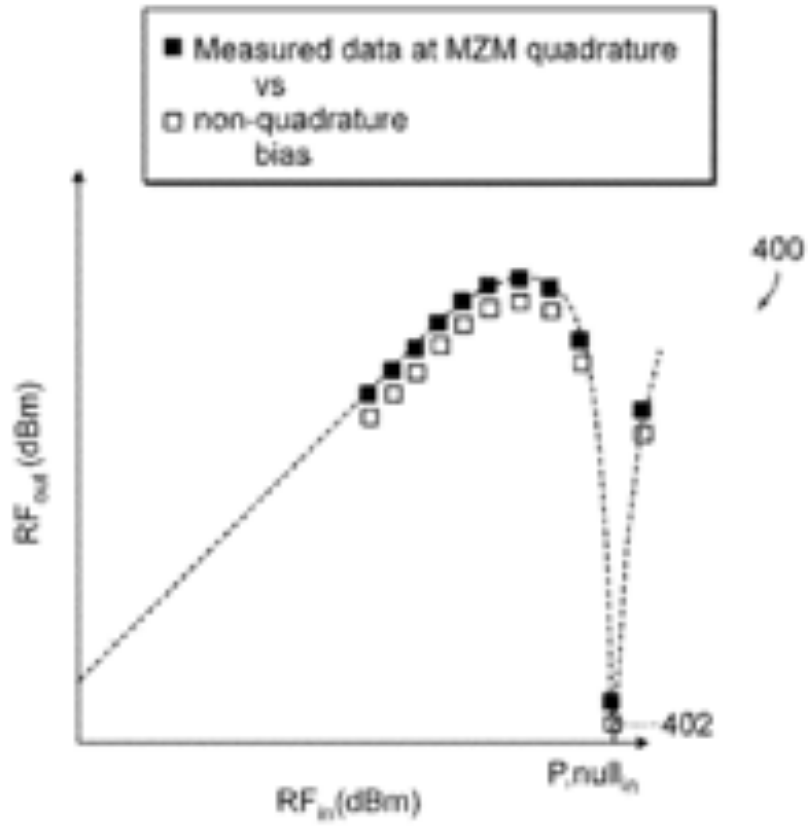


Figure 48: V_{π} measurement method [59]

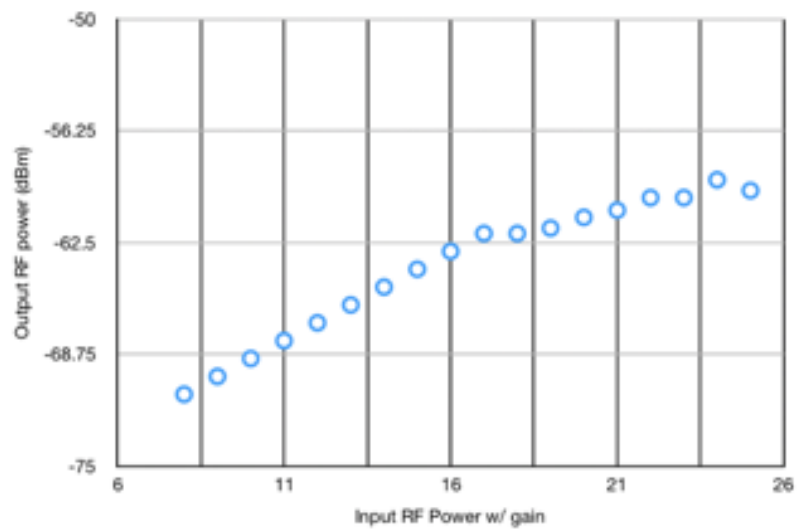


Figure 49: V_{pi} measurement

and -2 V bias on the photodiode. This link exhibits a flat frequency response from DC to 67 GHz and this is because both modulator and photodiode are designed to have around 50 GHz 3dB bandwidth. From this plot, the RF compression of the photodiode can be extracted assuming there is no power compression from the modulator. Figure 51 shows the extracted RF power compression and gain versus photocurrent for 4 different frequencies from 20 GHz to 61 GHz. The dotted line shows the calculated RF gain (equation 29) with the measured photocurrent and 4.3 V V_π . From these plots, it can be seen that the link and photodiode have not saturated yet at 61 GHz and 3.44 mA with 2 V reverse bias on the photodiode. This also shows the improvement of the new waveguide design shown in figure 31 for the power performance of the photodiode since comparing to previous RF saturation measurements of a stand alone photodiode with single mode (0.8 μm) straight waveguide, shows that a higher bias of 3 V at 3.44 mA was needed. This means the waveguide design for a more uniform distributed absorption profile helped to reduce the space charge impedance as we predicted [60] . Figure 52 shows the comparison of the new, optimized design and the previous narrow waveguide performance.

Figure 53 shows the low biasing technique mentioned before. By maintaining a low current level (0.1 mA), the link gain can be improved from moving downward the bias condition at the modulator transmission curve and increase the optical power to maintain this photocurrent. As shown in the plot, more than 20 dB improvement in gain can be achieved using this technique. Figure 54 shows the measurements of keeping the same input optical power and changing the bias condition of the modulator. It can be seen that at quadrature point where the photocurrent is half of its maximum, the gain reaches its maximum.

For gain improvement, low biasing offers a more relaxed requirement for the saturation power of photodiode since it can achieve higher gain at lower photocurrents. However, since it operates close to the minimum transmission, a lot of optical power is required to maintain a reasonable photocurrent. On the other hand, quadrature biasing offers the highest gain with the available optical power at a higher photocurrent level. To determine which bias condition to use, high power handling capability of PD and modulator and availability of high optical power source at the input should be taken in to consideration.

For our TFLN platform, the highest on-chip power test was around 22 dBm, with

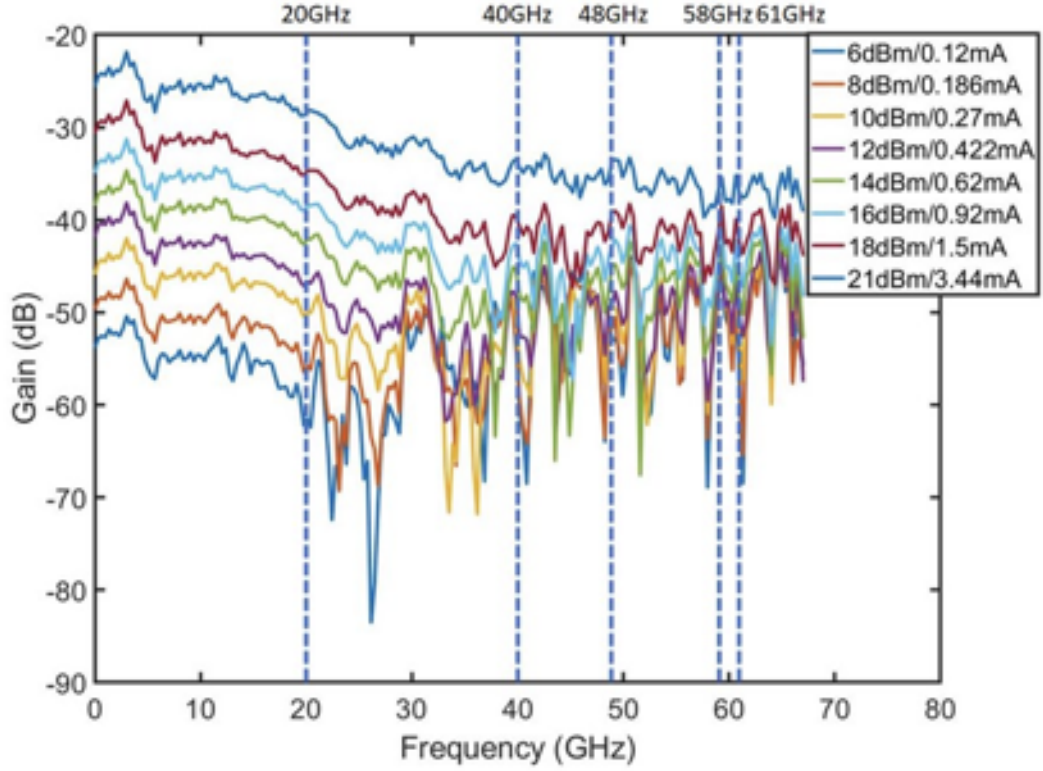


Figure 50: Link gain measurements at different optical input power/photocurrent levels at modulator quadrature point

higher optical power, it can risk damaging the input facet or the waveguide itself. Our measured saturation current for photodiode was around 10 mA. In conclusion, for link gain improvements, there are three limiting factors: available on-chip input optical power, passive waveguide power handling capability and photodiode saturation power. If photodiode saturation power is the limitation, then with low biasing technique, gain can be improved but requires higher optical power. If input optical power is limited, then quadrature bias will give the highest gain.

Figure 55 shows the projected gain and noise figure improvements with low biasing technique. By moving the bias point away from the quadrature both gain and noise figure can be improved by around 20 dB.

Figure 56 shows the measurement on a ESA with 1 GHz RF input applied on the modulator. The noise was measured to be -89 dBm on average.

Since the ultimate goal is to have the complete link on-chip which composes of laser, modulator and photodiode. After developing fabrication flow and robust device structures to integrate these three components together, with efforts from our collaborators at

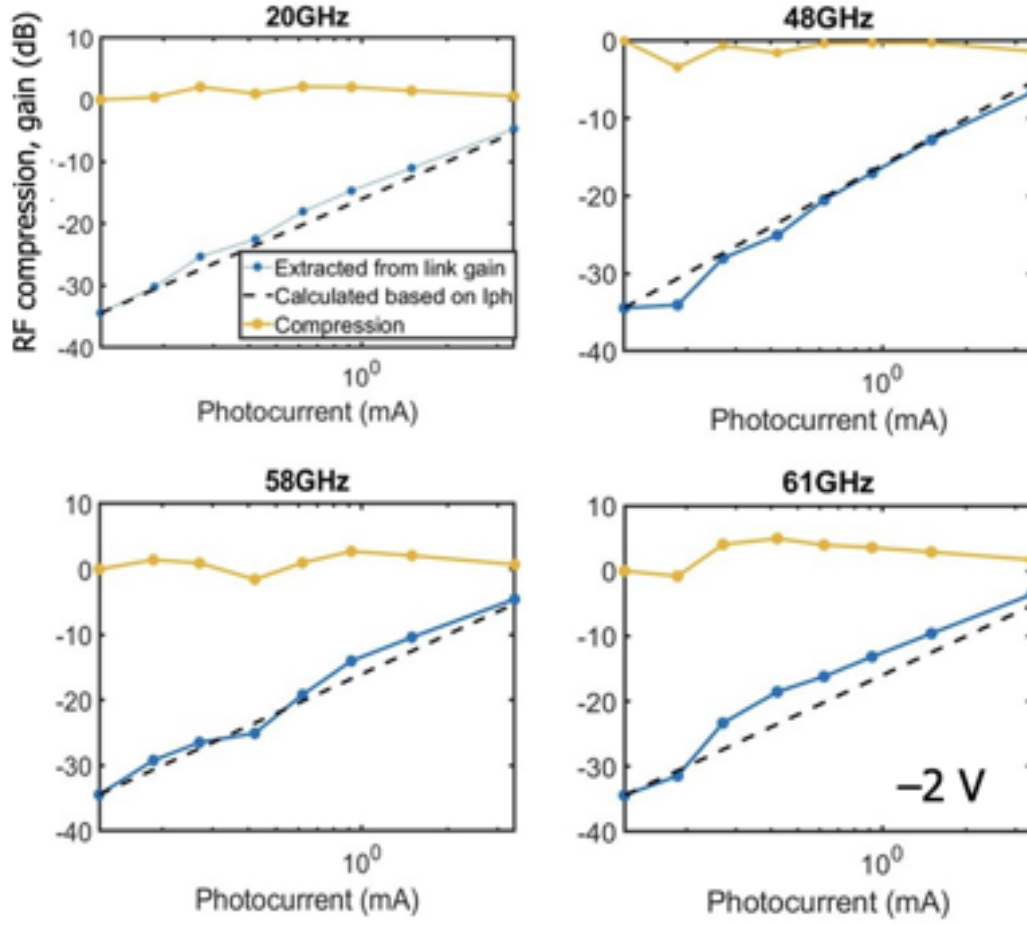


Figure 51: Extracted photodiode compression curve and link gain at different frequencies with 2 V PD reverse bias

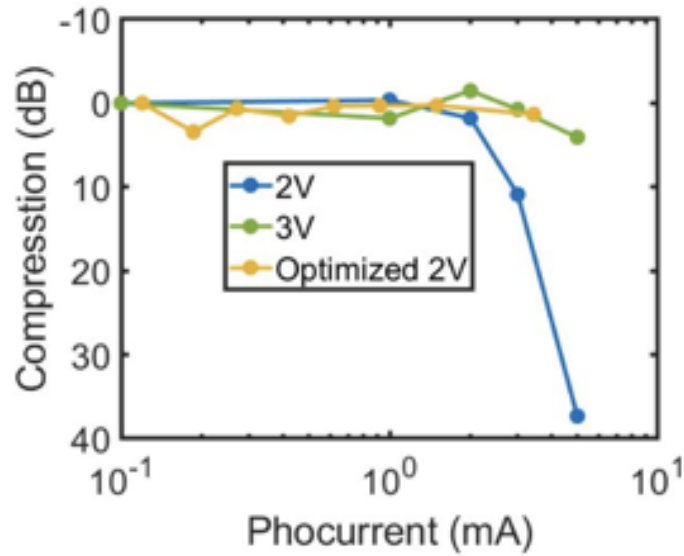


Figure 52: Photodiode RF power compression curves at 48 GHz. Strong compression can be seen for the previous waveguide design at 2 V and 3 mA, while the optimized design remains flat.

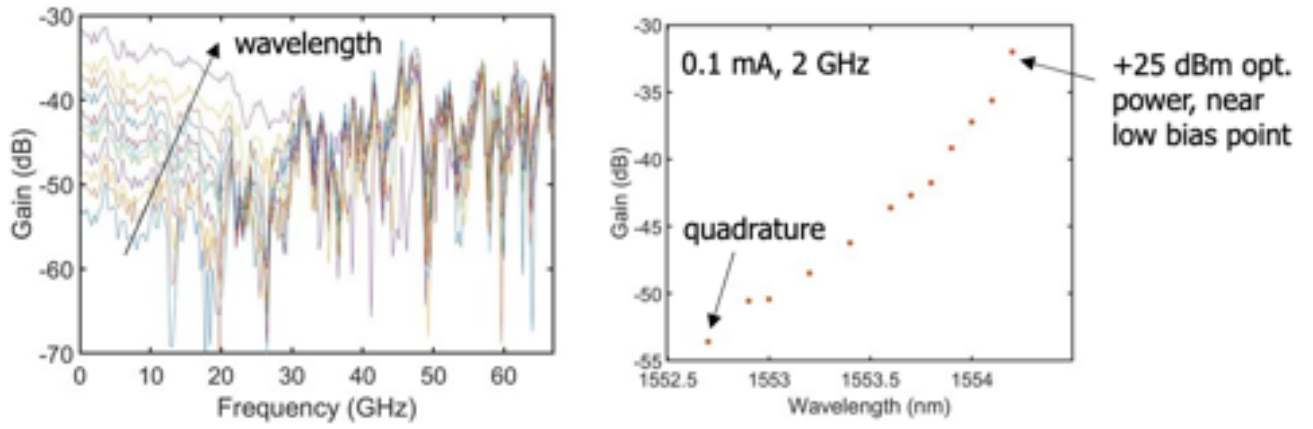


Figure 53: Gain measurements with 0.1 mA constant photocurrent at different wavelengths (=modulator bias points)

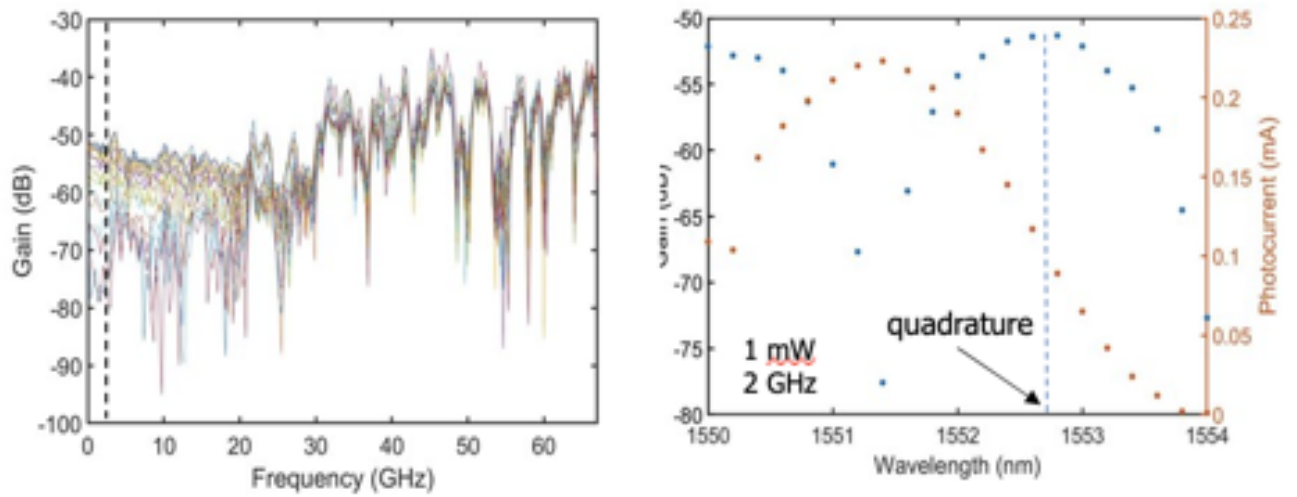


Figure 54: Gain measurements with 1 mW optical input power at different modulator bias points

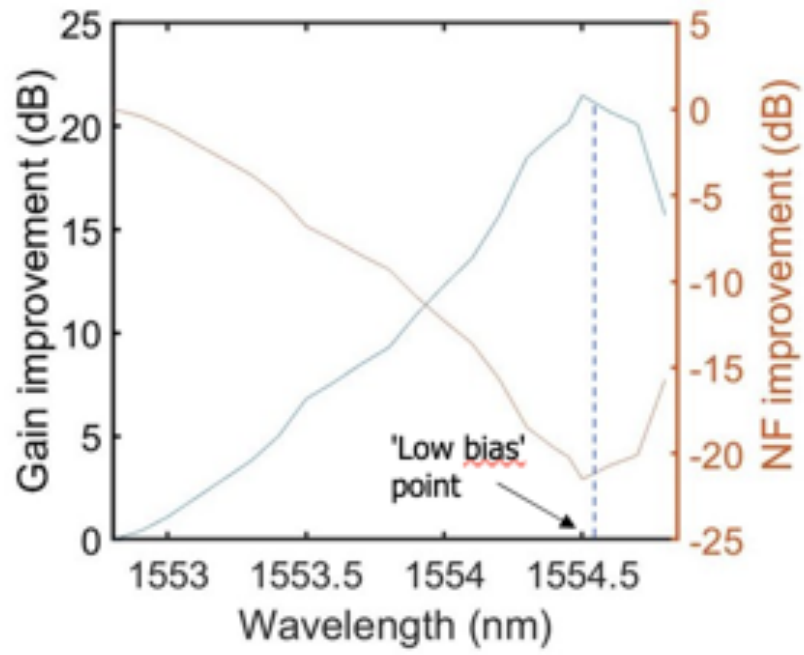


Figure 55: Gain and noise figure improvements with low biasing technique

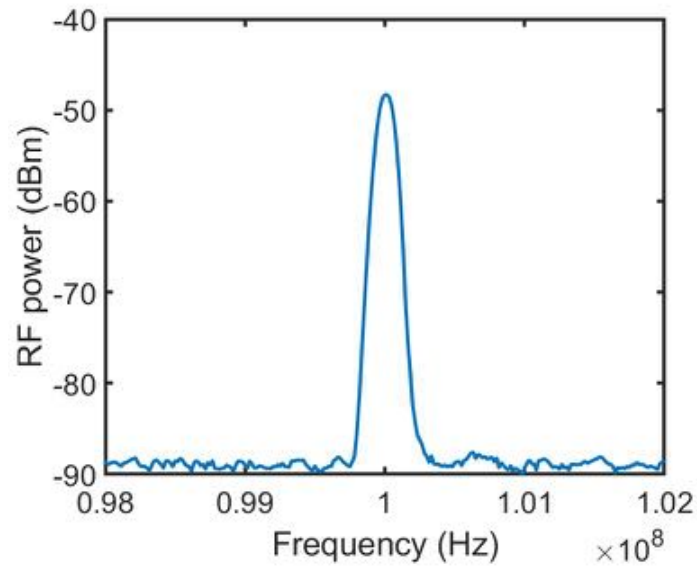


Figure 56: Noise measurement at quadrature bias point

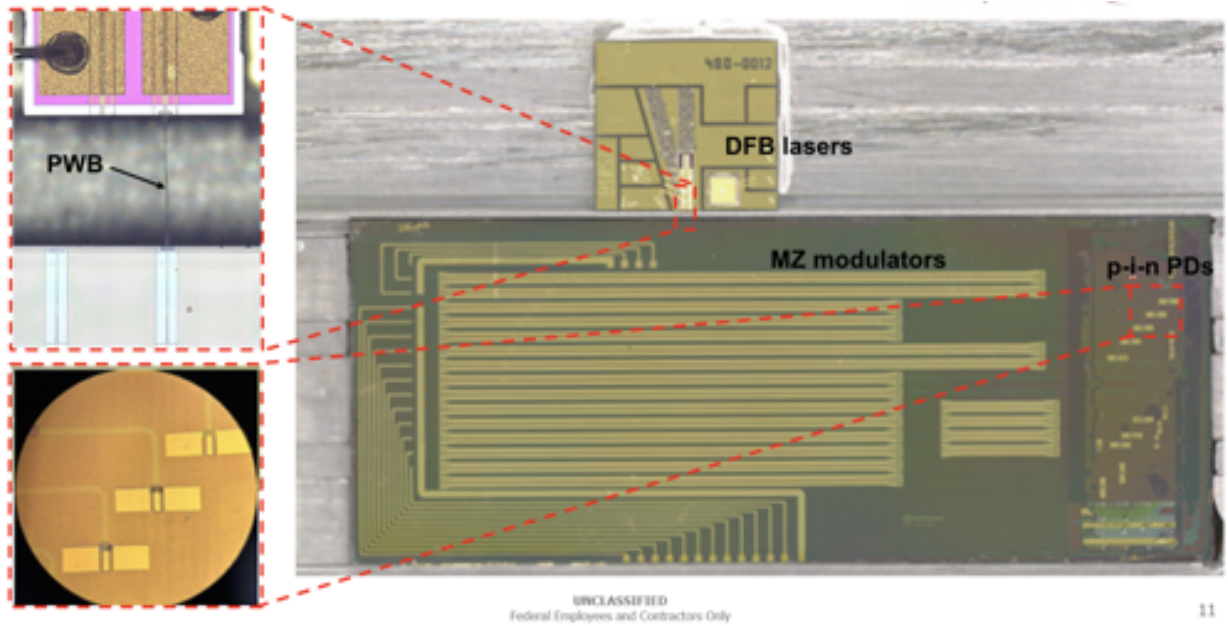


Figure 57: Pictures of laser, modulator and photodiode integrated on the same TFLN chip

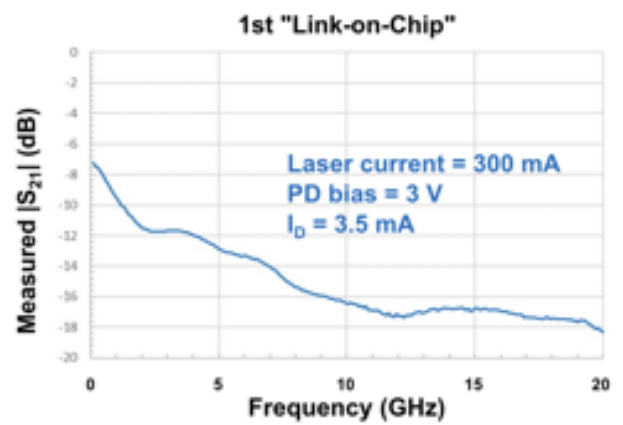
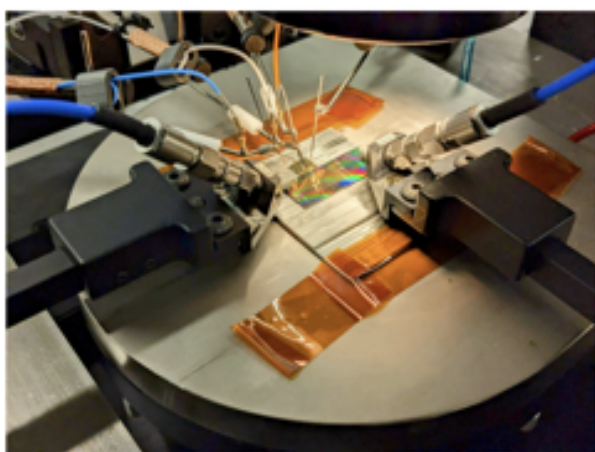


Figure 58: First integrated RF link on chip with gain measurement

Hyperlight (modulator) and Freedom Photonics (laser), the first complete on-chip photonic RF link is shown in figure 57. This chip was first fabricated at Hyperlight on TFLN with optical layer, modulator electrodes and heaters. Then this chip was sent to UVA for photodiode integration using adhesive bonding. In the end, this chip was sent to Freedom Photonics for laser integration using photonic wire bonding.

Figure 58 shows the link gain curve done at PSI with all electrical inputs. This is the fundamental step with huge impact in two aspects. The first one is that this is the first demonstration of the capability of integrating optical source (laser), optical processing (modulator) and optical detection (photodiode) on the same TFLN chip, pushing photonic integrated circuits on TFLN to the next chapter. The second one is that this is the first photonic RF link that has all electrical inputs, meaning the RF input signal is boosted by a DC electrical signal applied to the laser.

In summary, this work demonstrated the first integrated photodiode with TFLN modulator on-chip with link gain measured up to 67GHz. The first fully integrated photonic RF link with laser, modulator and photodiode on-chip was enabled by this work.

7 High quantum-efficiency photodiode and homodyne detection

7.1 Introduction

Balanced photodiodes for homodyne detection are essential in many applications that involve sensitive measurements both in quantum and classical fields. However, in the measurement of deterministic two-mode-squeezed quantum microcomb, minimizing the system loss and classical noise are essential [11].

Photonic integrated circuits (PICs) have gained significant attention due to their small foot print, versatile functionalities, and low loss. With the realization of low propagation loss SiN waveguides now available, SiN/Si has been proposed as one major photonics platform for quantum applications such as squeezed quantum microcomb [61]. Because quantum signals like squeezed light are very sensitive to loss, it is important to reduce the loss between the generation of the signal and detection and therefore on-chip detection with high quantum efficiency is most ideal. Tasker et al. demonstrated measurements of

9G Hz squeezed light by detecting the signal using a silicon photonic chips with integrated electronics [62]. Bruynsteen et al. also reported integrated balanced homodyne detectors on a silicon photonic chip featuring up to 20 GHz bandwidth [63]. This work presents the design of integrating high quantum efficiency balanced photodiodes with a squeezed quantum microcomb on silicon nitride platform for the first time.

7.2 Design and fabrication

For a pair of balanced photodiodes in operation, two optical signals are detected by two almost identical photodiodes as shown in figure 59. The output photocurrent of this circuit is the difference of the two photocurrents generated in each PD. The goal is to enhance the differential signal and rejecting the common part. The ability of a pair balanced photodiodes to perform this rejection is quantified as the common mode rejection ratio (CMRR). It is determined by the ratio of the output power when both photodiodes are illuminated in differential mode to the output power when both photodiodes are illuminated in common mode.

$$CMRR = \frac{|I_1 - I_2|^2}{|I_1 + I_2|^2} \quad (34)$$

Where I_1 is the photocurrent from the photodiode one and I_2 is the photocurrent from the photodiode two.

Figure 59(c) shows the waveguide layout design for the photodiodes. (The SiN chip layout also included microresonators designed by X. Yi's group that are not shown in the figure.) The right side of the chip has a thinned cladding (100 nm) for PD bonding. The output signal from the resonators is fed to a 2x2 multimode interferometer (MMI) splitter and mixed with the local oscillator (LO). The outputs of the MMI are routed to a pair of balanced photodiodes. In order to achieve higher saturation power and thus accommodate 10 dBm LO power, waveguides are tapered from 1 μm to 10-50 μm width. The taper length is 12 times the taper ratio from beginning to end, to minimize the transmission loss.

Figure 59(a) shows the balanced photodiode pair under different illumination conditions when measuring CMRR and figure 59(b) shows the schematic physical structure of a pair of balanced photodiodes. Figure 59(c) shows the layout design. Figure 60 shows

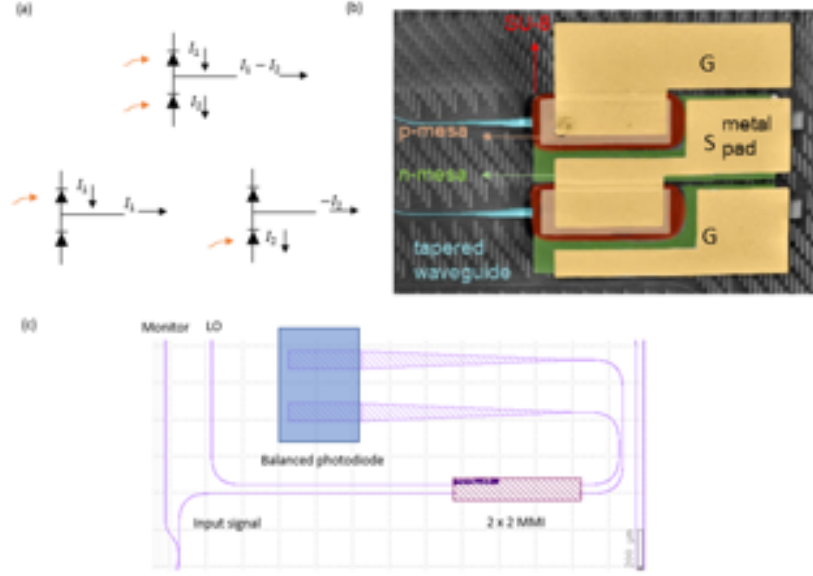


Figure 59: Balanced photodiodes (a) Circuit with three different illumination conditions. (b) Schematic of the physical structure of a pair of balanced photodiodes. (c) Layout of homodyne receiver for detection of squeezed quantum comb as input signal with integrated balanced photodiodes.

the simulation of a waveguide tapered from $1 \mu\text{m}$ to $10 \mu\text{m}$ before entering photodiode. Comparing to the previous result of a narrow single mode waveguide [Fig 24(c)] the e-field is more evenly distributed along the direction parallel to the width of the waveguide. This increases the uniformity of optical intensity in the absorber which results in the improvement of saturation power as discussed in chapter 5. The length difference in two waveguides at the output of the MMI is around $170 \mu\text{m}$ long which negligible in the frequency range we are measuring at.

7.2.1 Loss mechanisms

The loss mechanisms can be separated into four different categories based on different origins of the loss. The first one is the optical loss in the waveguides and MMI before being absorbed by the photodiode. The second one is the process that light couples from the optical waveguide into the photodiode absorber. The third one is related to the process that photogenerated carriers may not be completely collected. In addition, and for PD testing purposes, the fourth one is related to the off-chip loss and loss between the fiber tip and the photonic waveguide chip.

The optical power at the lensed fiber tip was measured by inserting it into an in-

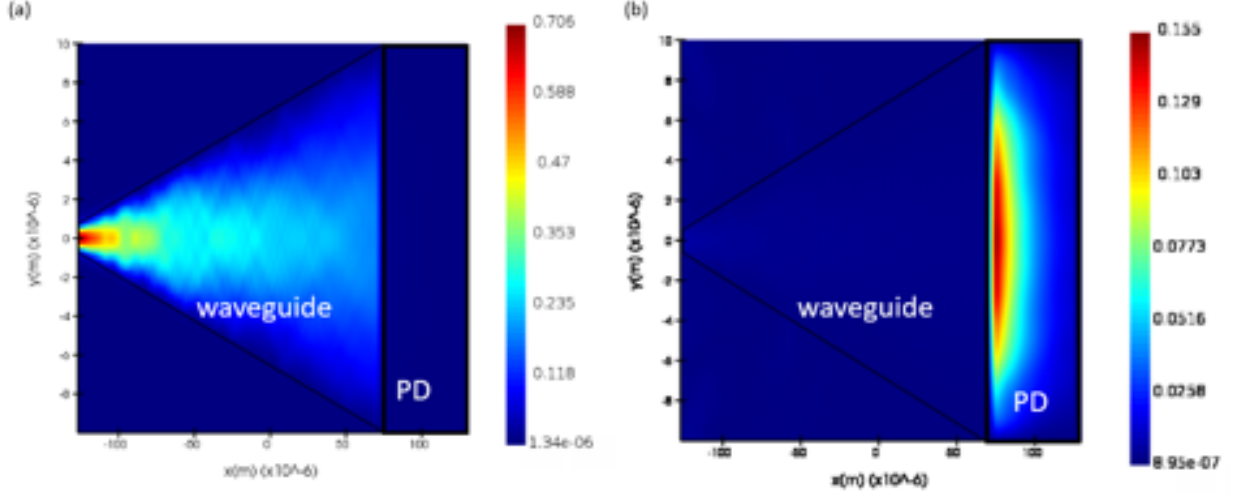


Figure 60: E-field simulation of photodiode with wide waveguide underneath (a) Cross section view in the middle of the waveguide. (b) Cross section view in the middle of the photodiode's absorber.

tegrating sphere optical powermeter for 1550 nm wavelength light and the difference between this number and the reading on the VOA in the measurement setup is fixed and determined as the loss between the VOA and fiber tip.

The fiber-to-chip coupling loss was measured on the same chip with photodiodes integrated on it. This was done on two test structures with identical designs consisting of a straight waveguide and a 2x2 MMI and another straight waveguide with $1 \mu m$ width. Since there is a transition step for the top cladding oxide thickness from the $3.3 \mu m$ in the quantum signal region on the chip to $0.1 \mu m$ in the PD bonding region, there is a reflection loss of 0.2 dB plus or minus 0.1 dB at this oxide transition interface provided by the photonic foundry. The loss of the 2x2 MMI is also provided by the photonic foundry being 0.22 dB at 1550 nm plus or minus 0.15 dB. The edge-coupler at the facet on the left side of the chip has a loss of 1.08 dB when using $2.5 \mu m$ spot-size fiber. The propagation loss of the waveguide is negligible for this scale of length (less than 1 cm). The edge coupler on the right side of the chip has thinned oxide and no taper design. The simulated coupling transmission from the fiber tip to the $1 \mu m$ wide waveguide is 0.34.

The coupling loss was simulated using the numerical FDE solver. The structure of the silicon nitride waveguide cross-section consists of a 800 nm tall and 1000 nm wide silicon nitride waveguide with $4 \mu m$ oxide underneath and 100 nm oxide on top of it. The mode

inside of this waveguide structure is simulated using FDE solver at 1550 nm wavelength. The optical mode from the fiber is approximated as a Gaussian beam source generated using lumerical. From the datasheet provided by the fiber company, the spot diameter at the 14 μm working distance is 2.5 μm . Thus, a Gaussian beam with a radius of 1.25 μm can be created with the refractive index of 1.44.

The coupling loss of the optical mode in the waveguide to the photodiode absorber can be separated into the power being reflected, the power being absorbed, and the power being radiated. Since the mode profile changes when the light 'sees' the photodiode, resulting in a change in the effective refractive index difference, there is a reflection. The reflection was simulated by putting a power direction monitor right before the light couples into the photodiode and monitor the percentage of light traveling/being reflected to the opposite direction. The simulated reflection is 0.9 %. The light being absorbed by the photodiode absorber is simulated by setting a 3D monitor that includes the entire absorber region. In order minimize the metal absorption loss, the contact was designed to be far away from the absorber. Free carrier absorption in n-type doped InP has a coefficient between 1/cm to 10/cm [64] which is less than 0.05% with 470 nm thickness in our structure. Grading layers were added in the epitaxial layers to ensure smooth bandgap difference transition to reduce the carriers being trapped and ensure carriers generated in the absorption region can be all collected with designed bias condition.

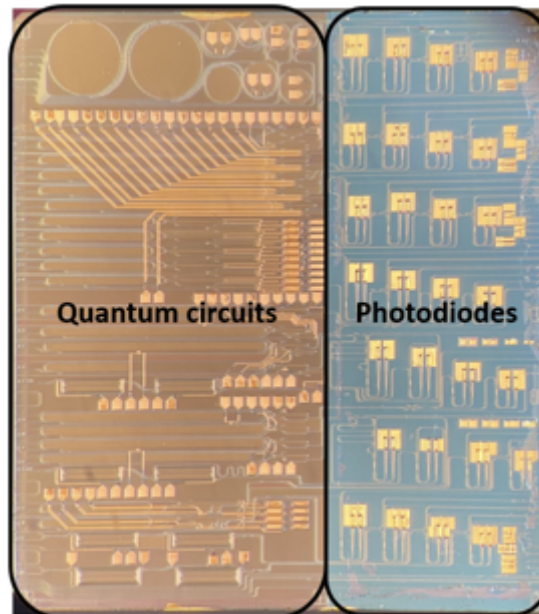


Figure 61: Microscope picture of the SiN chip

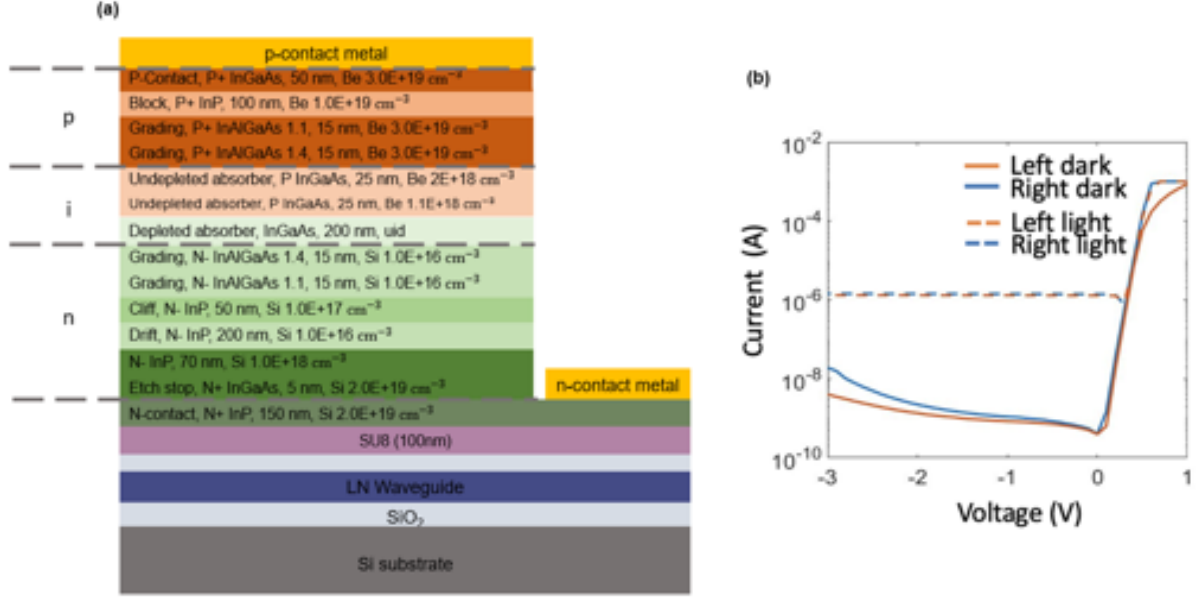


Figure 62: Device structure and DC measurements. (a) Device cross section and epitaxial structure of the integrated PD. (b) Dark current and photoresponse to microscope light characteristics of a pair of 14000 μm² devices.

Figure 61 shows the fabricate SiN photonic chip with quantum circuits (designed by X. Yi's group) on the left and photodiodes on the right. Figure 62(a) shows the cross sectional view of the photodiode on top of the SiN waveguide along with the epitaxial structure stack. The material stack was optimized for high quantum efficiency to the 800 nm thick SiN waveguide layer. Since the light is evanescently coupled from the waveguide to the photodiode absorber, the e-field distribution in the absorber is related to the mode area in the SiN waveguide. In order to improve the saturation power performance, the waveguide underneath the photodiodes were tapered to 20 μm wide [60]. This is because by tapering the single mode waveguide to a wider shape gradually, the mode becomes larger gradually with no loss. This larger mode when absorbed by the photodiode absorber will results in a more spread out distribution in e-field in the photodiode and this has a lower maximum e-field magnitude comparing to the single mode waveguide coupling while maintaining no scarification in quantum efficiency. The SiN waveguides were fabricated by a commercial foundry (Ligentec). Figure 63 (c) shows the false-colored scanning electron microscope (SEM) picture. The blue shows the silicon nitride passive waveguide and MMI structures, the green shows the bottom n-mesa of the photodiode, the orange shows the top p-mesa, the red shows the SU8 passivation around the top

mesa and the yellow shows the metal contacts. The photodiodes were heterogeneously integrated on the SiN photonic chip using SU8 adhesive bonding. The 2x2 MMI made of SiN was provided by the foundry PDK. The epitaxial structure was first bonded as a large die on to the photonic silicon nitride chip using adhesive bonding with SU8. Then using lithography patterning, the III-V material were defined into two identical photodiode mesas. Since the mesa size is relatively large, mesas were processed by wet etchant HCl and H_3PO_4 to have smoother sidewall, simpler process and higher yield. SU8 passivation was added to insulate the p and n mesa sidewalls for non-airbridge structures to increase the yield. Metal pads were then added for 150 μm pitch GSG probe. The n-mesa of the top photodiode is connected to the p-mesa of the bottom photodiode by directly depositing metal pad on them to form the balanced configuration. This connecting two mesas on-chip minimizes the imbalance between the two photodiodes.

Figure 62 (b) shows the IV characteristics of the device from 1 Volt forward bias to 3 Volt reverse bias. The dark current is around 10 nA and the dark current and response to light are similar for both PDs in the balanced detector.

7.3 Measurements

7.3.1 Common mode rejection ratio and radio frequency response

Figure 63 shows the radio-frequency measurements. The single illumination mode was measured by probing the balanced photodiodes separately (only connecting one ground pad and the signal pad, but leaving the second ground pad open) and send one modulated optical signal to one input of the 2x2 MMI. As shown in figure 63 (c), the two single illumination curves overlap with each other well representing that the two PDs have high similarity in their RF response. The common mode was measured using the setup in figure 63 (a), a continuous-wave (CW) laser light was modulated by an analog signal generator using a lithium niobate optical amplitude modulator and sent into one input of the 2x2 MMI. Since the p-mesa of the bottom photodiode is connected to the n-mesa of the top photodiode, the balanced photodiodes were probed using a single 150 μm -pitch ground-signal-ground (GSG) probe. The differential mode was measured directly by heterodyning two CW lasers (one external cavity laser and one distributed feedback laser, DFB) at two different wavelengths on-chip using both inputs of the 2x2 MMI as shown in figure 63 (a).

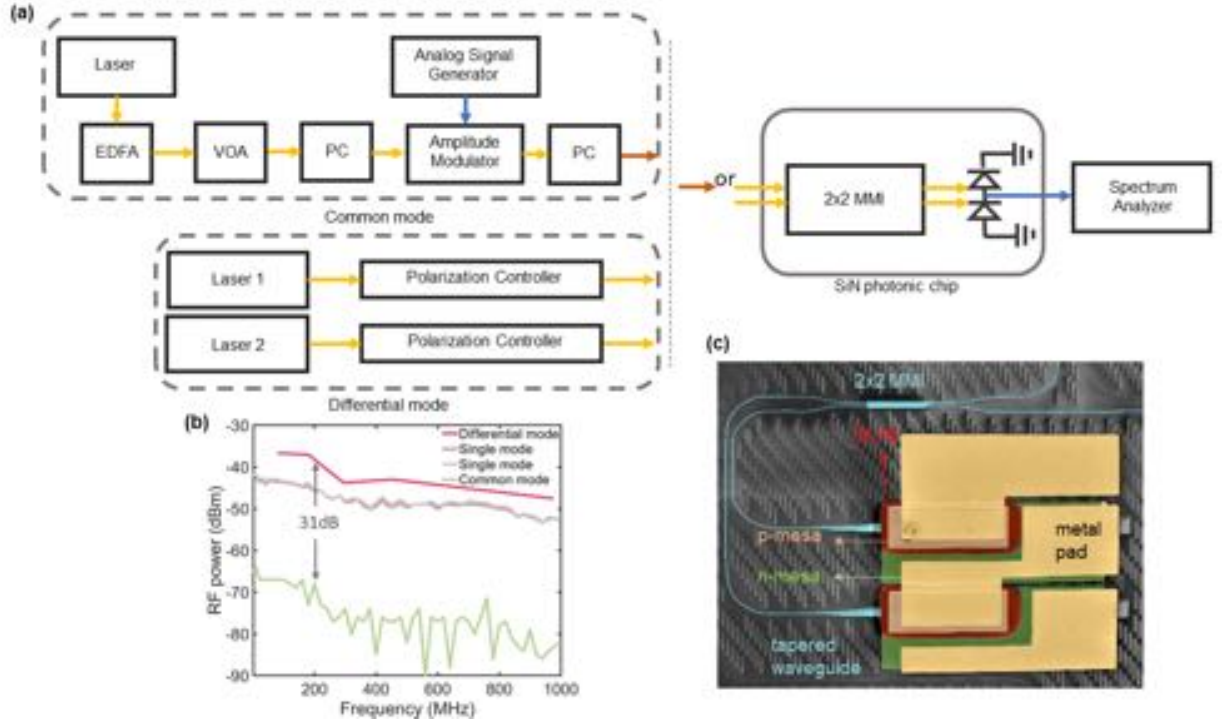


Figure 63: RF measurements. (a) Measurement system setup for common mode and differential mode. (b) False colored SEM picture. (c) Frequency response measured in of differential mode, common mode, and single illumination mode with 1V reverse bias.

By tuning the temperature controlled DFB laser, the RF frequency of the beat signal changes. Since the resolution in wavelength of the temperature controlled laser is large when converted into RF frequency step, therefore there were only few points collected in between DC to 1 GHz. The distance between the differential mode and the common mode is the CMRR which was 31 dB. The source of the imbalance can come from the imbalance from the MMI splitting ratio, the imbalance of the photodiode mesa, the imbalance in silicon nitride waveguide after MMI and before entering the two photodiodes.

The single illumination mode measurement also shows the 3dB bandwidth of the single PD which is around 200 MHz and the common mode measurement shows the 3dB bandwidth of one pair of balanced-photodiode which is around 100 MHz. This is expected since this device is RC limited and by connecting to photodiodes together into a balanced photodidode pair, the active area doubles which leads to half the 3dB bandwidth. The differential mode measurement is expected to have the same bandwidth as the common mode measurements since the active area is the same for both measurements.

7.3.2 Quantum efficiency

Figure 64(a) shows the cross sectional view of the single mode silicon nitride waveguide with 1 μm wide and 800 nm thick waveguide core. (b) shows the cross sectional view after gradually tapering the waveguide into 20 μm width. (c) shows the top view of the gradual long taper and (d) shows the top view of e-field distribution in the photodiode absorber. (e) shows the side view of how the light couples from the waveguide to the photodiode absorber with metal included. As shown in the plot, the light is absorbed in the first 50 μm length of the photodiode. (f) shows the quantum efficiency dependency of the photodiode length which agrees in figure (d) and (e).

Three devices with 200 μm length were measured. They are test devices that only connected to waveguides that couples light from the left side of the chip without going through any MMI or top cladding step. The lower data points in figure 64 (g) shows the external quantum efficiency of these three devices by taking the ratio of the optical power at the fiber tip and the photocurrent reading as measured at both 0 V and 1 V reverse bias with on average 29 % and 33 % external quantum efficiency respectively. The bottom line represents the theoretical 100% internal QE line with fiber-chip coupling loss being the only loss mechanism.

Since the fiber-to-chip coupling loss can not be measured directly when there is a photodiode, three 'through' waveguides with six facets were measured on the same photonic chip using the same setup. The top line in figure 64 (g) shows when the coupling loss was corrected to the external QE to get the internal QE. The average of these 6 coupling loss was used and the error bar represent the standard deviation of the coupling loss.

7.3.3 Shot noise and quantum clearance

The shot noise of the photocurrent is caused by the randomness of the arriving photons, and cannot be eliminated in the homodyne detection configuration since shot noise current fluctuations in the two PDs are not correlated and cannot be suppressed by current subtraction. Since the shot noise at practical photocurrents in the PD is low and below the noise floor of our electrical spectrum analyzer, a low noise amplifier is needed.

Quantum clearance is defined as the ratio between the system electronic (thermal) dark noise and the highest shot noise level measured with CW light. There are two ways to improve quantum clearance. One is to reduce the system dark noise and the other one

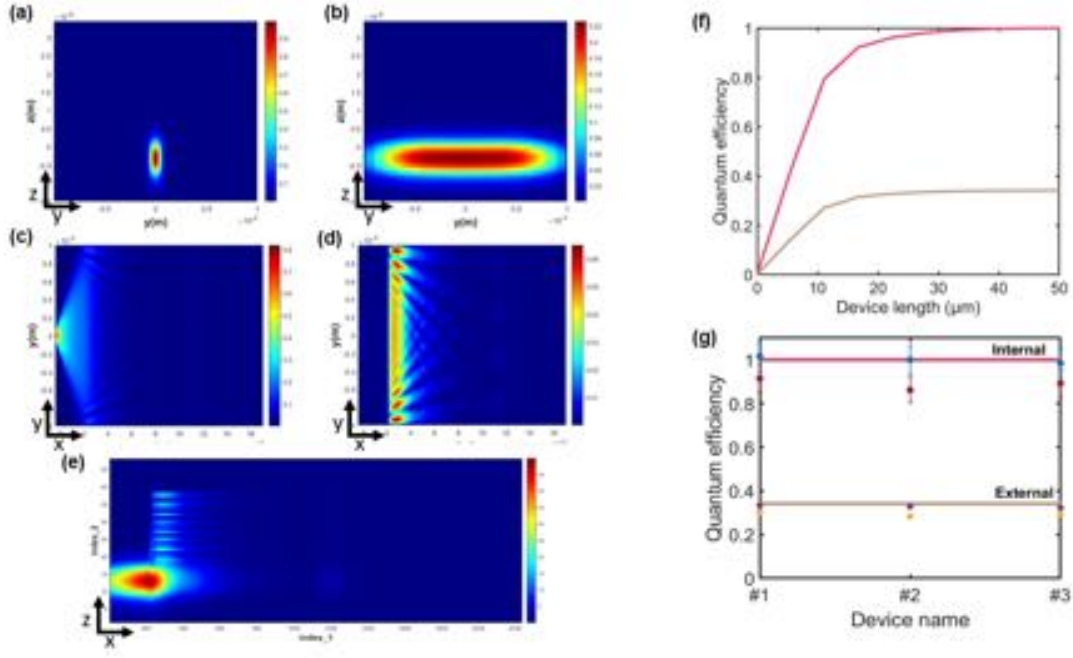


Figure 64: Quantum efficiency. (a) Mode profile at the beginning of the waveguide taper. (b) Mode profile at the end of waveguide taper before entering PD. (c) E-field in the waveguide. (d) E-field in the absorber. (e) Cross section view. (f) Quantum efficiency versus device length. (g) Measured quantum efficiency.

is to increase the photocurrent which requires high saturation power of the homodyne detector.

Efforts in reducing the system dark noise are limited by amplifiers. There are three main source for system thermal dark noise. (Since noise power is related to the resolution bandwidth, in order to unify unit conversion, all measurements of noise power in this section are referred to at 100 kHz resolution bandwidth). The first one is the loss in the cable, this is minimized through using the shortest cable connections and the least amount of connections possible after the signal generated from the photodiode and before entering the spectrum analyzer. The second source is the noise floor of the spectrum analyzer noise floor. There are two ESAs in our lab Agilent E4440A and FSU, and the Agilent (E4440A) has 10 dB lower noise floor. The dark curves in figure 19 shows the noise floors using FSU ESA and the dark curves in figure 66 shows the noise floor using the Agilent ESA under the same settings with 100 kHz RBW, 1kHz VBW and zero dB input attenuation. The third and the most major source is the amplifier noise. One thing to notice is that when the measurement goes beyond the bandwidth of the amplifier, it acts as a low pass filter for noise and therefore that noise measurement. It is essential

for noise measurements that noise was measured within the bandwidth limitation of the amplifier. The amplifier used in the measurements of figure 66 was ZFL-500LN+ from Mini-circuits and the amplifier used in measurements in figure 65 was DHPCA-100 from Femto.

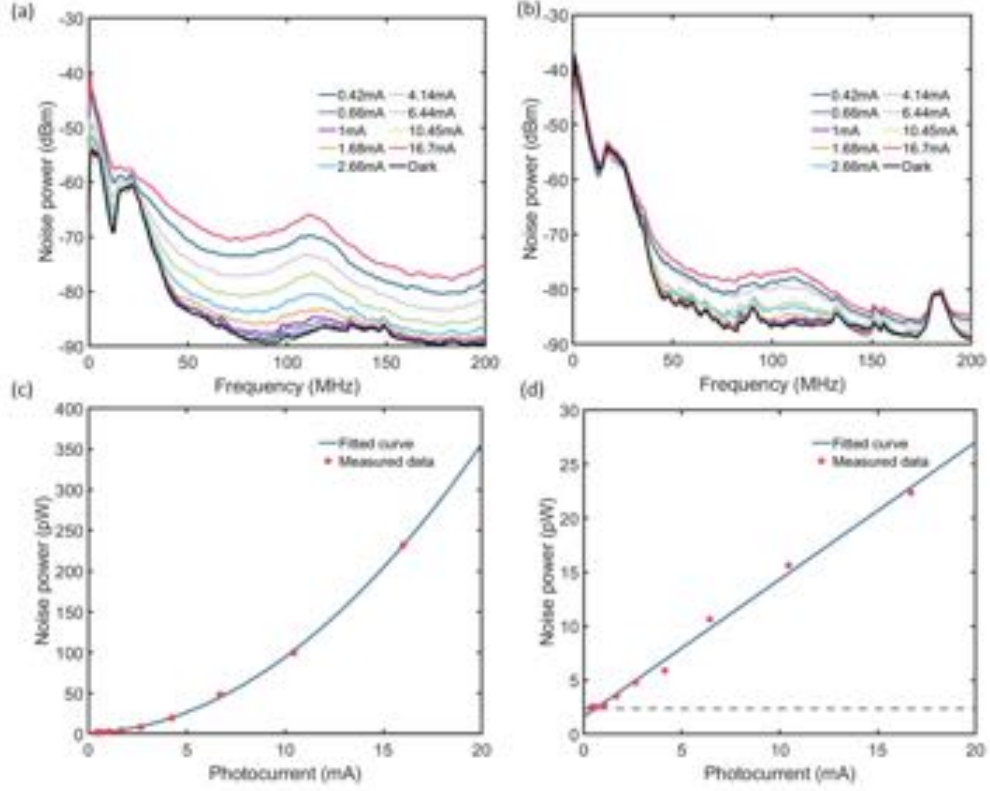


Figure 65: Noise measurements. (a) Noise spectrum of a single photodiode in the balanced detector at different photocurrents. (b) Noise spectrum of the balanced photodiodes at different photocurrents. (c) Noise power vs. photocurrent of the single photodiode at 93 MHz. (d) Noise power vs. photocurrent of the balanced photodiodes at 93 MHz. All measurements were done with 100 kHz RBW.

Figure 65 shows the noise measurements of the photodiode with FSU ESA. (a) and (c) are plots for a single photodiode and (b) and (d) are plots for balanced photodiodes. These measurements were done by sending a CW light into the MMI at one input and measure the output of the photodiode at different optical input power with increasing bias from 1V to 4V starting from dark which is the system noise. For the single photodiode, only two of the three electrodes were probed meaning only probing one photodiode and for the balanced photodiodes measurements, all there electrodes were probed at the same time. The dark curves in (a) and (b) show the system dark noise. Since the single photodiode configuration does not suppress any noise, therefore all classical noise including the laser

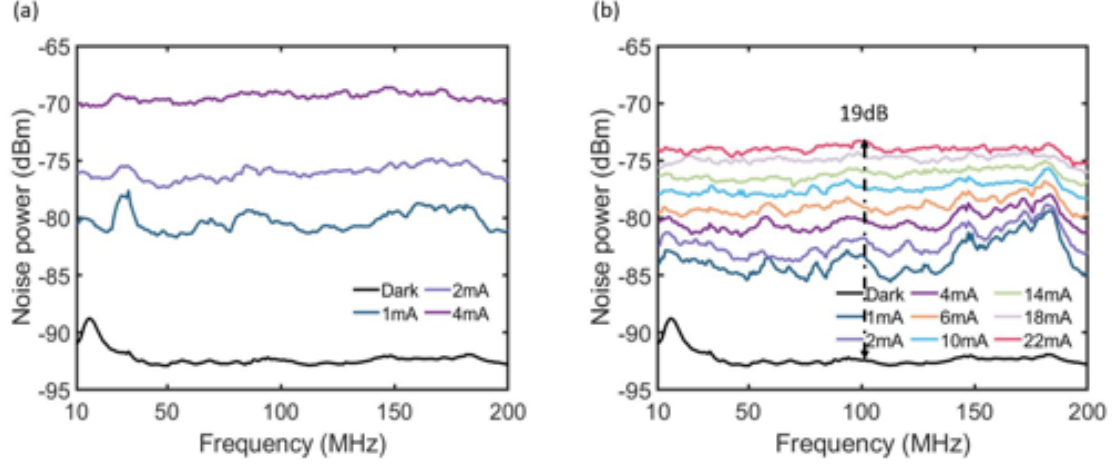


Figure 66: Noise measurements with FSU ESA (a) Noise spectrum of a single photodiode in the balanced detector at different photocurrents. (b) Noise spectrum of the balanced photodiodes at different photocurrents. All measurements were done with 100 kHz RBW.

relative intensity noise, EDFA amplified spontaneous emission and other technical sources are shown in the result. In (b) and (d), since it is in the homodyne configuration, classical noise from the system is significantly suppressed, except for the shot noise. (c) shows the relationship between photocurrent and noise power in the single photodiode measurement being quadratically related, as expected, indicating it is not shot noise limited. (d) shows a linear fit between the noise power and the photocurrent, indicating this balanced photodiode is operated in the shot noise limited regime. The distance between the highest photocurrent corresponding noise power and the system dark noise is the quantum clearance for this homodyne detector. This quantum clearance decreases at higher frequency because of the bandwidth limitation of the device and amplifier. Figure 66 shows the similar measurements as in figure 65 but using the Agilent ESA with lower noise floor and with 22mA photocurrent, 19dB quantum clearance was achieved as shown in the fig 66 (b).

In summary, this work demonstrated fully integrated balanced photodiodes for quantum signal homodyne detection on the same SiN photonic chip with 19 dB quantum clearance at 100 MHz, 31 dB CMRR and 100 MHz 3dB bandwidth.

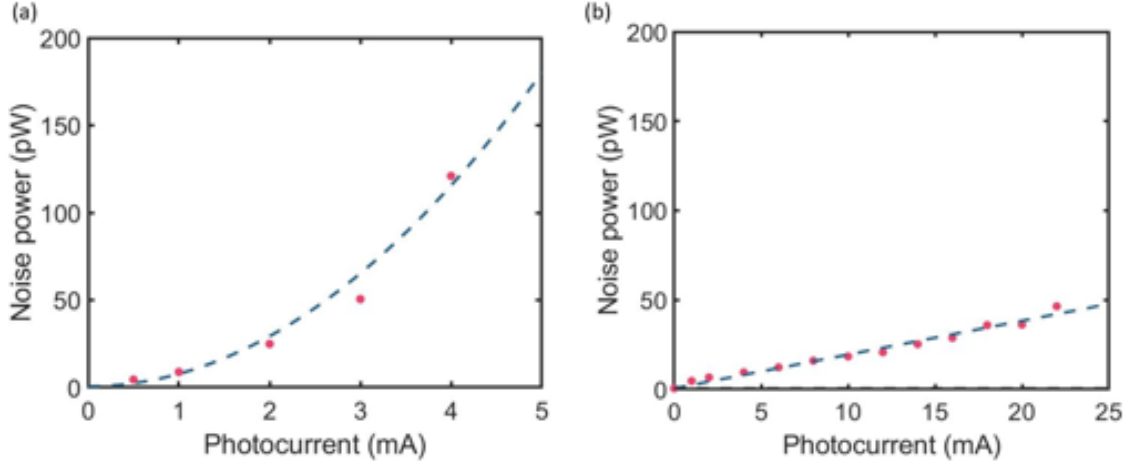


Figure 67: Noise measurements with Agilent ESA (a) Noise power vs. photocurrent of the single photodiode at 100 MHz. (b) Noise power vs. photocurrent of the balanced photodiodes at 100 MHz. All measurements were done with 100 kHz RBW.

8 Towards photon number resolution single photon counting detectors

An ideal single photon detector should be operated in room temperature, with high QE and photon number resolution (PNR). PNR here means that the detector produces a signal proportional to the number of incident photons. Photonnumber-resolving detectors have been realized with superconducting transition-edge sensors (TES), silicon photomultipliers, superconducting nanowires, linear mode avalanche photodiodes (SPADs), and quantum-dot field-effect transistors [65]. Moreover, methods based on spatial-multiplexing of non-PNR detectors have been proposed for PNR measurements using SPADs operated in Geiger mode. The following chapter describes two designs to tackle spatial multiplexing using integrated photonics and Geiger mode avalanche photodiodes (APDs), which do not provide intrinsic PNR. The first one is the tree structure, this design splits the photon stream several times using directional couplers arranged in a tree like structure. At each leaf node, and with proper scaling of the tree, only one photon is left which is then absorbed by an APD with near 100 %QE. Since directional couplers can achieve very low excess loss and with APDs approaching 100% QE, this architecture is a promising candidate for PNR. Another design is the segmented waveguide photodiode [66]. This structure also utilizes the lossless coupling behavior of light between two waveguides. Here one waveguide consists of a very thin layer of absorber

and thus each time when light is coupled to the absorber waveguide a small percentage is absorbed and being detected [65].

8.1 Segmented waveguide SPAD

8.1.1 Design

By utilizing the structure of segmented photodiodes, photon number resolution detectors can be realized on an integrated photonics platform using Geiger mode APDs [[66], [65]]. The length of the photodiode is chosen to be equal to the beat length

$$l = 2/(\Delta\beta_r) \quad (35)$$

in order to have lossless coupling between the photodiode and the waveguide. The waveguide is chosen to be $1.2 \mu\text{m}$ wide for single mode operation. This is because the beat length is calculated using the fundamental modes and by having higher order modes, the coupling between the photodiode and the waveguide may become less efficient. Figure 68 shows the epitaxial structure that designed by former student Qianhuan Yu. Figure 69 illustrates the fabrication process flow of this segmented waveguide photodiode for single photon counting and the floor plan of a single chip.

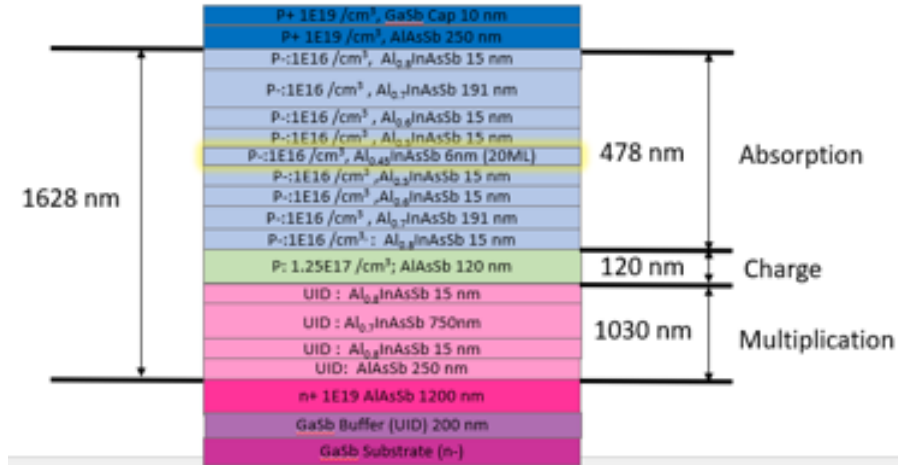


Figure 68: Epitaxial structures of the segmented photodiodes designed by Qianhuan Yu

Figure 70 shows the fabricated chip with $10 \mu\text{m}$ wide waveguide along with other circular test structure using wet etch.

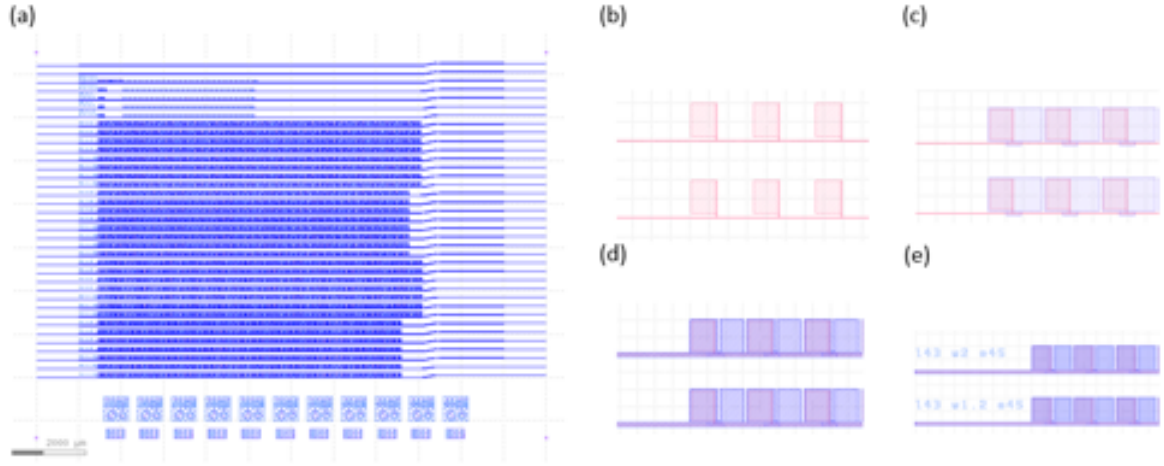


Figure 69: Segmented photodiodes layout and fabrication process. (a) Floor plan. (b)-(e) Fabrication process (b)Waveguide etch, (c)P-mesa etch, (d) N-mesa etch, and (e)Metal contact.

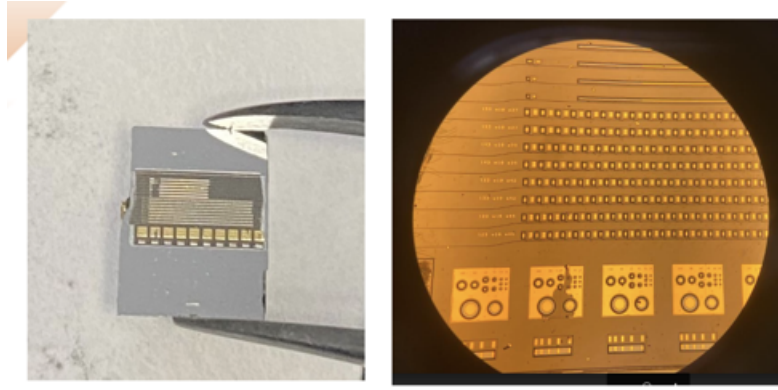


Figure 70: Fabricated segmented waveguide photodiode chip with 10 μm wide waveguides

8.1.2 Preliminary measurements

Figure 71 shows the IV and gain characteristics of different device areas. The punch through voltage is around 31 V and the break down voltage is around 61 V.

Figure 72 shows the CV characteristic with doping profile extracted. Comparing to the epitaxial structure design, it can be seen that the doping profile is relatively close to the design.

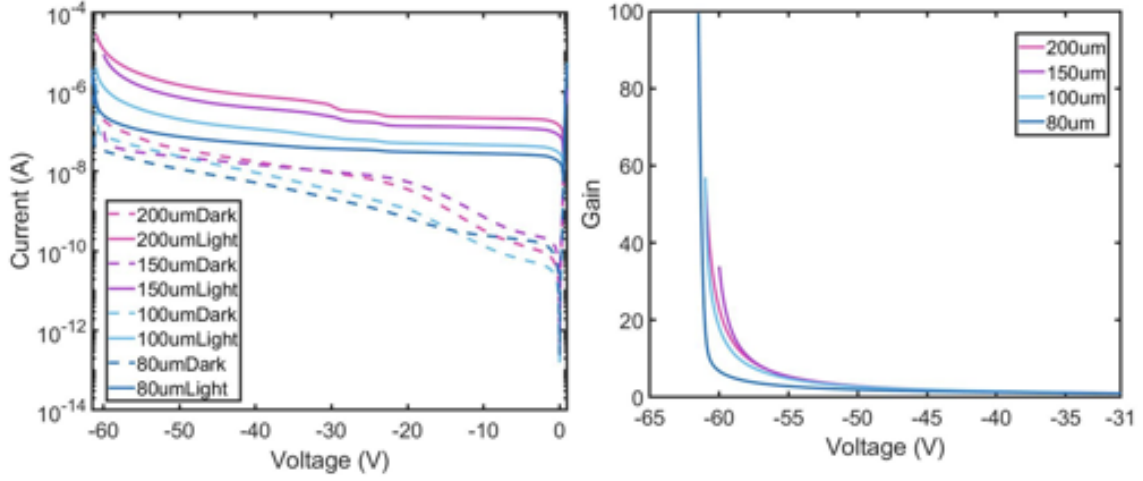


Figure 71: IV and gain characteristic

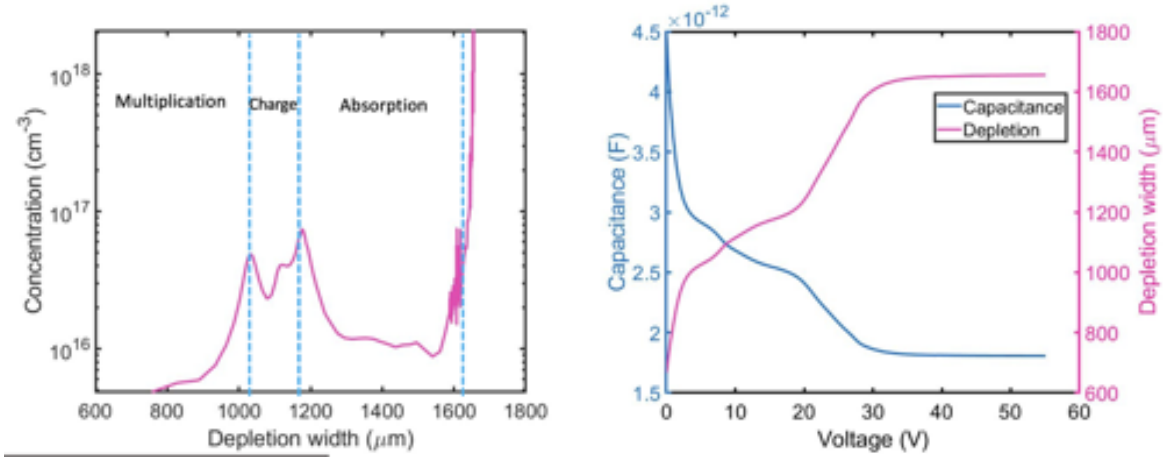


Figure 72: CV characteristic and extracted doping profile

8.2 SACM APD integrated on SiN photonic platform

8.2.1 Design

As shown in figure 73(a), this design utilizes a binary tree structure to split light into 2^N branches where N is the total number of levels in the tree. Each split is done by a 50/50 directional coupler to minimize the loss. The probability of one photon hitting one of the end nodes is $1/2^N$. Here, the photodiode is designed to be 100% QE. Each metal pad is designed to be wire bonded to a PCB to process the output current signal.

Figure 73(b) shows the half tree structure, each PD is connected to different level of the tree structure. With an N level half tree, the probability of one photon hitting the middle PD is $1/2$, and the node next to the middle with the probability of $1/2^m$ (where $1 < m < n$).

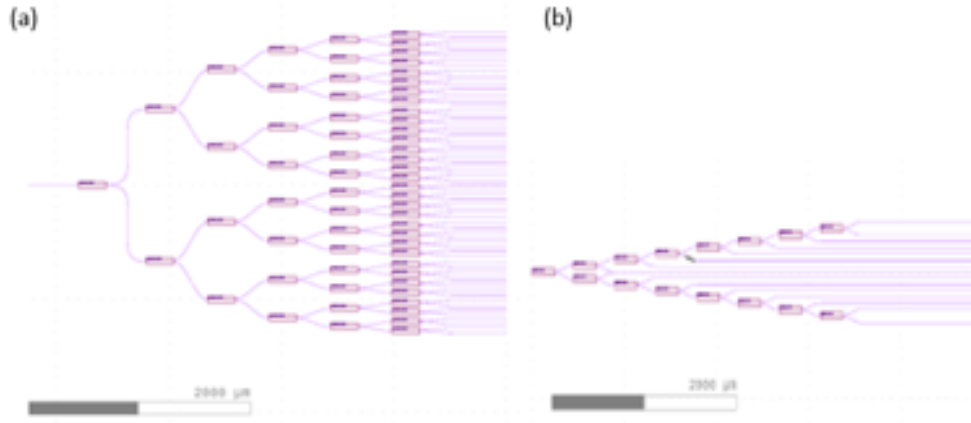


Figure 73: Tree structure (a) Full tree (b)Half tree

Since for avalanche photodiodes, it is important to have low multiplication noise and low dark current especially for single photon counting purposes. Separated absorber charge and multiplication (SACM) structure has been widely used for decades. This structure utilizes a doped charge layer in between the multiplication and absorber to control the electric field in them making the electric field in the multiplication high to reach break down and the electric field in the absorber layer low since this is the primary source of dark current [67]. Besides quantum efficiency, there are two more parameters to optimize, one is the punch through voltage and the other one is the breakdown voltage. The punch through voltage can be quantified as the minimum applied voltage when the field in the absorber starts to rise. Any bias voltage below this punch through voltage will result in close to zero electric field in the absorber and therefore the generated carriers can not be collected resulting close to zero quantum efficiency. The breakdown voltage refers to the applied bias needed for the field in the multiplication layer reaches the material break down field. It is important to make sure that the multiplication layer breaks down before the absorber. For single photon counting, usually the device is cooled to decrease the dark count rate. Since the material break down field decreases with temperature, the break down voltage also follows this trend. Therefore, it is important to keep the break down voltage relatively far away from the punch through voltage to ensure there is enough room for temperature decrease and the APD can punch through before break down.

Avalanche multiplication can be described in terms of the electron and hole ionization

coefficients, α and β respectively. The ionization coefficient ratio k is defined as, $k = \frac{\beta}{\alpha}$. To reduce excess noise, one of the earliest approaches is to select the semiconductor with favorable impact ionization coefficients [67]. For InP, the k is 0.4 which means a hole multiplication process should result in lower excess noise. Figure 74 shows the designed epitaxial structure for hole impact ionization favored, 17V punch through voltage and 30V break down voltage.

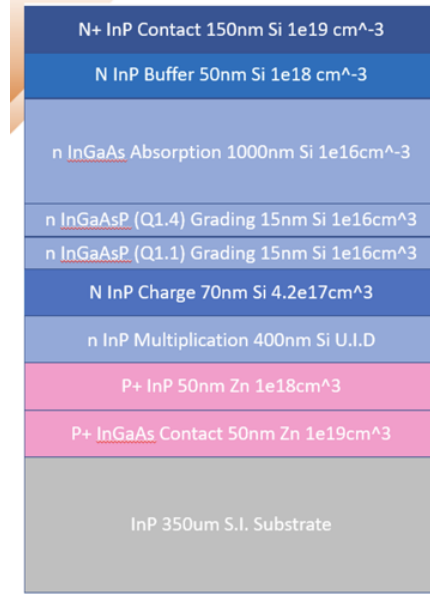


Figure 74: Epitaxial structure of an SACM APD

8.2.2 Measurements

Figure 75 and figure 76 show the IV and CV characteristic of simple mesa devices for top illumination.

Figure 77 shows the IV characteristic of both dark and the total photocurrent with SiN waveguide coupled light. The gain was extracted at from this curve as shown in figure 78

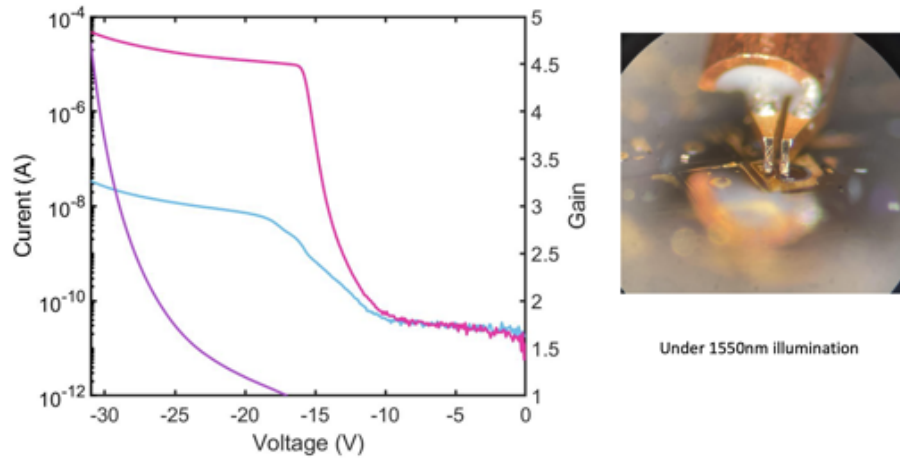


Figure 75: IV and gain characteristic under 1550nm top illumination

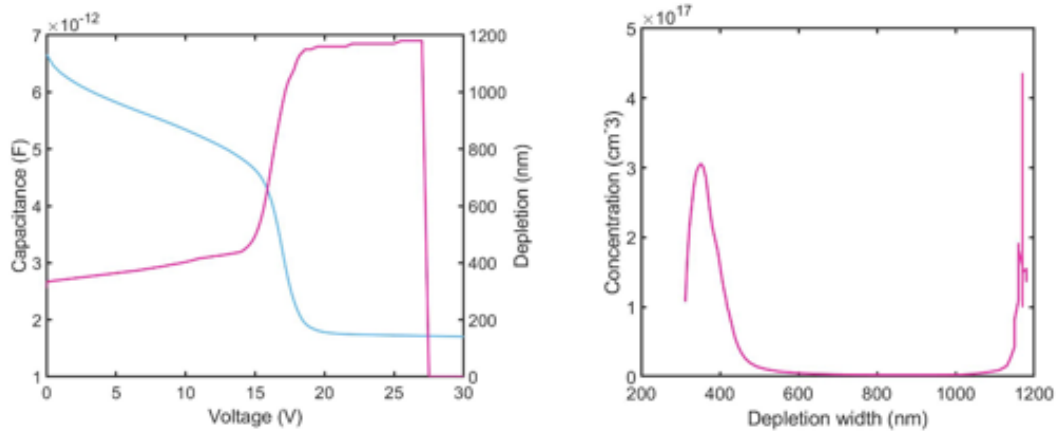


Figure 76: CV characteristic and extracted depletion depth

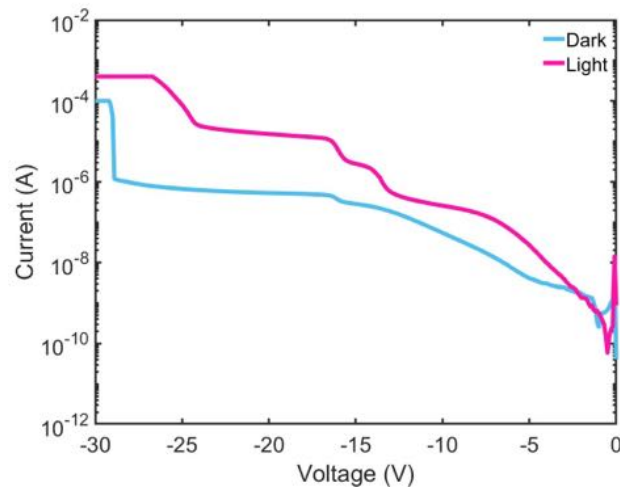


Figure 77: IV characteristic of integrated SACM APD on SiN waveguide

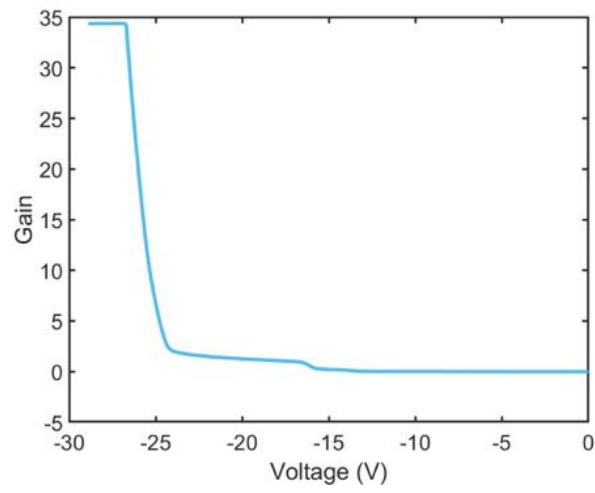


Figure 78: Gain characteristic of integrated SACM APD on SiN

9 List of publications

1] **X. Guo**, L. Shao, L. He, K. Luke, J. Morgan, K. Sun, J. Gao, T.-C. Tzu, Y. Shen, D. Chen et al., “High-performance modified uni-traveling carrier photodiode integrated on a thin-film lithium niobate platform,” *Photonics Research*, vol. 10, no. 6, pp. 1338–1343, 2022.

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[3] **X. Guo** and X. Yi, “High precision atom-referenced and fully frequency- stabilized soliton microcomb,” *Science China Physics, Mechanics & Astronomy*, vol. 67, no. 2, p. 224263, 2024.

[4] **X. Guo** and A. Beling, “High-quantum efficiency and high-quantum clearance homodyne-detector integrated on SiN,” manuscript in preparation.

[5] **X. Guo** and A. Beling, “Fully integrated high-speed on-chip photonic analog RF link on thin-film lithuim niobate,” manuscript in preparation.

[6] Y. Hu, Y. Song, X. Zhu, **X. Guo**, S. Lu, Q. Zhang, L. He, C. Franken, K. Powell, H. Warner, D. Assumpcao, D. Renaud, Y. Wang, A. Shams-Ansari, X. Li, R. Cheng, K. Luke, K. Yang, G. Barbastathis, M. Zhang, D. Zhu, A. Beling, N. Sinclair, and M. Loncar, “Integrated photonic matrix-vector-multiplication accelerator in a thin-film lithium niobate circuit,” submitted to *Nature*.

[7] B. Guo, M. Schwartz, S. H. Kodati, K. M. McNicholas, H. Jung, S. Lee, J. Konowitch, D. Chen, J. Bai, **X. Guo** et al., “Low-noise ingaas/alnassb avalanche photodiodes on inp substrates,” in *2023 IEEE Photonics Conference (IPC)*. IEEE, 2023, pp. 1–2.

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with low noise and strong temperature stability,” APL Photonics, vol. 8, no. 11, 2023.

[11] J. Bai, Y. Shen, P. Yao, D. Chen, M. Konkol, B. Guo, **X. Guo**, V. Carey, J. C. Campbell, and D. Prather, “Thermal dissipation enhancement in flip- chip bonded uni-traveling carrier photodiodes,” Optics Letters, vol. 48, no. 19, pp. 5157–5160, 2023.

10 Appendix

10.1 Epitaxial structures

10.1.1 High-speed epi with 167GHz transit-time limited bandwidth

Layer	Comment	Material	x	y	Thickness (Å)	Dopant	Level (/cm ³)	Type
14	N+ contact	InP			1,500	Si	2.0E+19	n+
13	Etch stop	In(x)Ga(1-x)As	0.53		50	Si	2.0E+19	n+
12	N-InP	InP			700	Si	1.0E+18	n-
11	Drift layer	InP			2,000	Si	1.0E+16	n-
10	Cliff layer	InP			500	Si	1.0E+17	n-
9	Grading 4(Q1.1)	In(x)Al(y)Ga(1-x-y)As	0.53	0.28	150	Si	1.0E+16	n-
8	Grading 3(Q1.4)	In(x)Al(y)Ga(1-x-y)As	0.53	0.115	150	Si	1.0E+16	n-
7	Depleted absorb	In(x)Ga(1-x)As	0.53		2,000	-	UID	i
6	Absorber 2	In(x)Ga(1-x)As	0.53		250	Be	2E18-->1.1E18	p
5	Absorber 1	In(x)Ga(1-x)As	0.53		250	Be		p
4	Grading 3(Q1.4)	In(x)Al(y)Ga(1-x-y)As	0.53	0.115	150	Be	3.0E+19	p+
3	Grading 4(Q1.1)	In(x)Al(y)Ga(1-x-y)As	0.53	0.28	150	Be	3.0E+19	p+
2	p-InP	InP			1,000	Be	1.0E+19	p+
1	Contact layer	In(x)Ga(1-x)As	0.53		500	Be	3.0E+19	p+
0	Substrate	S.I. InP						

Figure 79: Epitaxial structure information for high-speed epi with 167GHz transit-time limited bandwidth

10.1.2 High-quantum efficiency epi

Layer	Comment	Material	x	y	Thickness (Å)	Dopant	Level (/cm ³)	Type
14	N+ contact	InP			1,500	Si	2.0E+19	n+
13	Etch stop	In(x)Ga(1-x)As	0.53		50	Si	2.0E+19	n+
12	N-InP	InP			700	Si	1.0E+18	n-
11	Drift layer	InP			2,000	Si	1.0E+16	n-
10	Cliff layer	InP			500	Si	1.0E+17	n-
9	Grading 4(Q1.1)	In(x)Al(y)Ga(1-x-y)As	0.53	0.28	150	Si	1.0E+16	n-
8	Grading 3(Q1.4)	In(x)Al(y)Ga(1-x-y)As	0.53	0.115	150	Si	1.0E+16	n-
7	Depleted absorb	In(x)Ga(1-x)As	0.53		9,000	-	UID	i
6	Absorber 2	In(x)Ga(1-x)As	0.53		500	Be	2E18-->1.1E18	p
5	Absorber 1	In(x)Ga(1-x)As	0.53		500	Be		p
4	Grading 3(Q1.4)	In(x)Al(y)Ga(1-x-y)As	0.53	0.115	150	Be	3.0E+19	p+
3	Grading 4(Q1.1)	In(x)Al(y)Ga(1-x-y)As	0.53	0.28	150	Be	3.0E+19	p+
2	p-InP	InP			1,000	Be	1.0E+19	p+
1	Contact layer	In(x)Ga(1-x)As	0.53		500	Be	3.0E+19	p+
0	Substrate	S.I. InP						

Figure 80: Epitaxial structure information for high-quantum efficiency

10.1.3 Hole SACM APD epi

Layer	Comment	Material	x	y	Thickness (Å)	Dopant	Level (/cm ³)	Type
8	Contact	InP			1,500	Si	1.0E+19	n+
7	Buffer	InP			500	Si	1.0E+18	n
6	Absorption	In(x)Ga(1-x)As	0.53		10,000	Si	1.0E+16	n
5	Grading	In(x)P(y)Ga(1-x-y)As	0.53		300	Si	1.0E+16	n
4	Charge	InP			700	Si	4.2E+17	n
3	Multiplication	InP			4,000	-	UID	i
2	p-InP	InP			500	Zn	1.0E+18	p+
1	Contact layer	In(x)Ga(1-x)As	0.53		500	Zn	1.0E+19	p+
0	Substrate	S.I. InP						

Figure 81: Epitaxial structure information for SACM APD

10.2 Fabrication recipes

10.2.1 Bonding and substrate removal

Bonding

1. Clean both passive waveguide chip and IIIV epi with acetone, methonal and IPA.
2. Surface activation by putting both chips in O2 plasma for more than 1 hour.
3. Spin SU-8 with 5% solid content on waveguide chip with 9600 rpm for 40S.
4. Softbake SU-8 60S at 100C.
5. Flood expose SU-8
6. Put IIIV epi on photonic chip
7. Use Suss bonder SU8 PVG recipe

Substrate removal with un-thinned substrate

1. Spin AZ5214 with 4000rpm for 40S

2. Softbake 60S at 100C
3. Expose pattern that has 200um overlap within the IIIV epi using the Microwriter with 200mJ/cm²
4. Develop using AZ 300mif for 30S until no resist is left
5. HCl:Water 3:1 for 1.5 hour until no remaining substrate.

10.2.2 Fabrication flow for non-airbridge structures

With metal hard mask for dry etch

1. Spin AZ5214 with 4000rpm for 40S
2. Softbake for 60S at 100C
3. Expose using Microwriter with 150mJ/cm² dose with pattern only on p-mesas.
4. Develop using AZ 300mif for 90S
5. Deposit Ti/Pd/Au with 10nm/80nm/20nm thickness.
6. Liftoff using Remover-PG overnight at room temperature
7. Spin AZ4620 with 4000rpm for 40S
8. Softbake for 5min at 90C
9. Expose using Microwriter with 600mJ/cm² dose with the same shape or smaller as the IIIV epi for protection from the dry etch.
10. Develop using AZ 300mif for 3min.

With photoresist mask for dry etch

1. Spin AZ4620 with 4000rpm for 40S
2. Softbake for 5min at 90C
3. Expose using Microwriter with 800mJ/cm² with negative pattern on p-mesas.
4. Develop using AZ 300mif for 3min.

Dry etch

Use Oxford ICP (find model number) Cl₂ 8.0, N₂ 2.0 ICP power 200 RF power 115W Between 4 minutes to 6 minutes. Etch until color blue. InP etch rate between 80nm/min to 170nm/min.

SU-8 Passivation

1. Spin SU-8 2002 with 5000rpm for 45S
2. Softbake for 1min 10S at 90C
3. Expose using Microwrtiter with 4800mJ/cm²
4. Postbake for 1min10S at 90C
5. Develop using SU-8 developer for 20S, rinse with IPA

RF pads

1. Spin LOR3A 3000rpm for 40S
2. Softbake 3min at 160C
3. Spin AZ5214 4000rpm for 45S
4. Softbake 1min at 100C
5. Expose using Microwriter with 200mJ/cm²
6. Develop using AZ 300mif for 2min, rinse with DI water
7. Deposit 8nm Ti and more than 400nm of Au.
8. Soak in Acetone for overnight
9. Rinse in NMP to remove LOR

Epi residue cleaning

1. Protect devices using AZ5214 following a hardbake 2min at 100C
2. Dip in InGaAs (H₃PO₄:H₂O₂:DI, 3:1:10) and InP (HCl:H₃PO₄, 5:1) wet etchant repeatably until all IIIV materials are gone

10.2.3 Fabrication flow for airbridge structures

Mesa formations are same as above for non-airbridge structures

Seed layer

1. Spin AZ5214 4000rpm for 40S
2. Softbake 1min at 110C
3. Expose using microwriter with 200mJ/cm²
4. Develop using AZ 300mif for 2min, rinse with DI water
5. O₂ plasma 200W 5min
6. Metal deposition Ti/Au 10nm/100nm

Plating layer

1. Spin AZ5214 4000rpm for 40S
2. Softbake 1min at 90C
3. Expose using Microwriter with 200mJ/cm²
4. Develop using AZ 300mif for 2min 30S, rinse with DI water

Plate

1. Pre-heat Au plating solution to 40C
2. Dip chip into DI water
3. Secure chip on to the probe
4. Use Keithley as voltage source to adjust the current reading to around 0.08mA
5. Typically takes 1 hour and 30min to plate 1.6um

Lift-off

1. Overnight O₂ plasma
2. Gold etchant for 1min or until no golden color

3. Ti etchant for 10S or until no metal left
4. Acetone bath for over night following IPA rinse
5. 10S sonicator clean

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